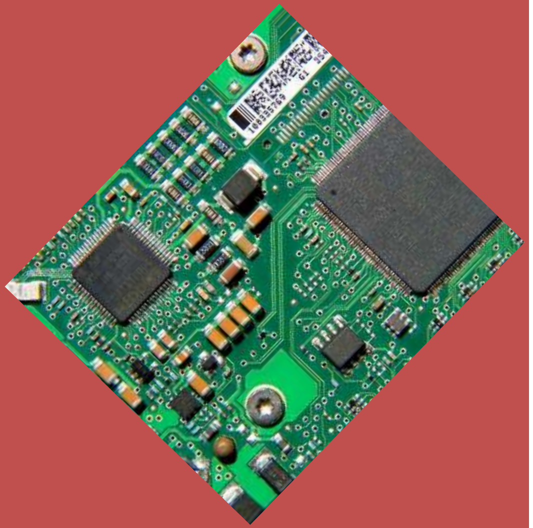


Design of printed circuit boards using Altium Designer

1.- Software Framework

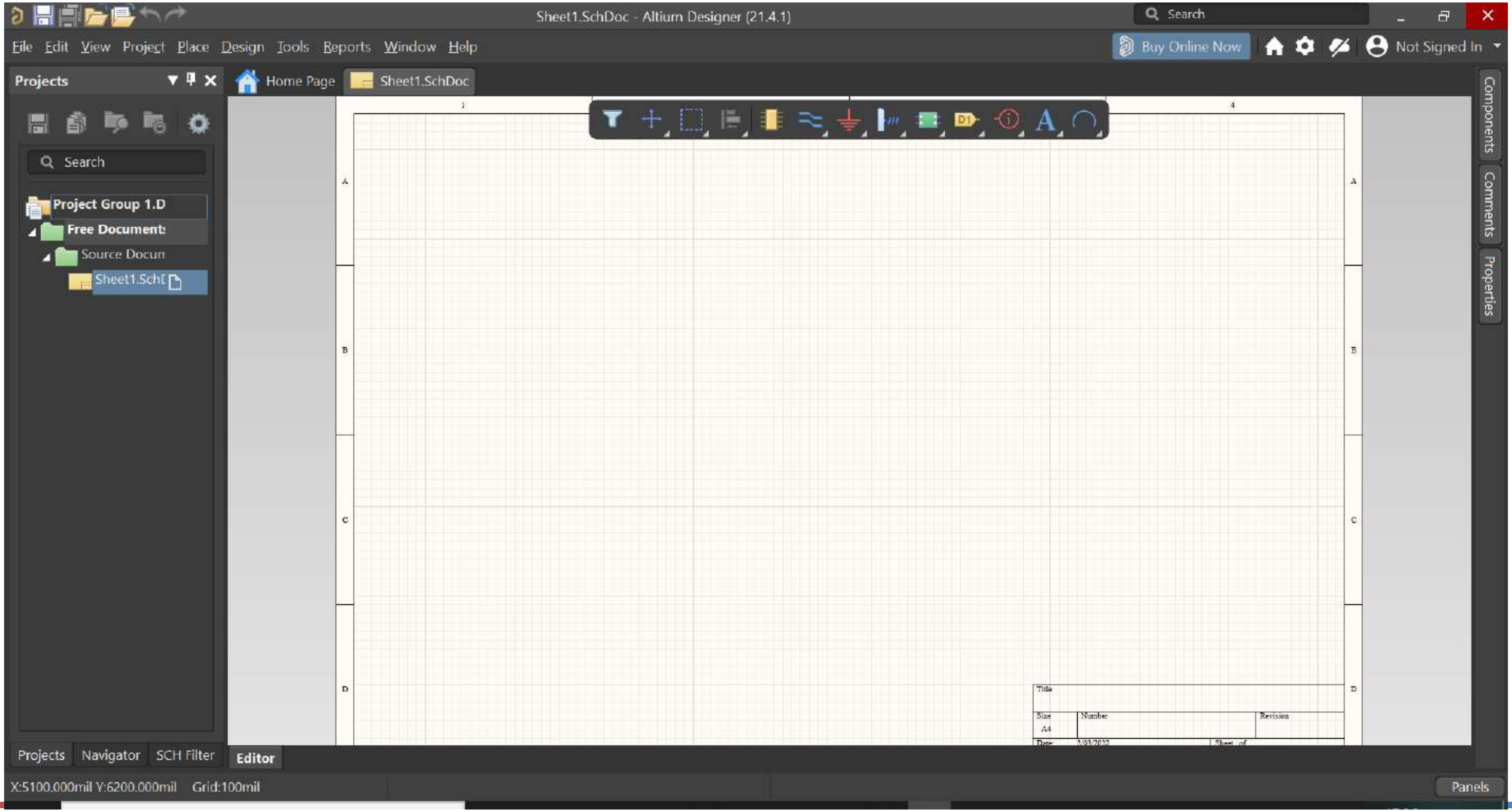
Dr. Carlos Cruz de la Torre
Dr. Ignacio Bravo Muñoz
Departamento de Electrónica
Univeresidad de Granada

Creating a project

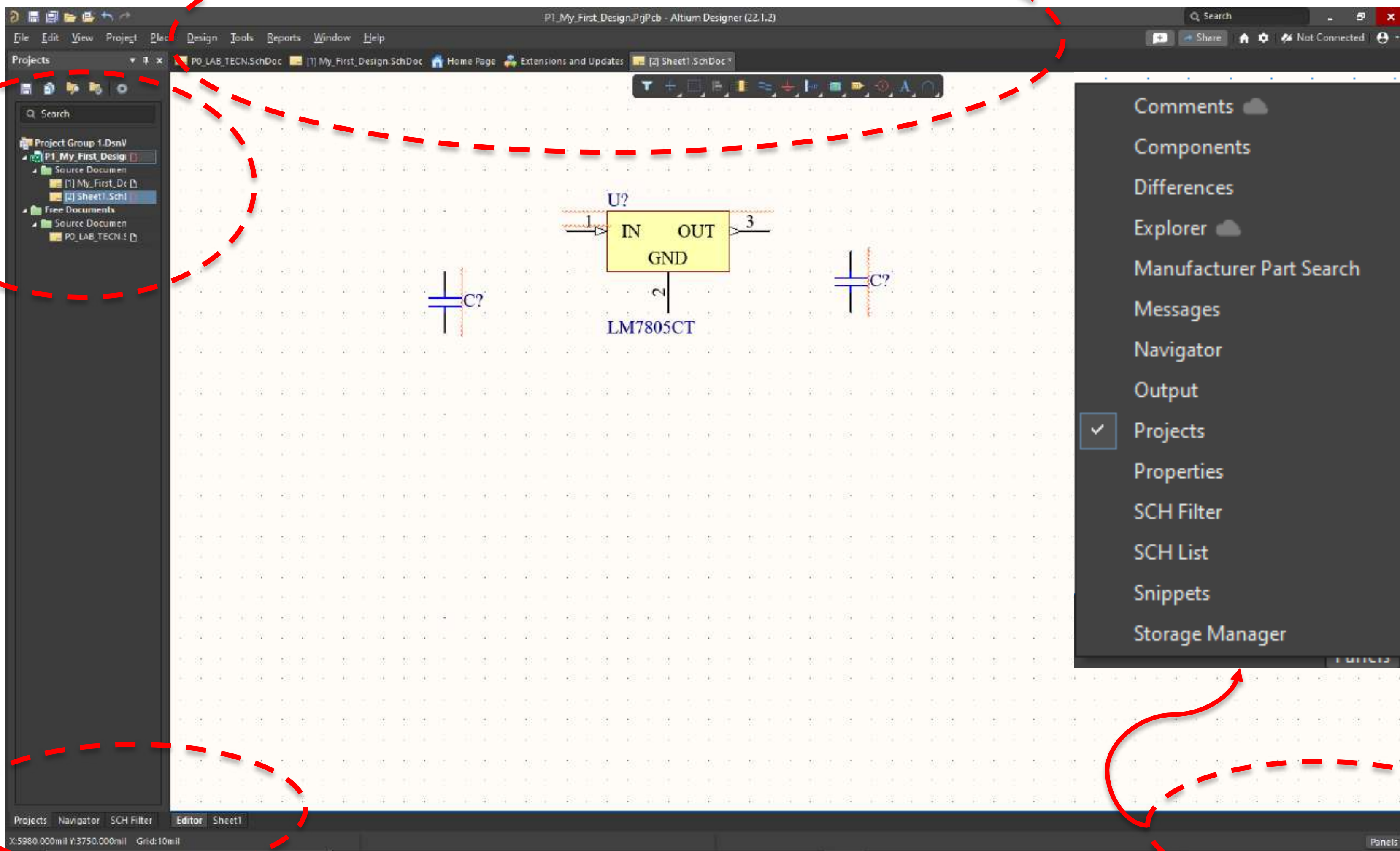
- A project can be composed by several parts:
  - Schematics
  - PCB
  - Documentation
  - Gerber files
  - Etc.
- Even we could create a Schematics (for instance) and included within a project later
- We will only address Schematics + PCB in this lab

Schematics

- Regular view of Schematics tool



Schematics



P1_My_First_Design.PrjPcb - Altium Designer (22.1.2)
File Edit View Project Place Design Tools Reports Window Help
Projects
Project Group 1.DsnW
P1_My_First_Design
Source Documents
Free Documents
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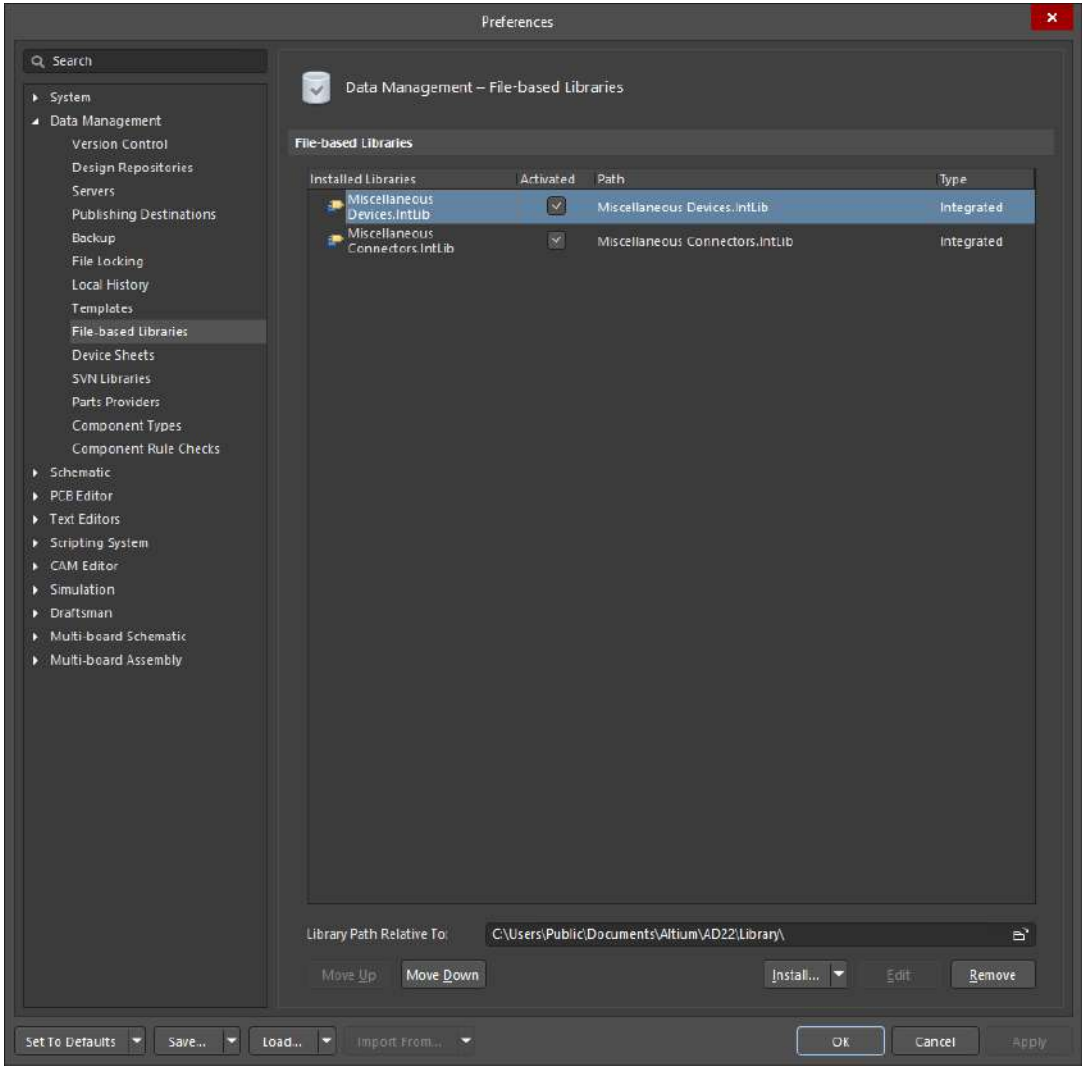
IN OUT
GND
LM7805CT
C?
C?
Comments
Components
Differences
Explorer
Manufacturer Part Search
Messages
Navigator
Output
Projects
Properties
SCH Filter
SCH List
Snippets
Storage Manager
Projects Navigator SCH Filter
Editor Sheet1
Panels

Schematics

- Schematics quick tool bar.
- It contains the most common elements to be used within a Schematics.
- We must distinguish between:
  - Physical devices (R’s, C’s, circuits, etc.)
  - Symbols (GND, Vcc, bus, entry bus, etc.)
  - Both are included inside components

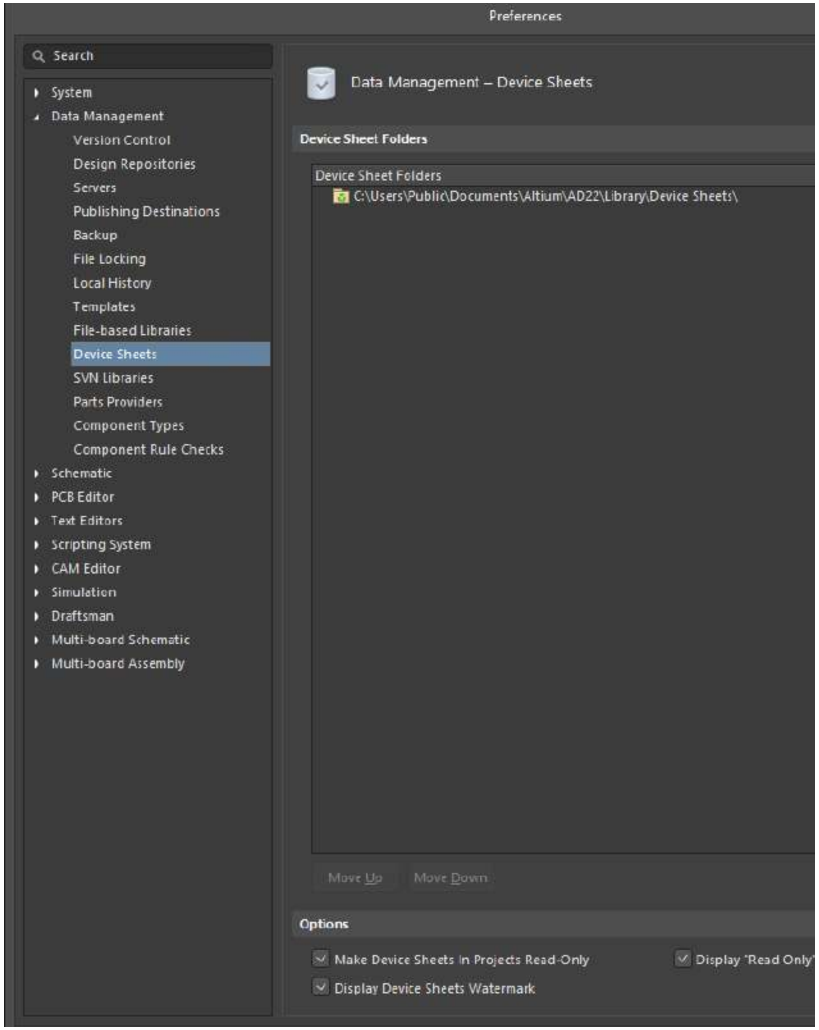
Schematics

- Tools ➔ Preferences
- Data Management ➔ File based libraries
- By default libraries included in panel of components.



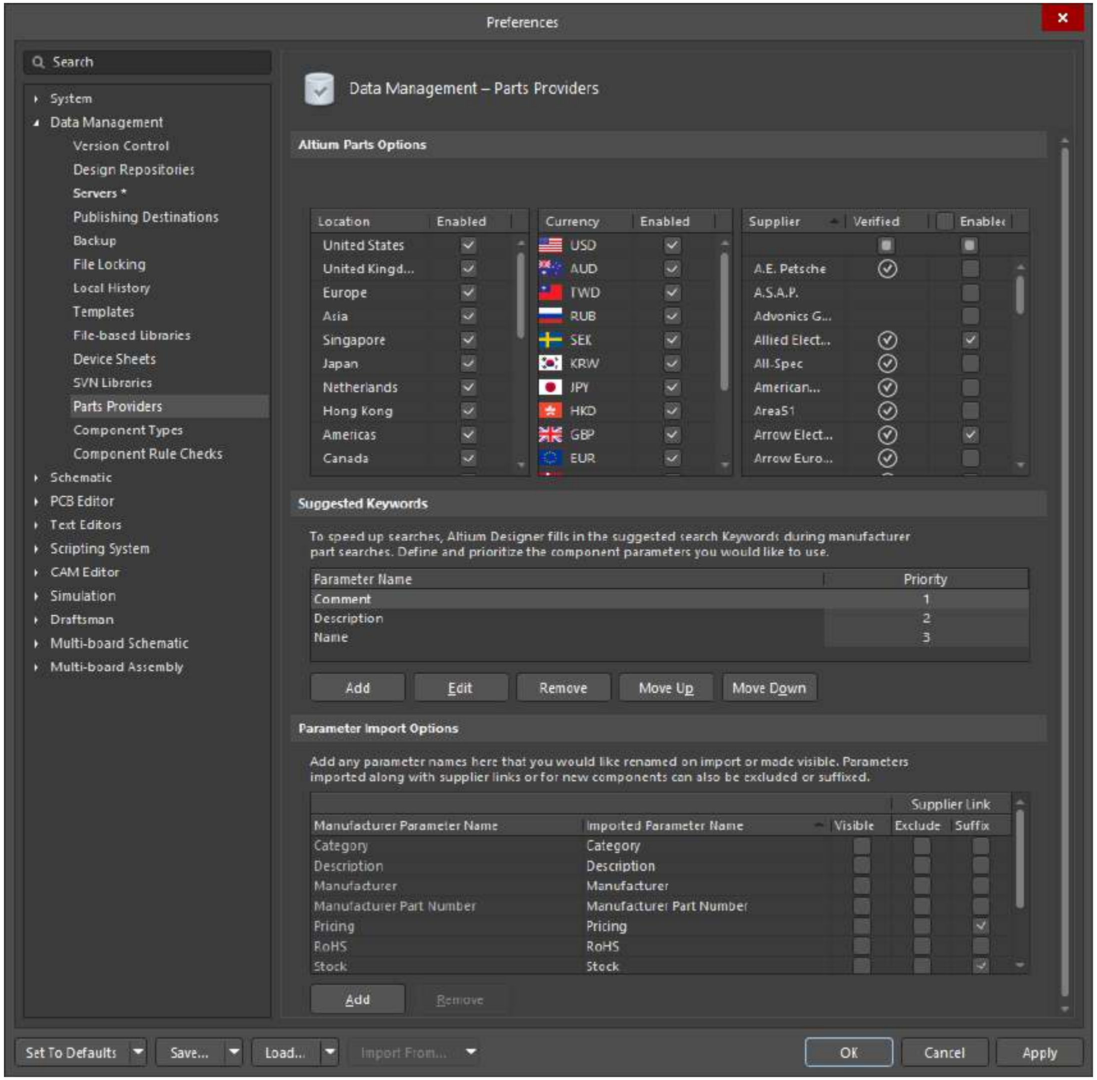
Schematics

- Data Management ➔ Device Sheets
- Interesting place for loading datasheets



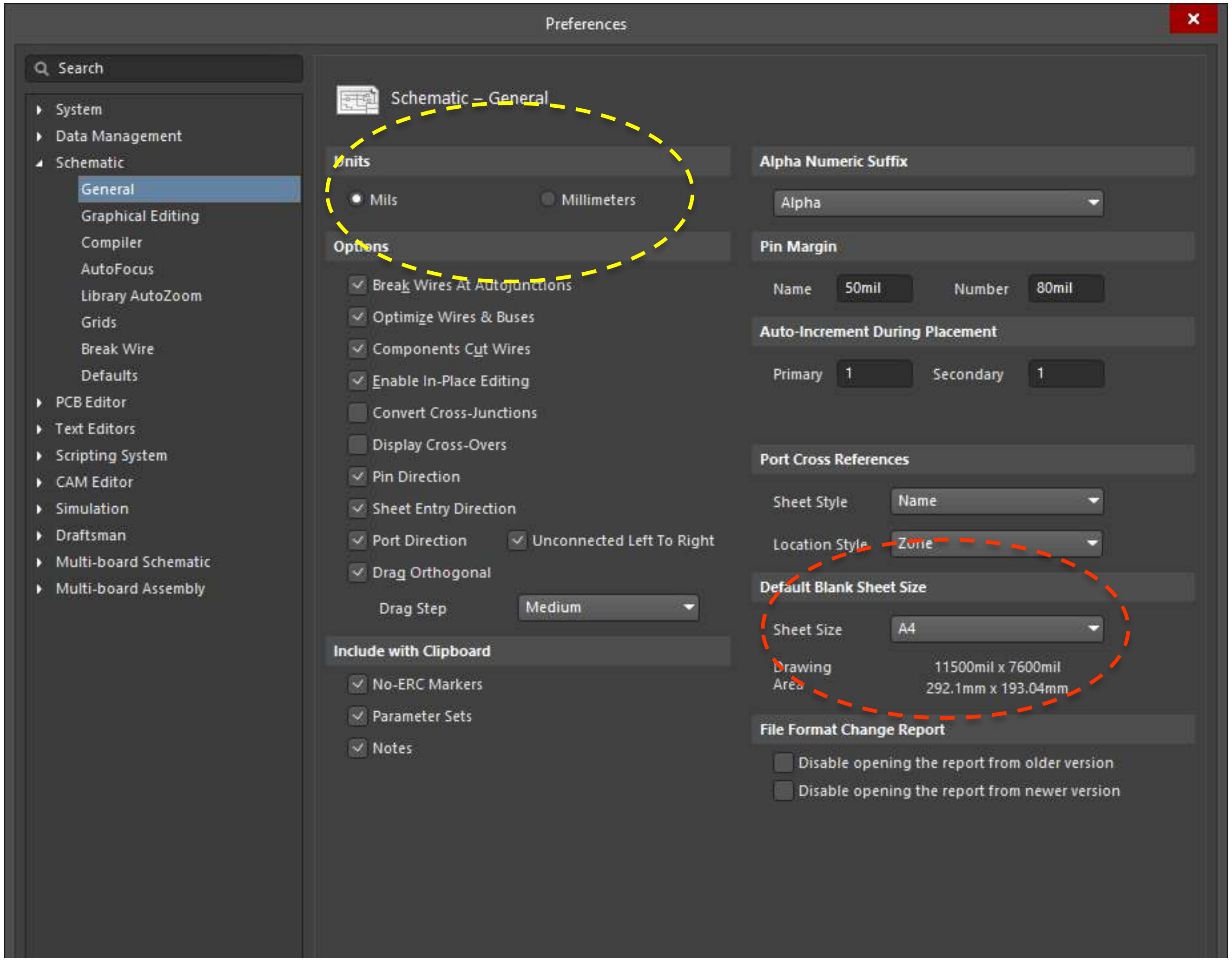
Schematics

- Path Providers: Device suppliers and currency. Relevant for BOM (Bill of Materials)



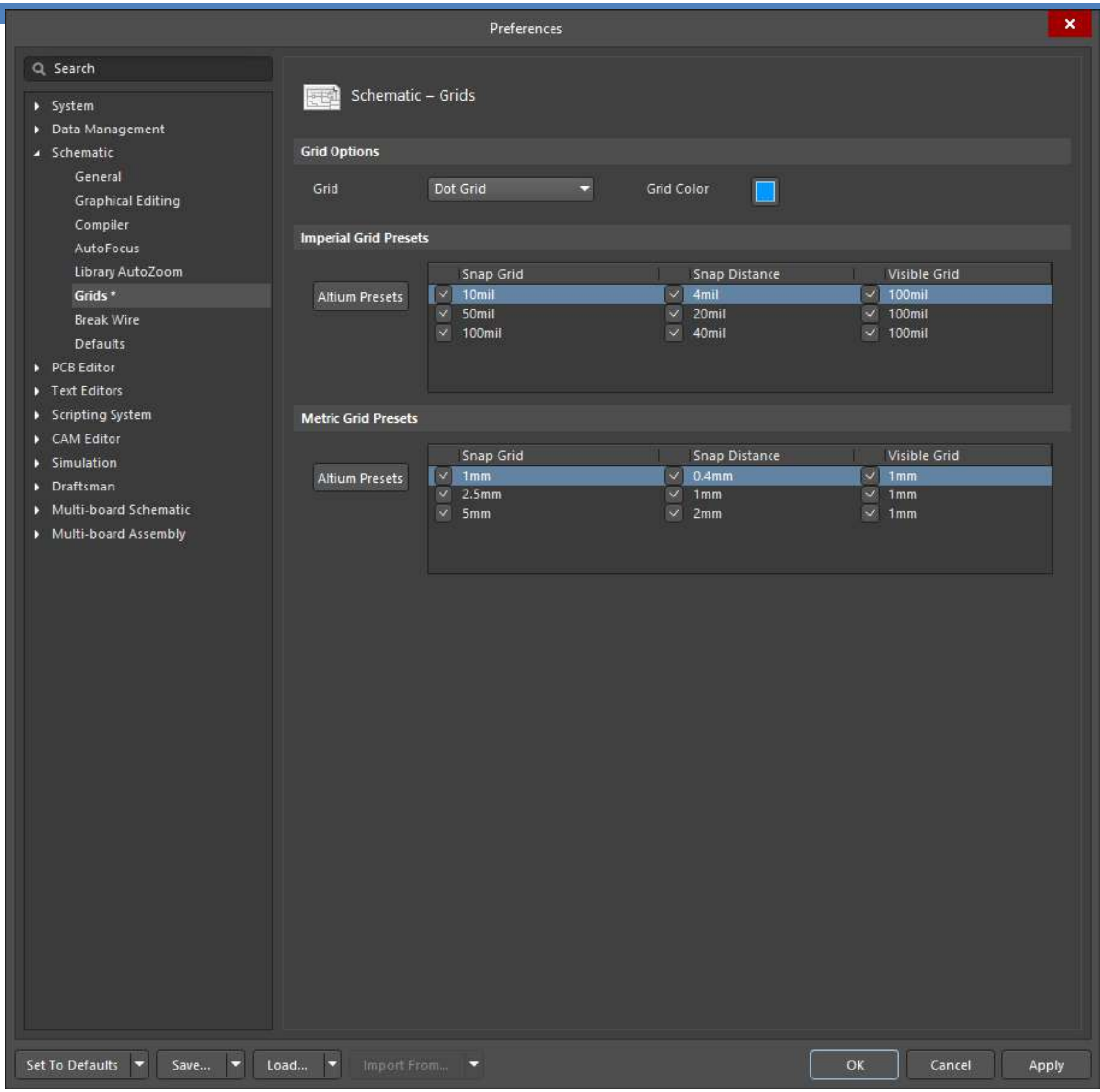
Schematics

- ☐ Schematic ➔ General
- ☐ Units: Mils (inches) or milometers
- ☐ Sheet size



Schematics

- Grid: Setup grid options.

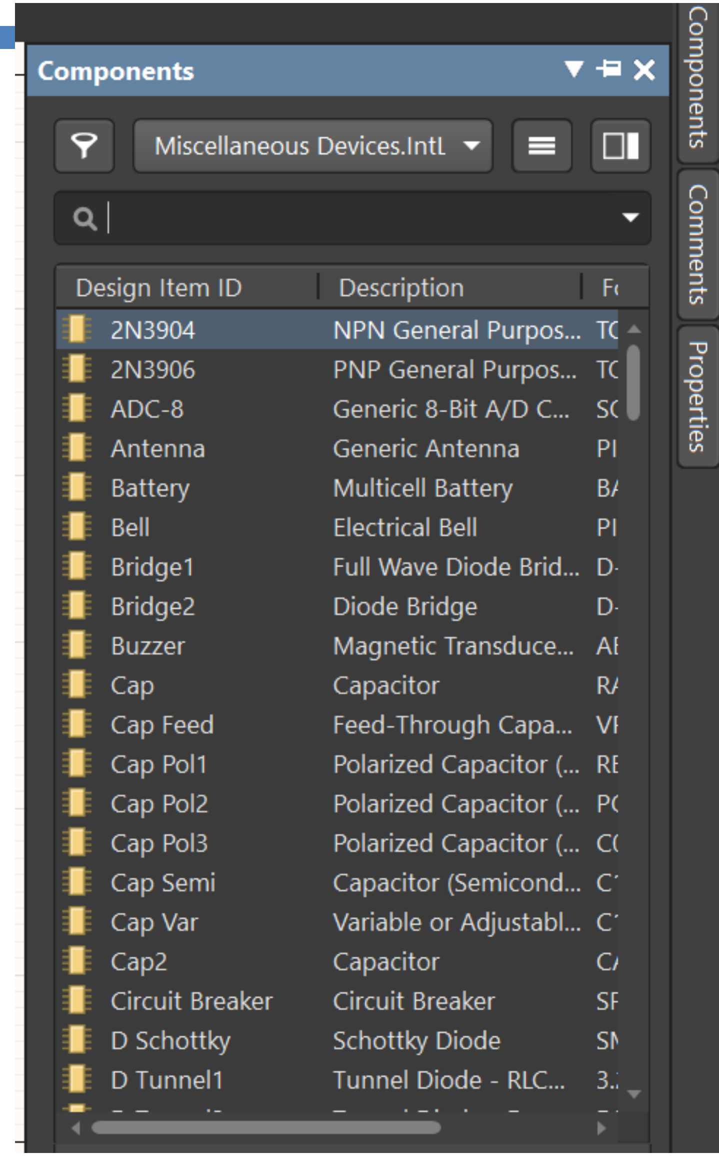


Schematics

- Some tips related to shortcuts:
  - Zoom in: CTRL + Up Wheel
  - Zoom out: CTRL + Down Wheel
  - Right click: Moving the grid / Escape (wiring)
  - Escape: Finishing mode
  - Space bar: Rotating

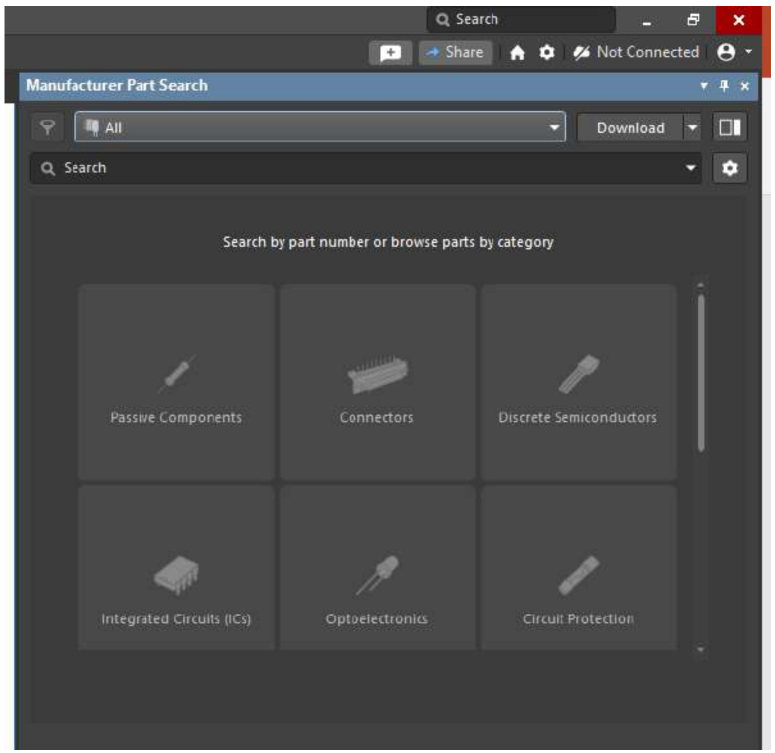
Schematics

- Components view:



Schematics

- Manufacturer Part Search



Schematics

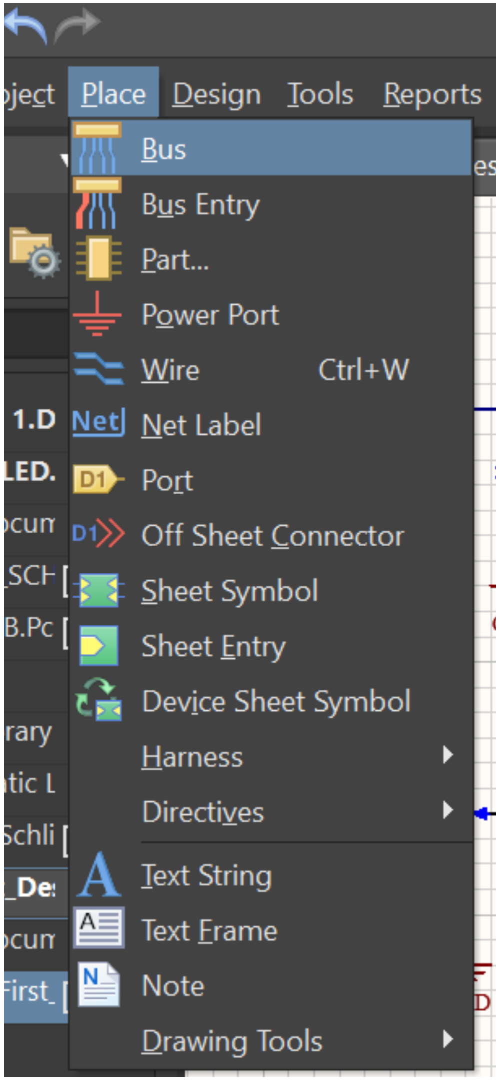
- What is a Library?
  - Place where a collection of components are included.
- May one component belong to different libraries?
  - Yes, it is. Even we can create our own library.
- What types of library can be managed?
  - Schematic Library
  - PCB Library
  - Project Library

Schematics

- What happen if we need a component and it is not included in our environment?
  - Option 1: To install the library
  - Option 2: To find the library in Altium community
  - Option 3: Create our own device/component

Schematics

- Before doing our first design:
- Difference between wire and bus:
  - Bus is a set of wires (usually related to digital signals) joined to ease the schematic view and design.
  - The connexion between the bus and the physical pin is done by a Bus Entry.
  - Wire is a single line to create an electrical connection between two points.

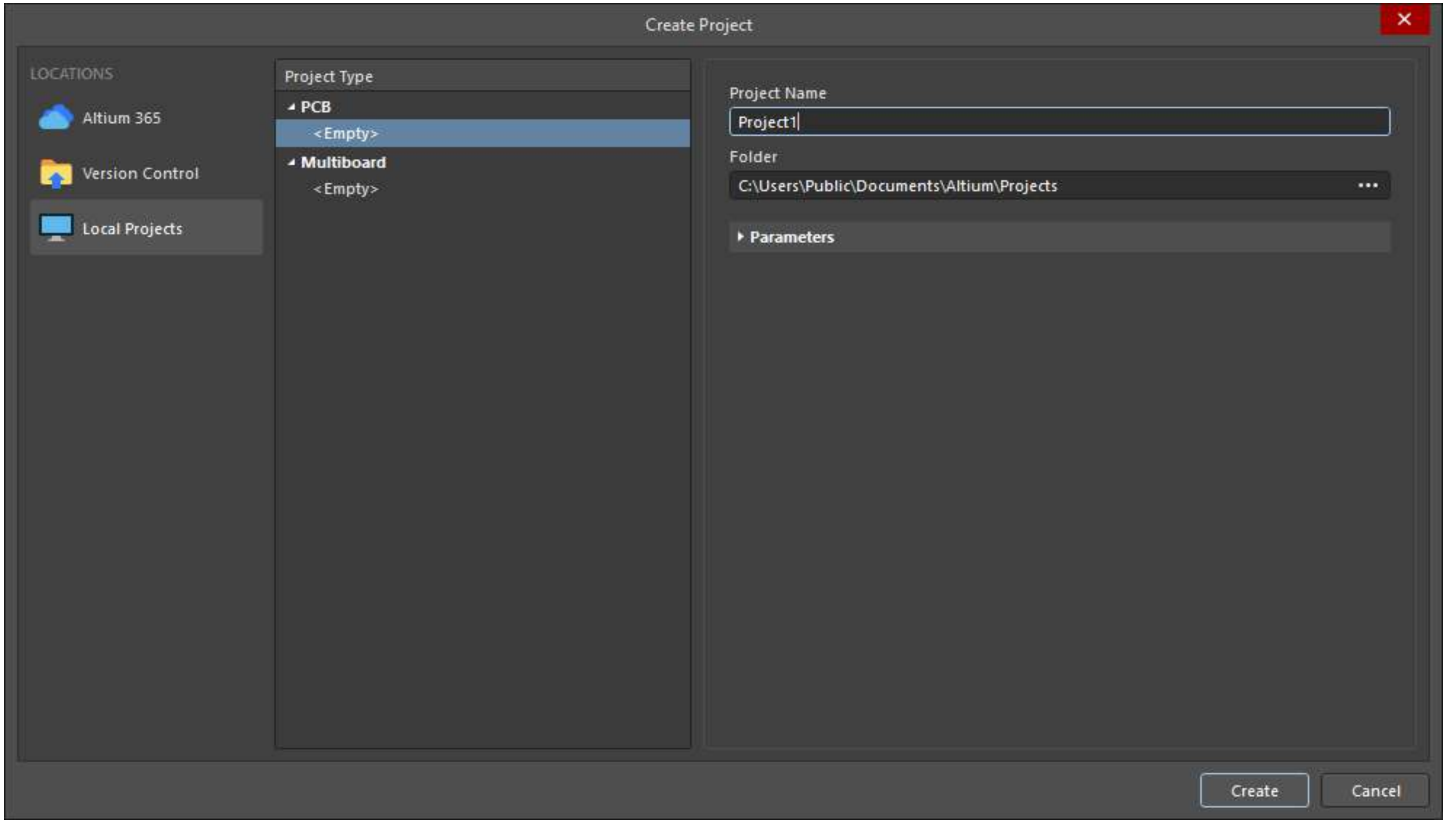


Schematics

- Other issues:
  - Net Labels: Beyond clarifying purposes, it works as an electrical connection between several parts instead of drawing a long wire.
  - V_{CC}: take care regarding the power supply. It will come from some battery, connector, etc. “It is not magical”
  - GND: it is used to have only one when we are designing an analog system. If we have digital and analog signals two grounds can be used.

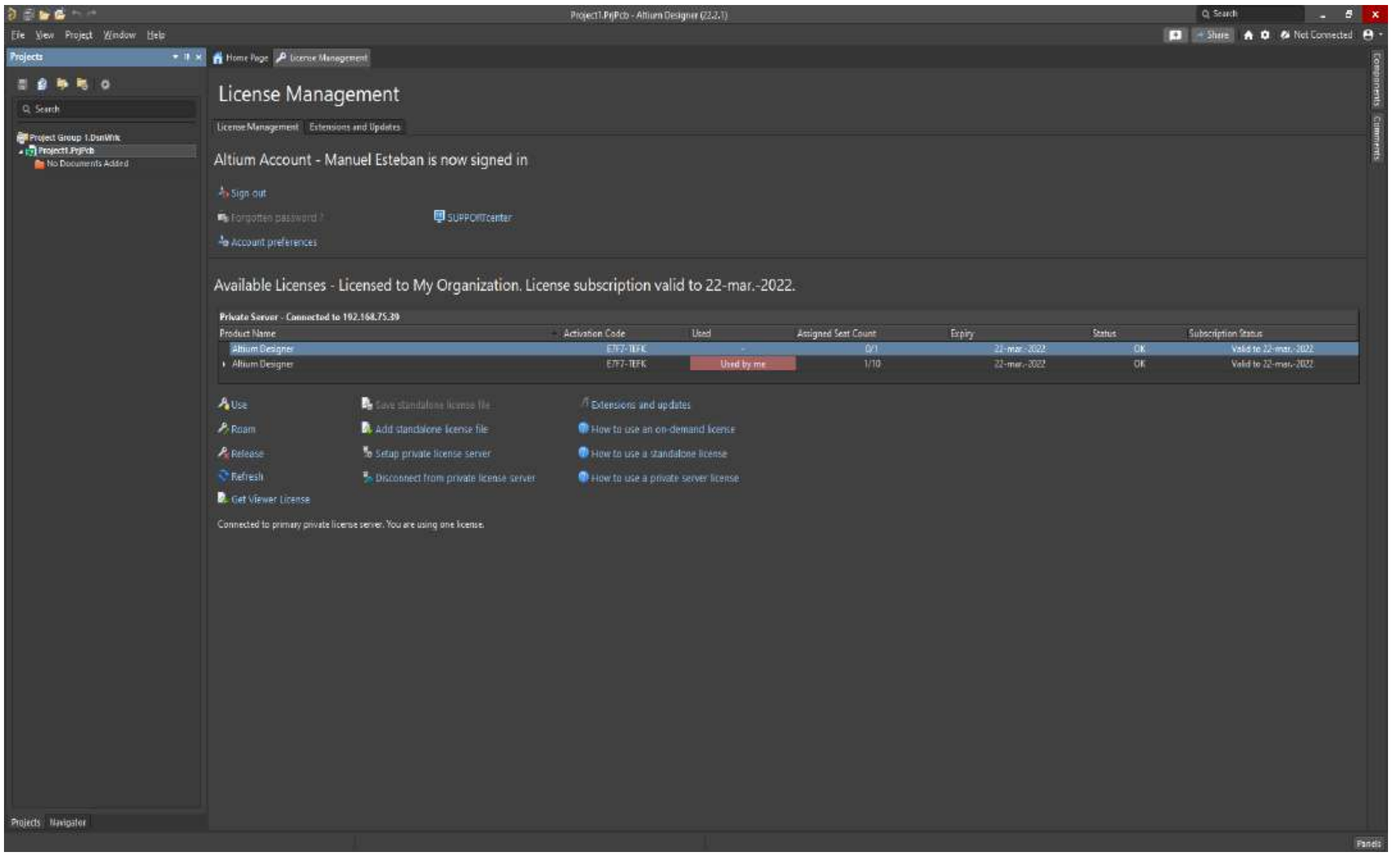
My first design

- ☐ Creating a new project



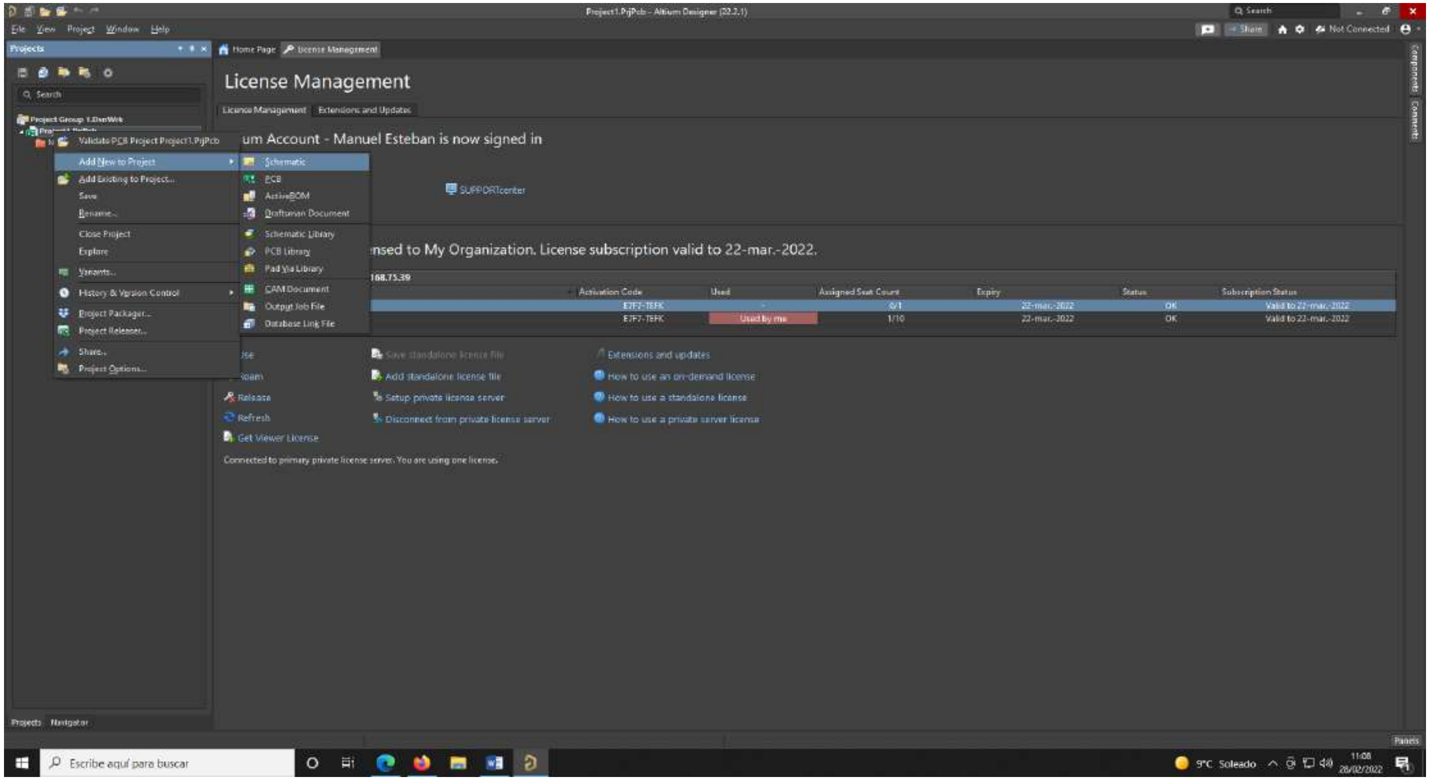
My first design

- Reviewing the license



My first design

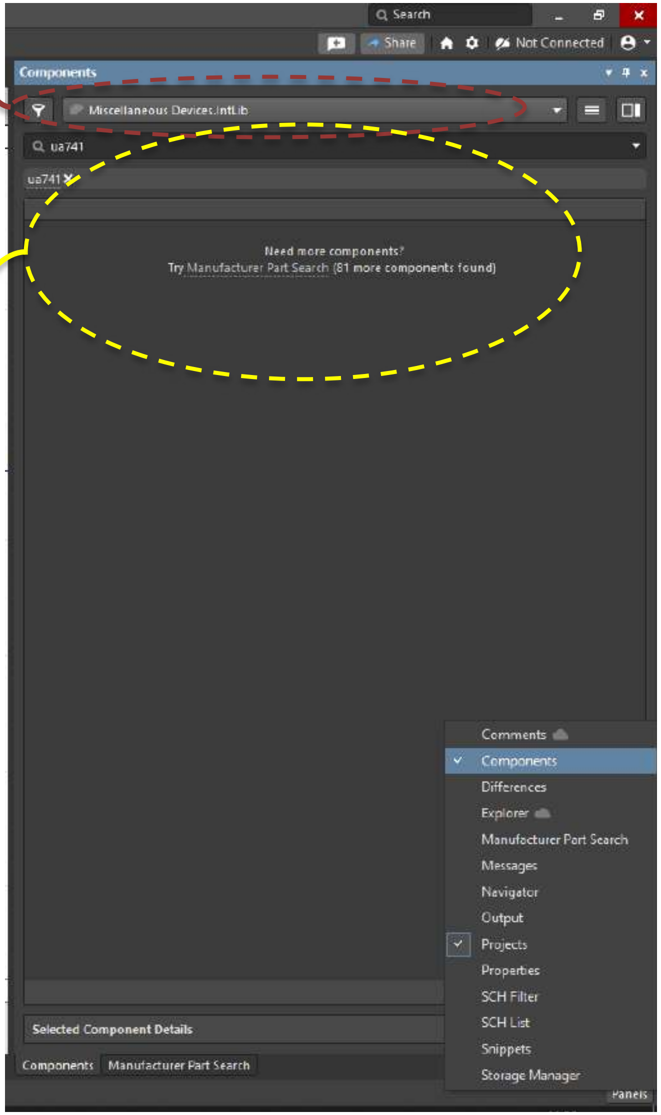
- Generating an empty schematic sheet



My first design

- ☐ Searching and selecting the components (Panels Tab ➔ Components)

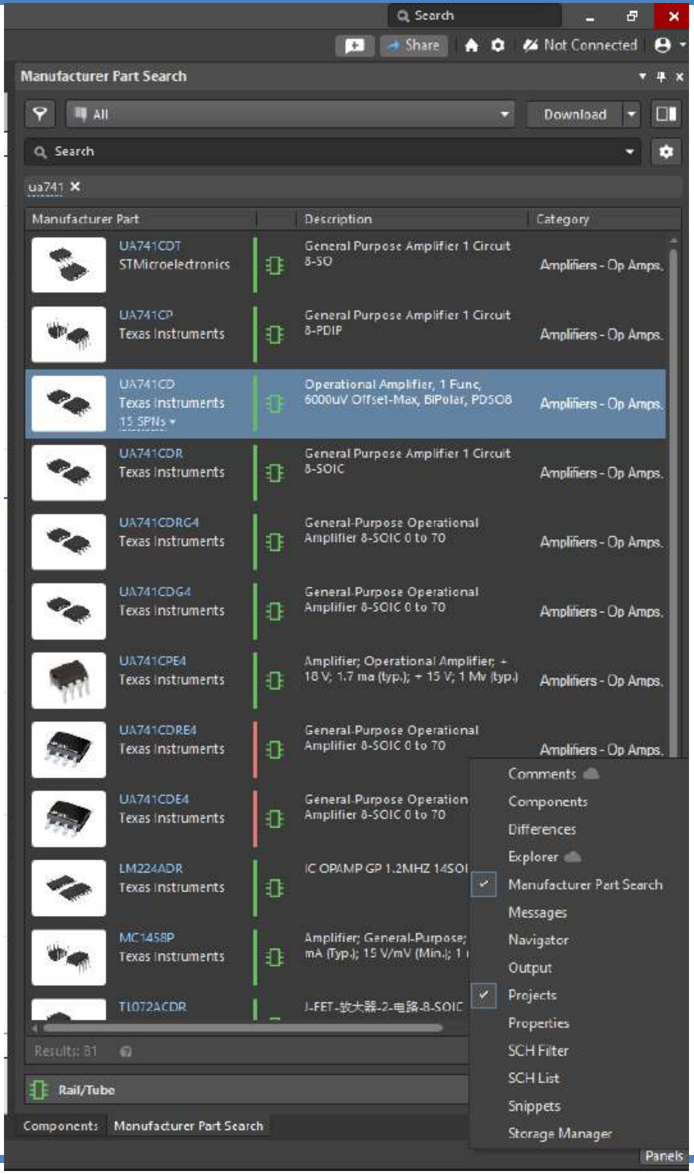
Installed libraries in the PC



Unlisted components can be searched through this panel or Manufacturer Part Search Tab

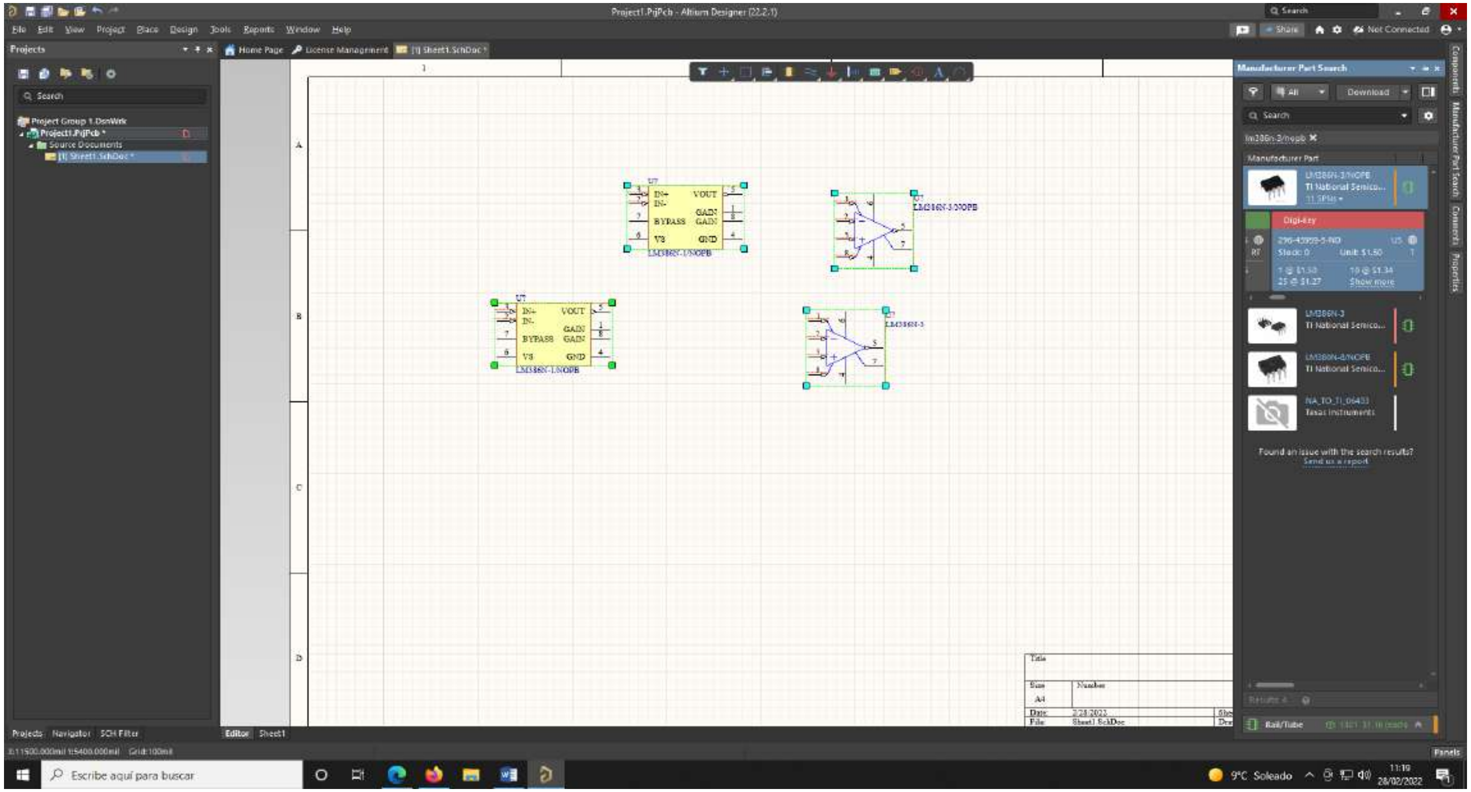
My first design

- Searching and selecting components not installed in local libraries.
- Several options can be got. Relevant issues such as availability, quantity or SCH symbol should be considered

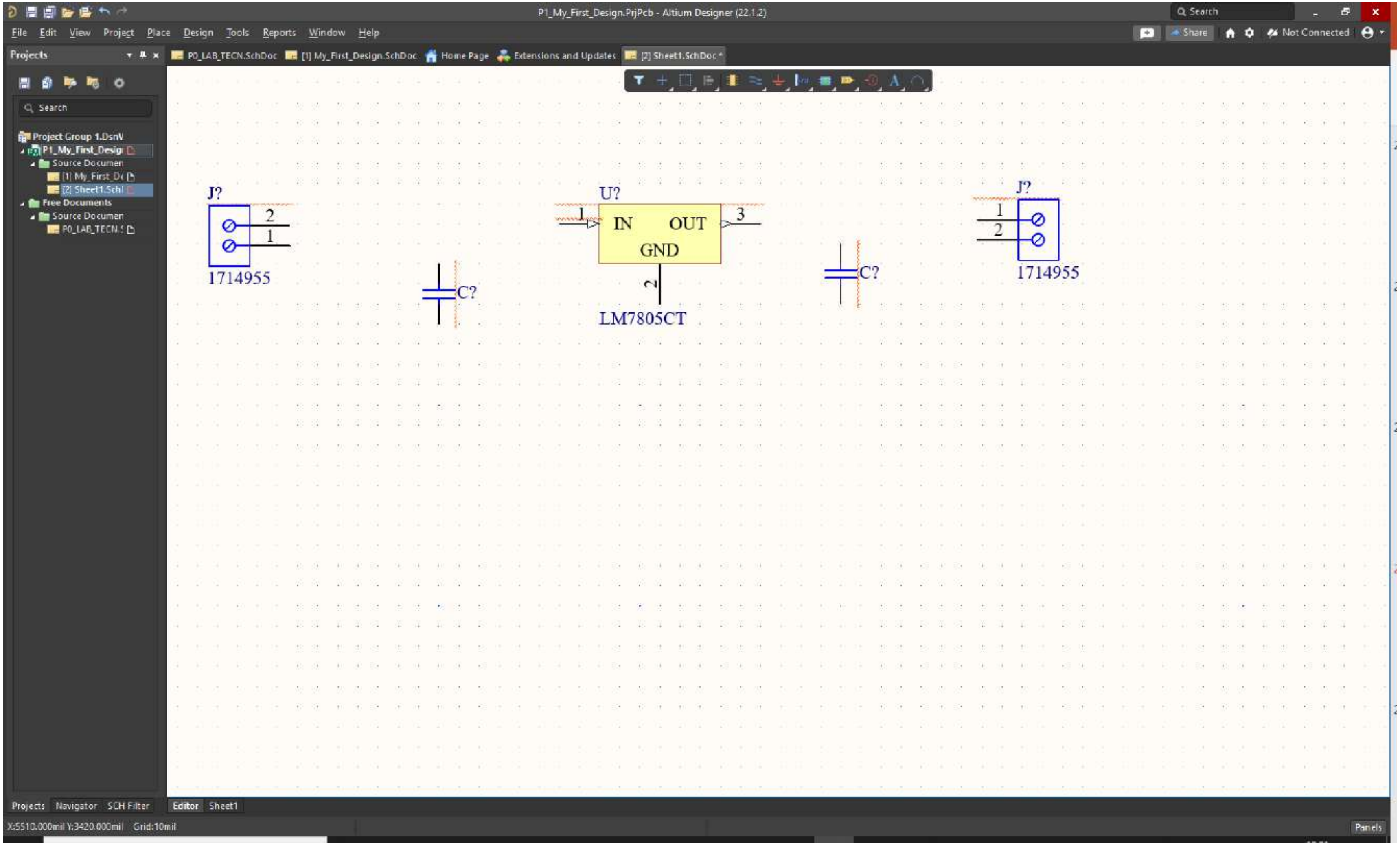


My first design

- Right click over the component and Place option must be selected to include in our design



“Hello World”



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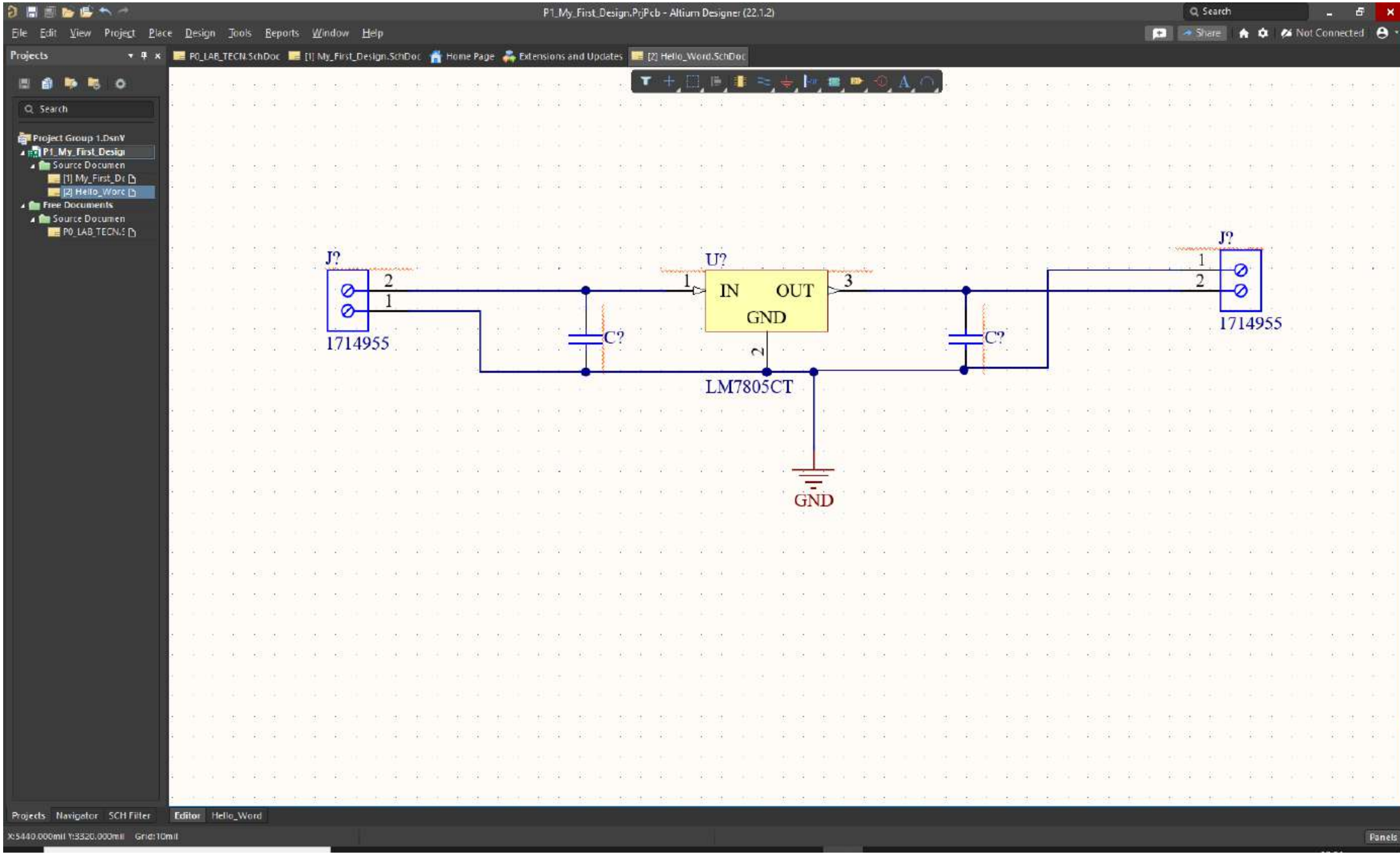
Panels

“Hello World”



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File Edit View Project Place Design Tools Reports Window Help
Projects
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[2] Hello_Word.SchDoc
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P1_My_First_Design
Source Documen
Free Documents
J?
1714955
U?
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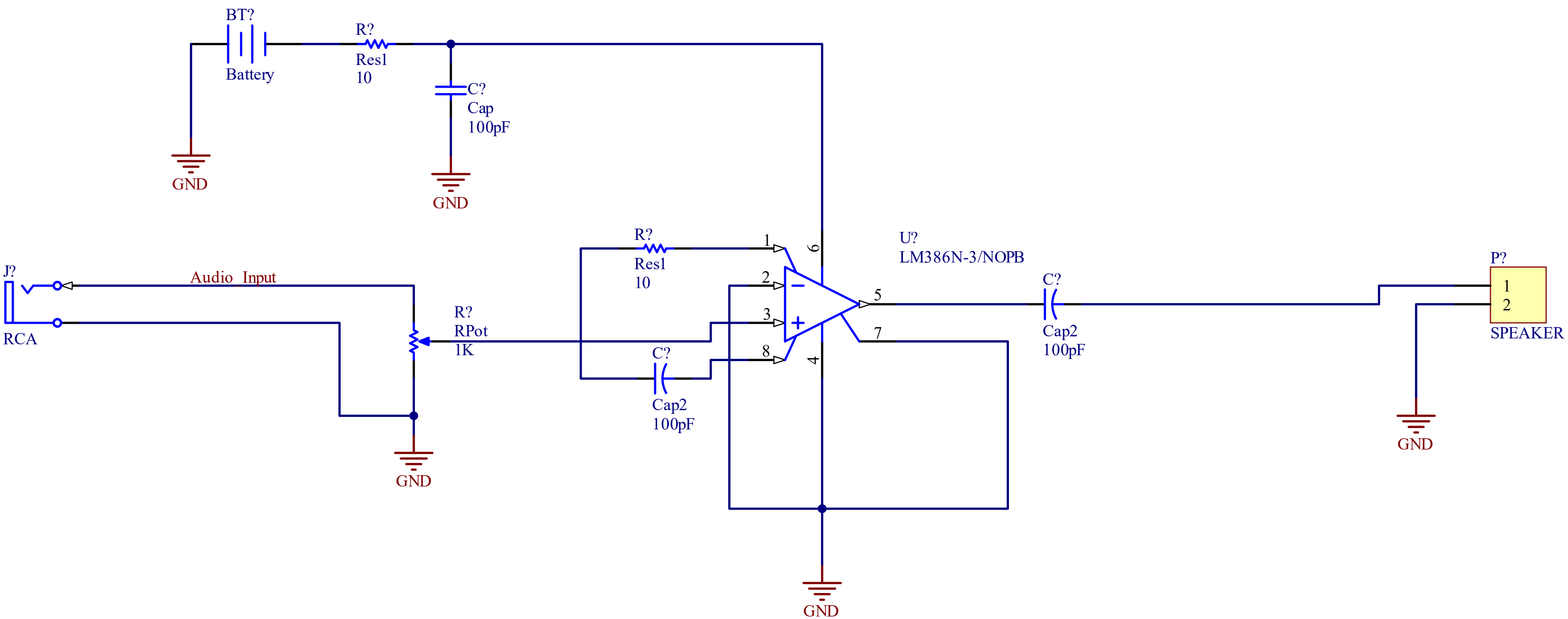
OUT
GND
C?
LM7805CT
GND
Projects Navigator SCH Filter
Editor Hello_Word
Panels

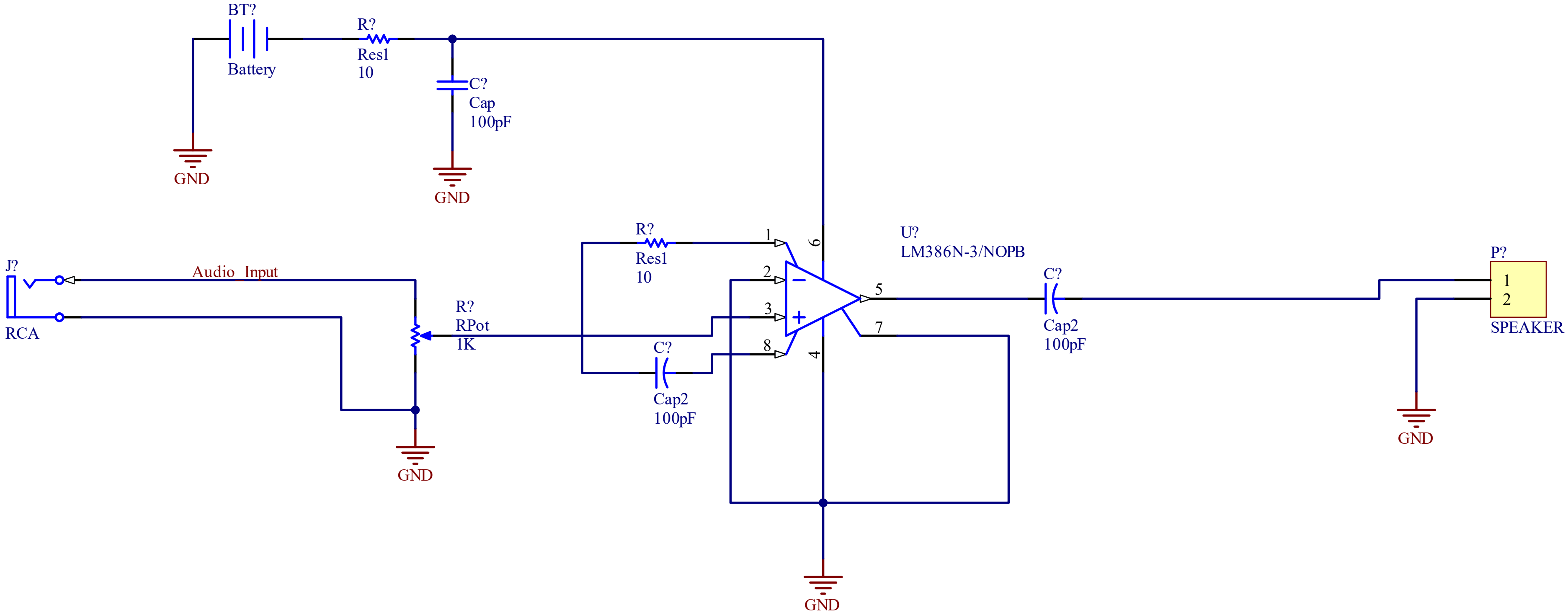
My first design

- Mi first design: An audio amplifier.



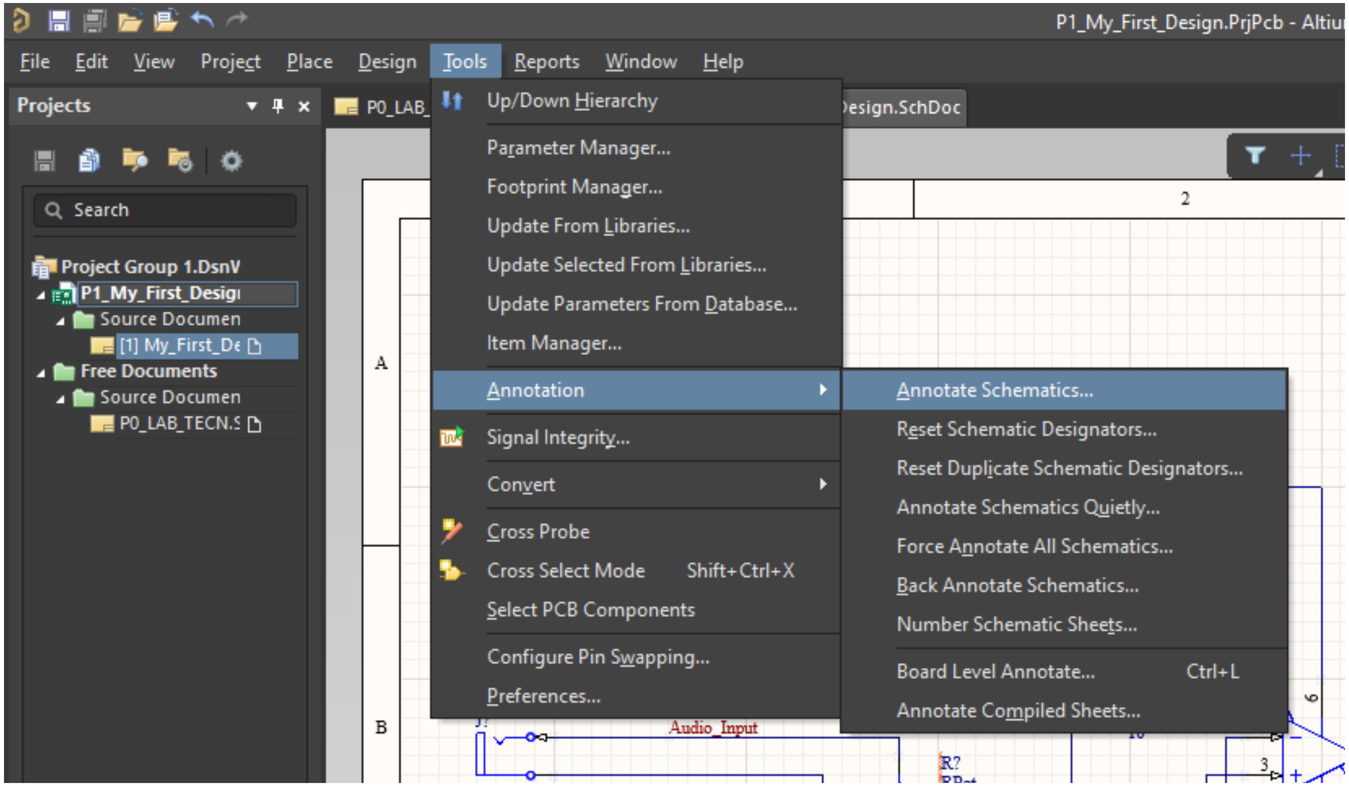
My first design

- With this example power options, wire and net label must be used.

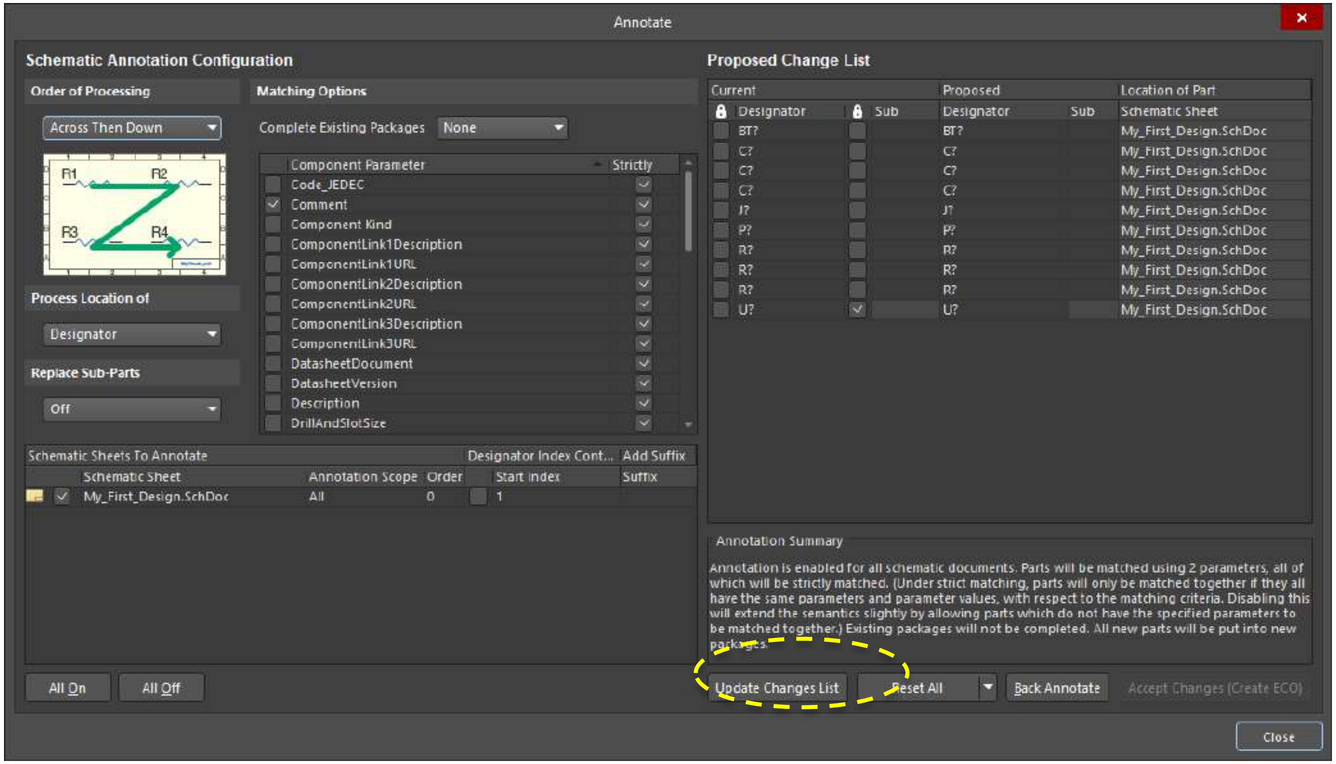


My first design

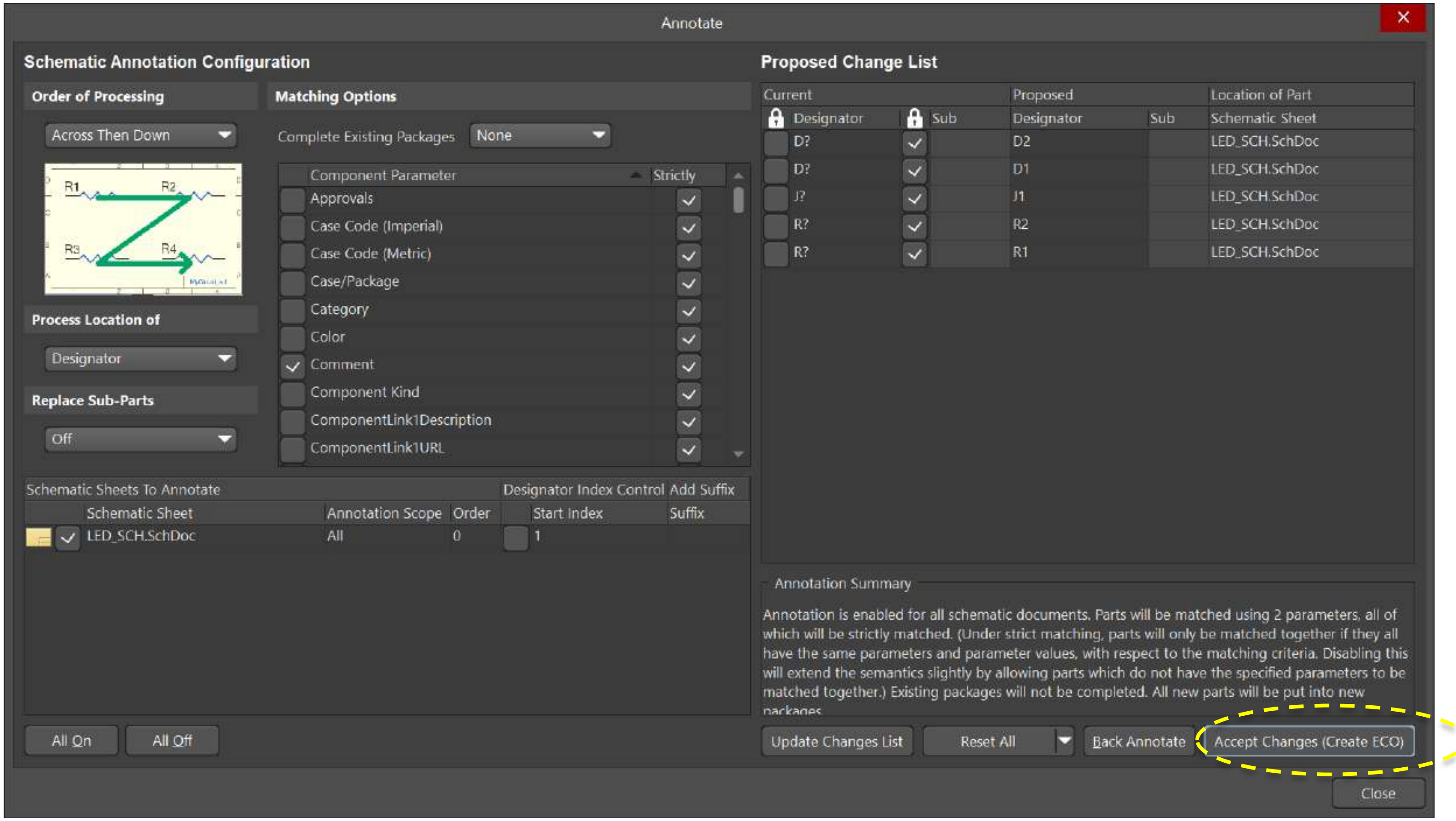
- Annotation (Tools Menu)



My first design



My first design



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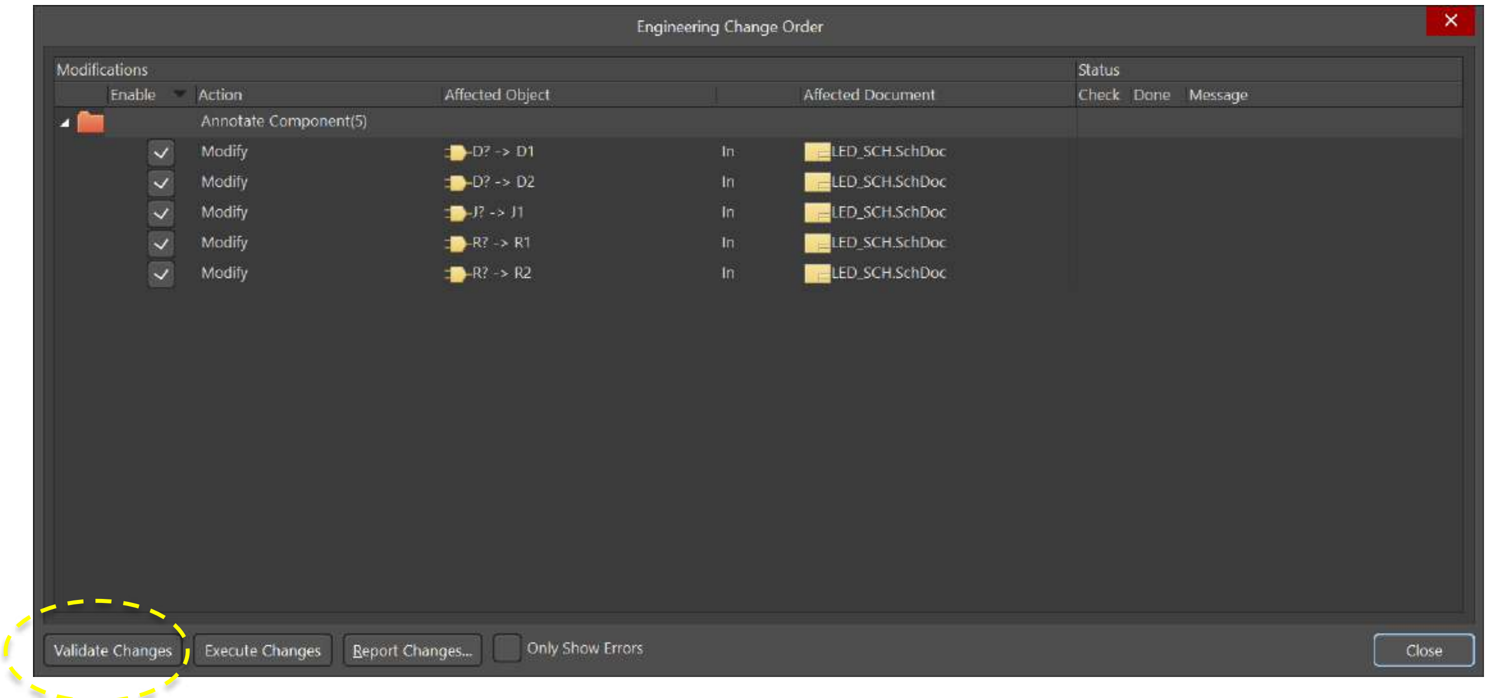
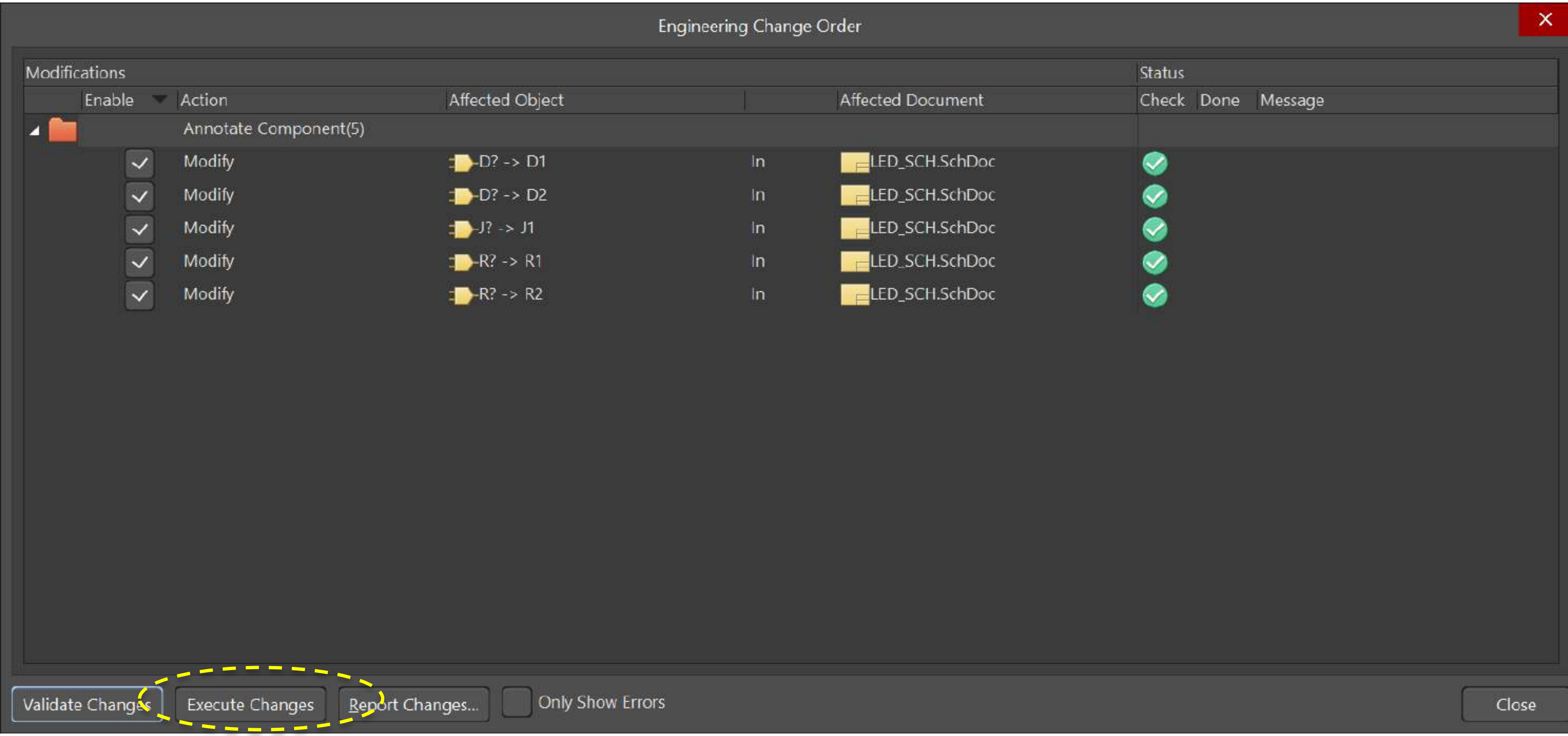
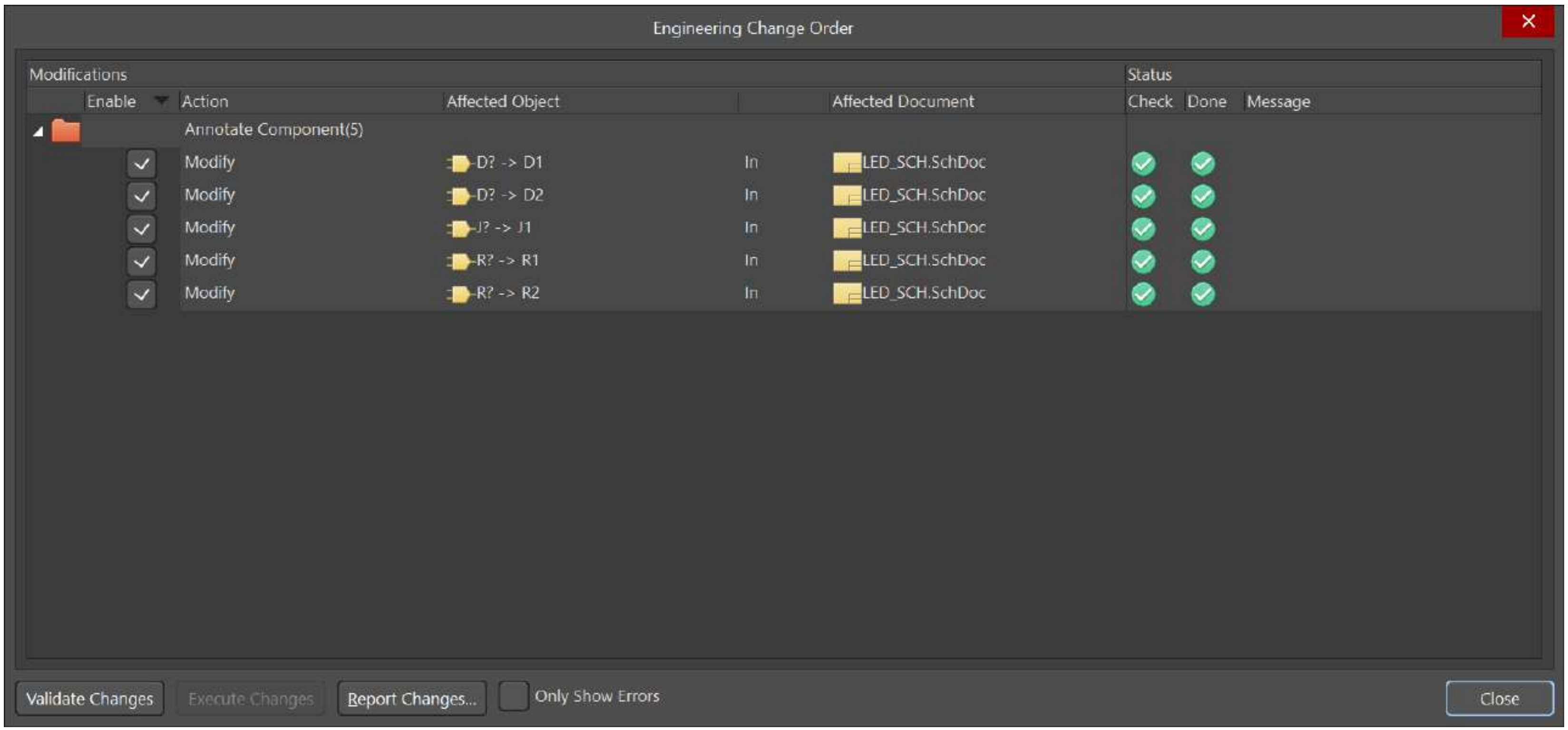


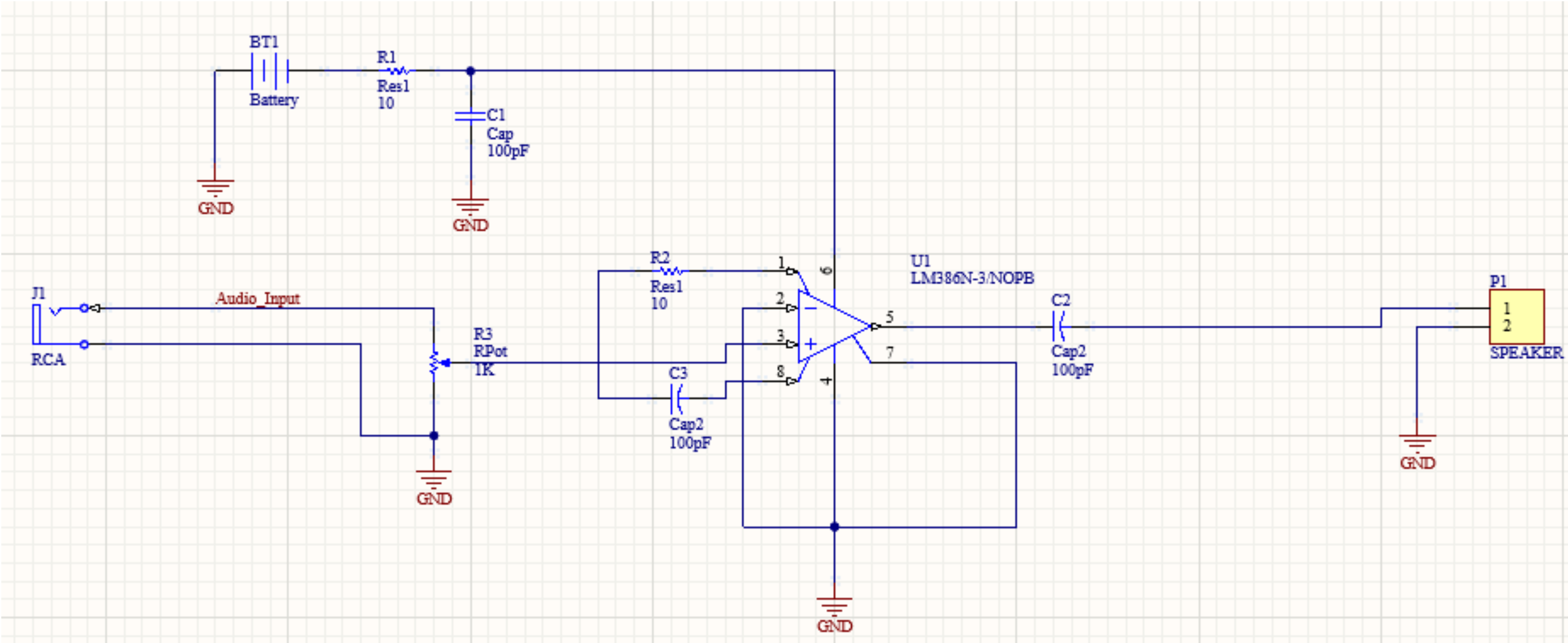

Engineering Change Order
Modifications
Status
Enable
Action
Affected Object
Affected Document
Check
Done
Message
Annotate Component(5)
Modify
D? -> D1
In
LED_SCH.SchDoc
Modify
D? -> D2
In
LED_SCH.SchDoc
Modify
J? -> J1
In
LED_SCH.SchDoc
Modify
R? -> R1
In
LED_SCH.SchDoc
Modify
R? -> R2
In
LED_SCH.SchDoc
Validate Changes
Execute Changes
Report Changes...
Only Show Errors
Close

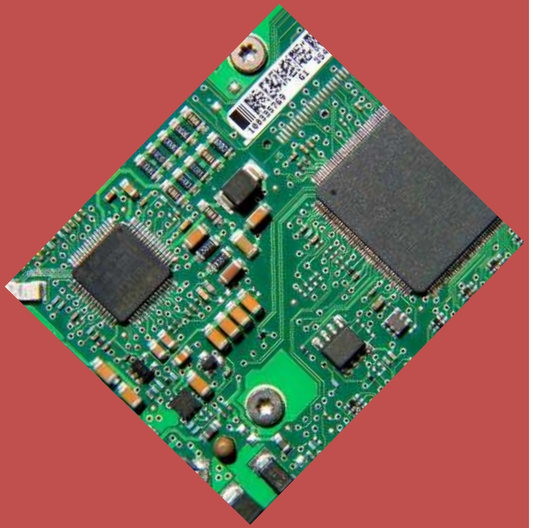
My first design



My first design

- Final design view

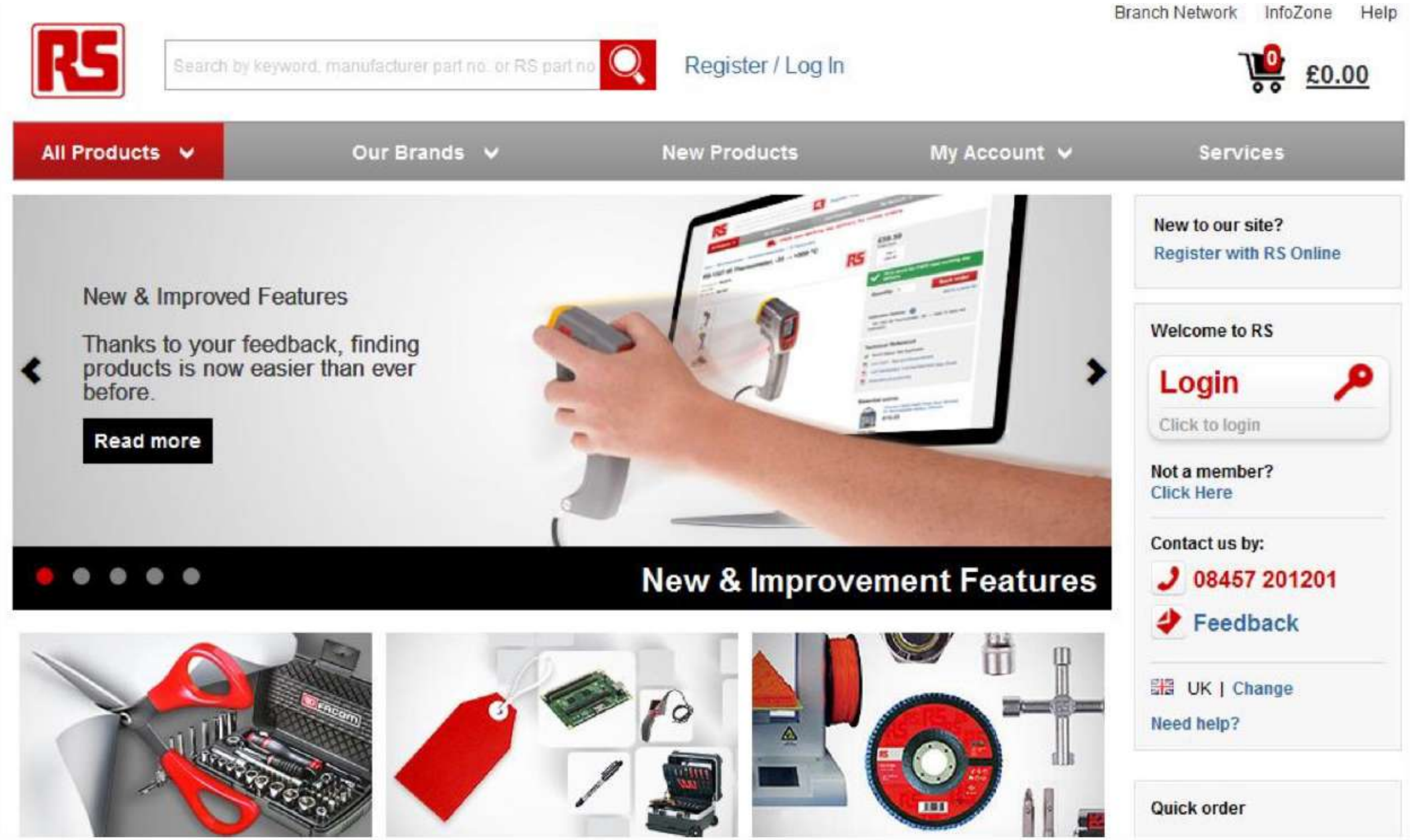




Introduction to Printed Circuit Board (PCB) design with ALTIUM

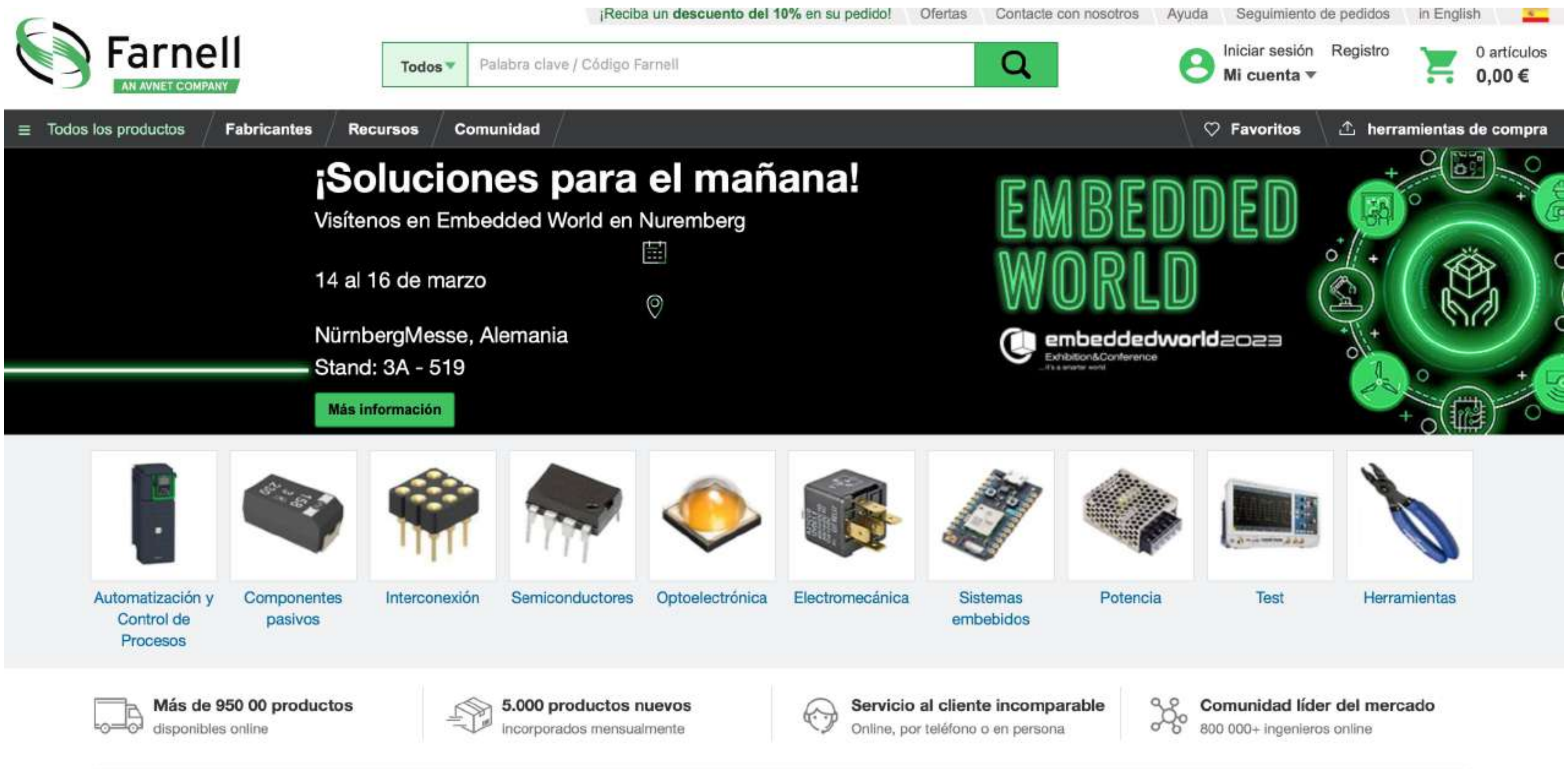
2.- Electronic Suppliers

Electronic Suppliers



https://es.rs-online.com/web/

Electronic Suppliers

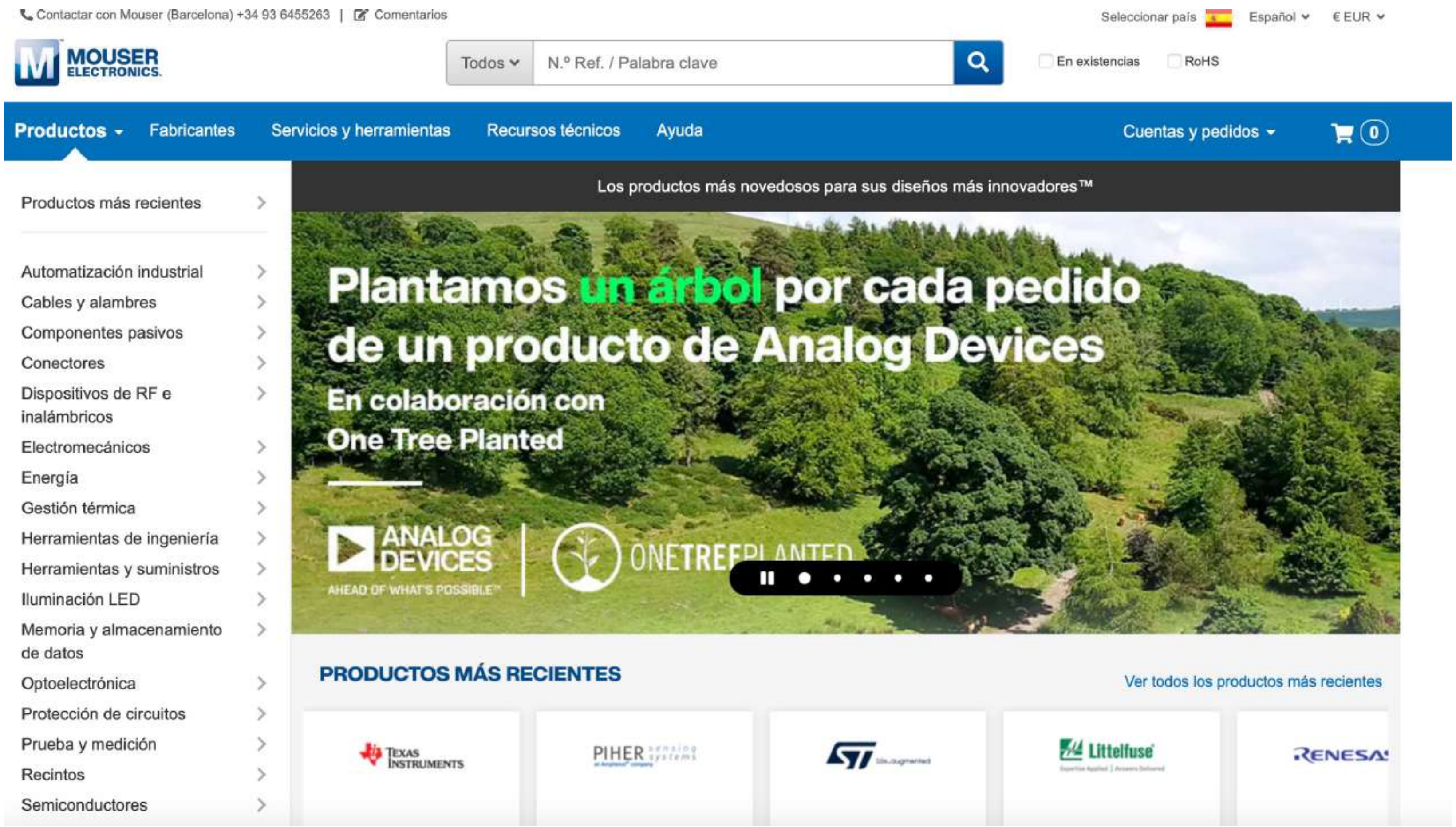


https://es.farnell.com/

Electronic Suppliers

https://www.digikey.es

Electronic Suppliers



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Electronic Suppliers

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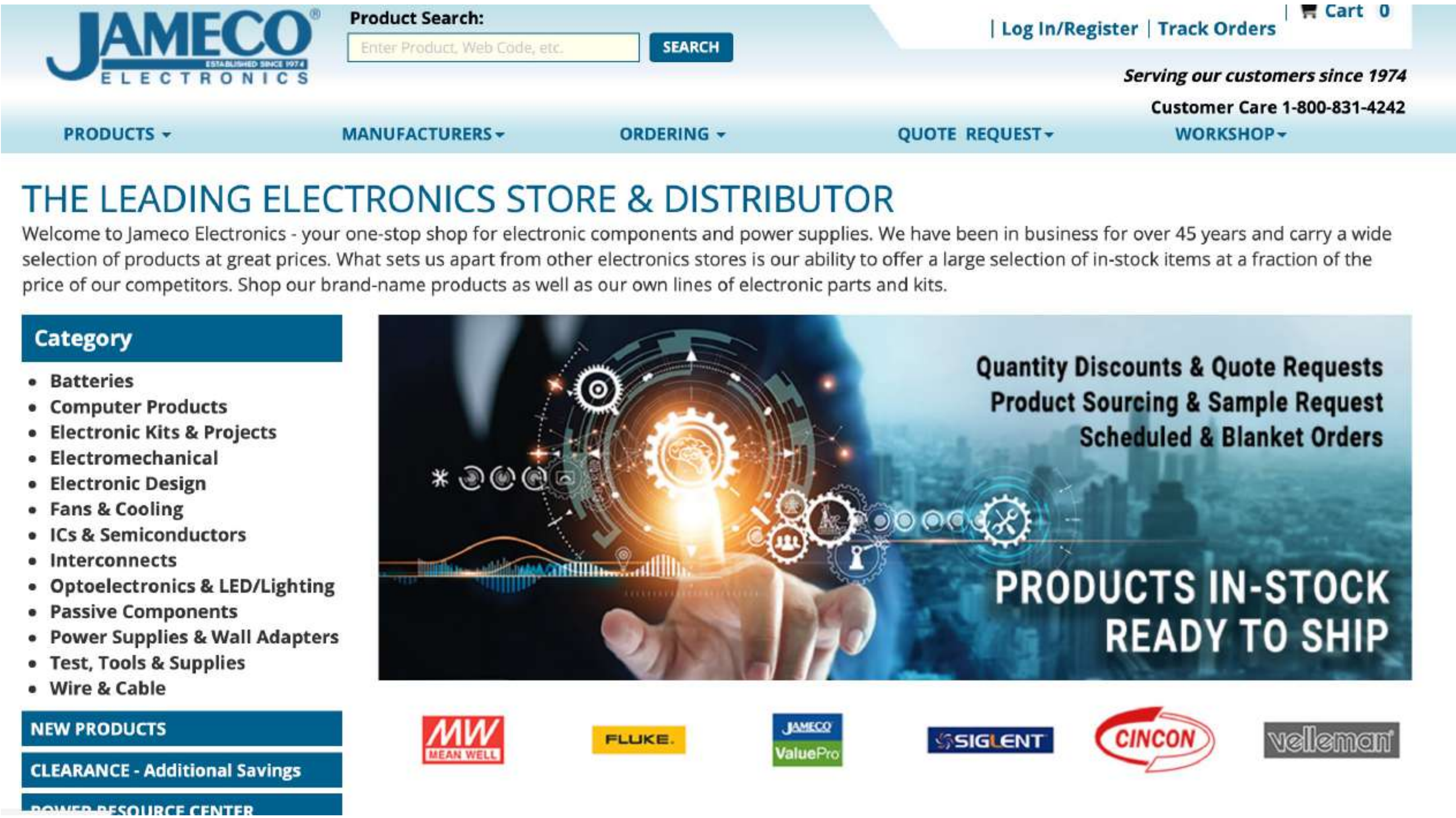
Electronic Suppliers



http://www.jameco.com

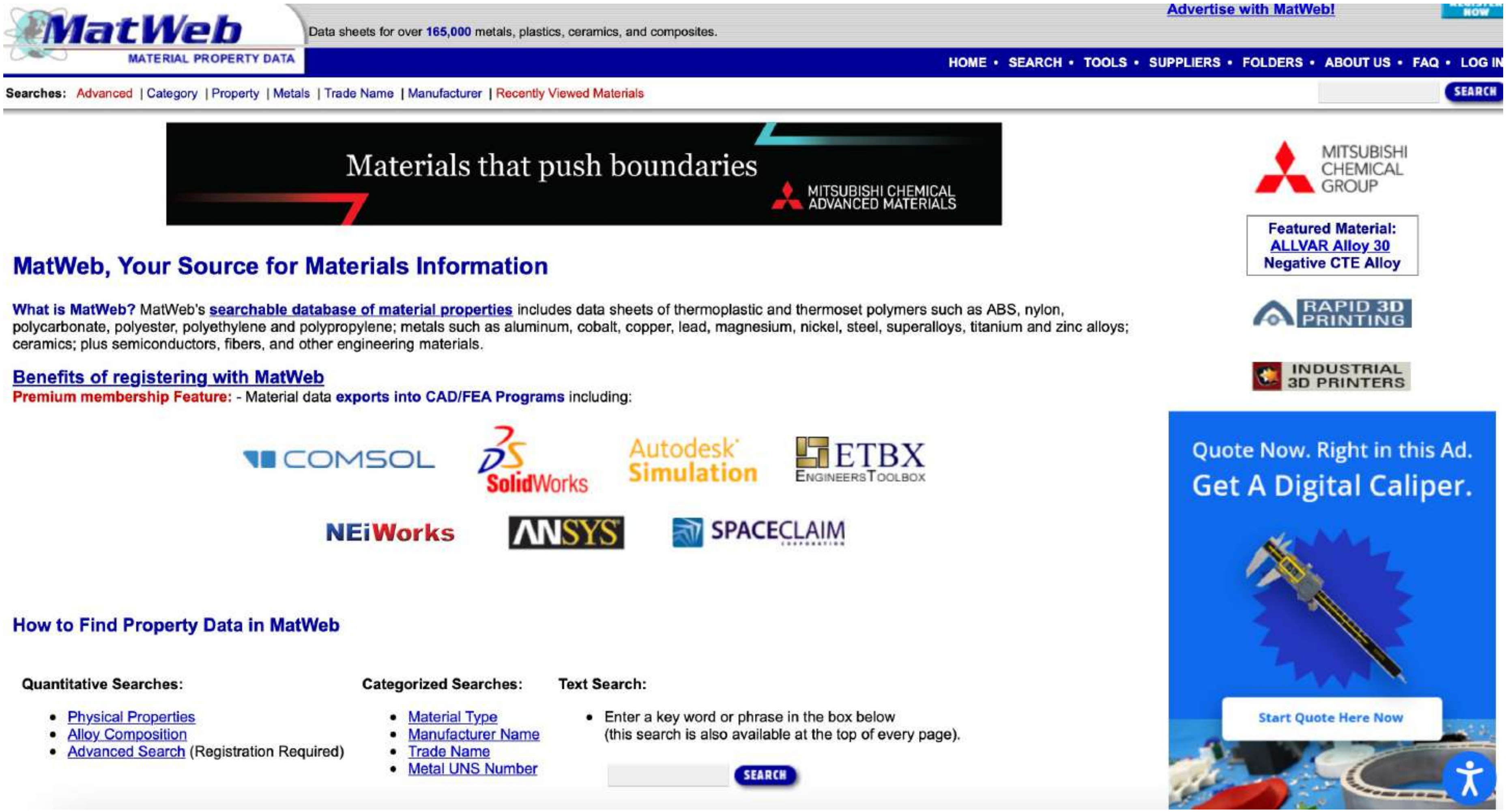
USA Electronic Suppliers

- Reynolds Electronics
- Hobby Engineering
- All Electronics
- BG Micro
- Jameco Electronics
- Radioshack
- Fry's Electronics
- Goldmine Electronics
- Futurlec

Industrial Suppliers

- On-line catalogs and directories
- ThomasNET
- A comprehensive directory of industrial suppliers. Look here to find sources for almost anything!
  GLOBALSPEC
  An "engineering search engine". It is similar to ThomasNET, and is useful to help find vendors and catalog items.
- SKF : An excellent interactive engineering catalog for bearing (rodamientos) selection.
  Stock Drive Components/Sterling Instruments : A comprehensive on- line catalog of mechanical components. You can download 3D CAD models to use in your CAD assembly.
  McMaster-Carr
  Over 465,000 products, the website is easily searchable with full technical specifications.

MatWEB



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Trade Name
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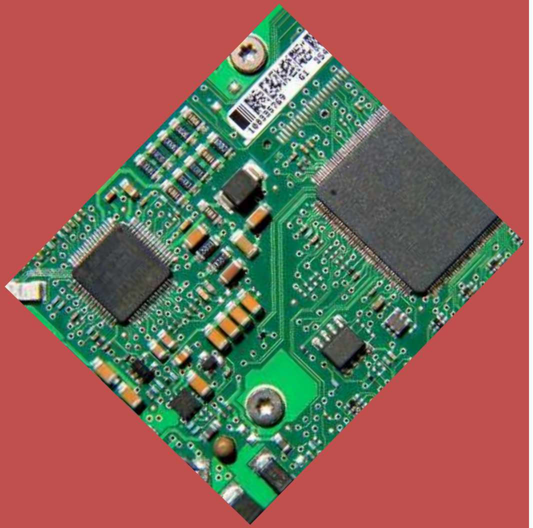
SEARCH

Introduction to Printed Circuit Board (PCB) design with ALTIUM



2.- Schematic Library

Library concept

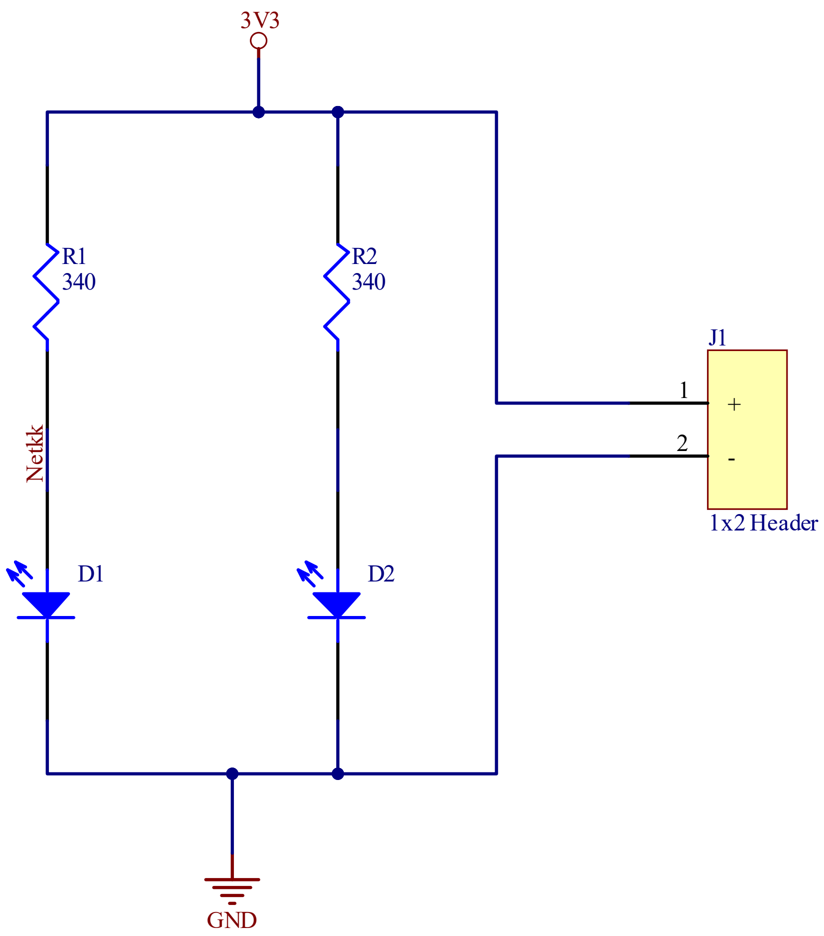
- A **library** is a repository where components are stored.
- Several type of library could be handled:
  - PCB and Schematic
  - Manufacturer Library
  - Working Library
  - Project Library
- For schematic purposes, we will use commercial components and will design customized elements linked with commercial products (linking with official information and/or models).

Working with libraries

- It is convenience to choose the components/device to be used in advance for the schematic design.
- So, we usually link the model, symbol, datasheet, footprint, etc. at the beginning of our design.
- Methodology:
  - Creating a SCH Library
  - Creating SCH symbols
  - Creating the SCH.

Our basic design

- This is the simple design that we are going to create.
- All the components will be drawn by us without using the official symbol.
- All of them (diode, resistor and header) will be linked with commercial units.



Devices to be used

- The methodology to select the appropriate devices will be as follows:
  - To find the right device (component) through a manufacturer supplier using filter search tool (web).
  - Identify part number or supplier number (web)
  - Finding the chosen device into Manufacturer Search Part (Altium).

Devices to be used

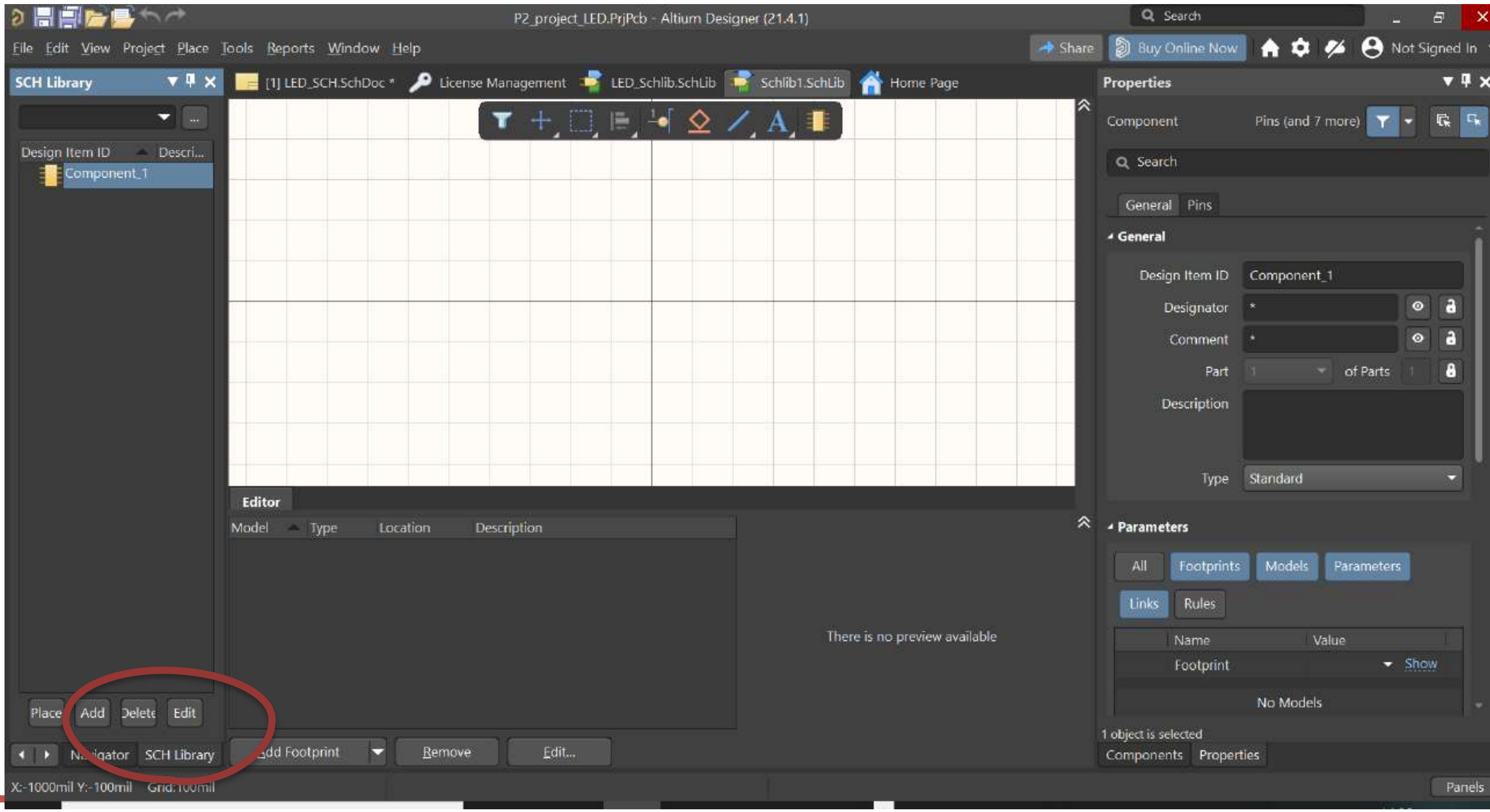
- For this example we will use the next devices:

Name	Manufacterer reference	Manufacturer	Digi-Key Part number
HEADER 2	0022272021	Molex	WM411-ND
RESISTOR	**CRCW0805340RFKEA**	Vishay Dale	541-340CCT-ND
LED	LNJ337W83RA	Panasonic	LNJ337W83RACT-ND

- Additionally, we will create the schematic symbol for each element.

Starting with the SCH Library

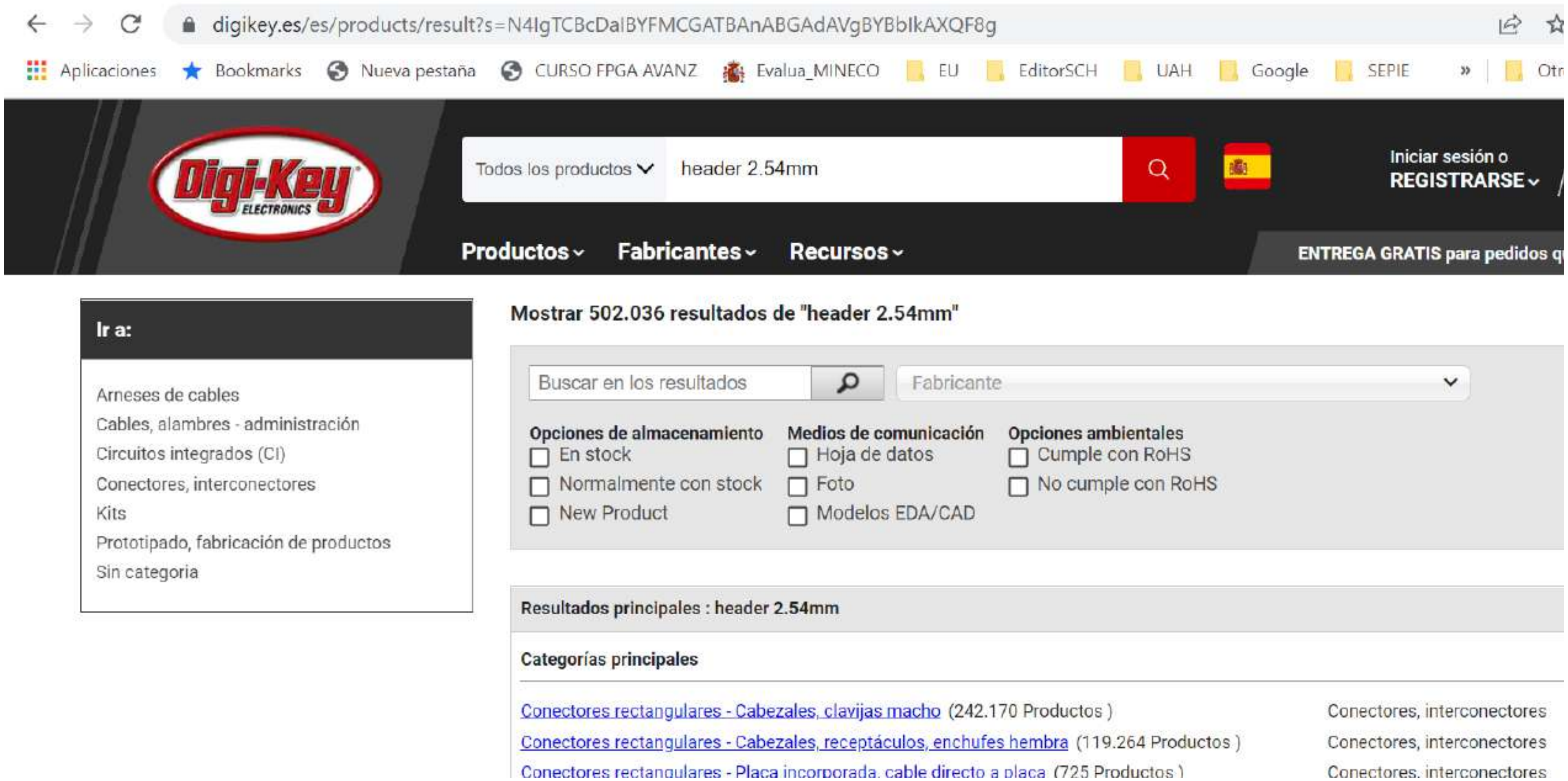
- The first step must be create a SCH Library (File → New → Library → Schematic Library).



Selection of header from a provider

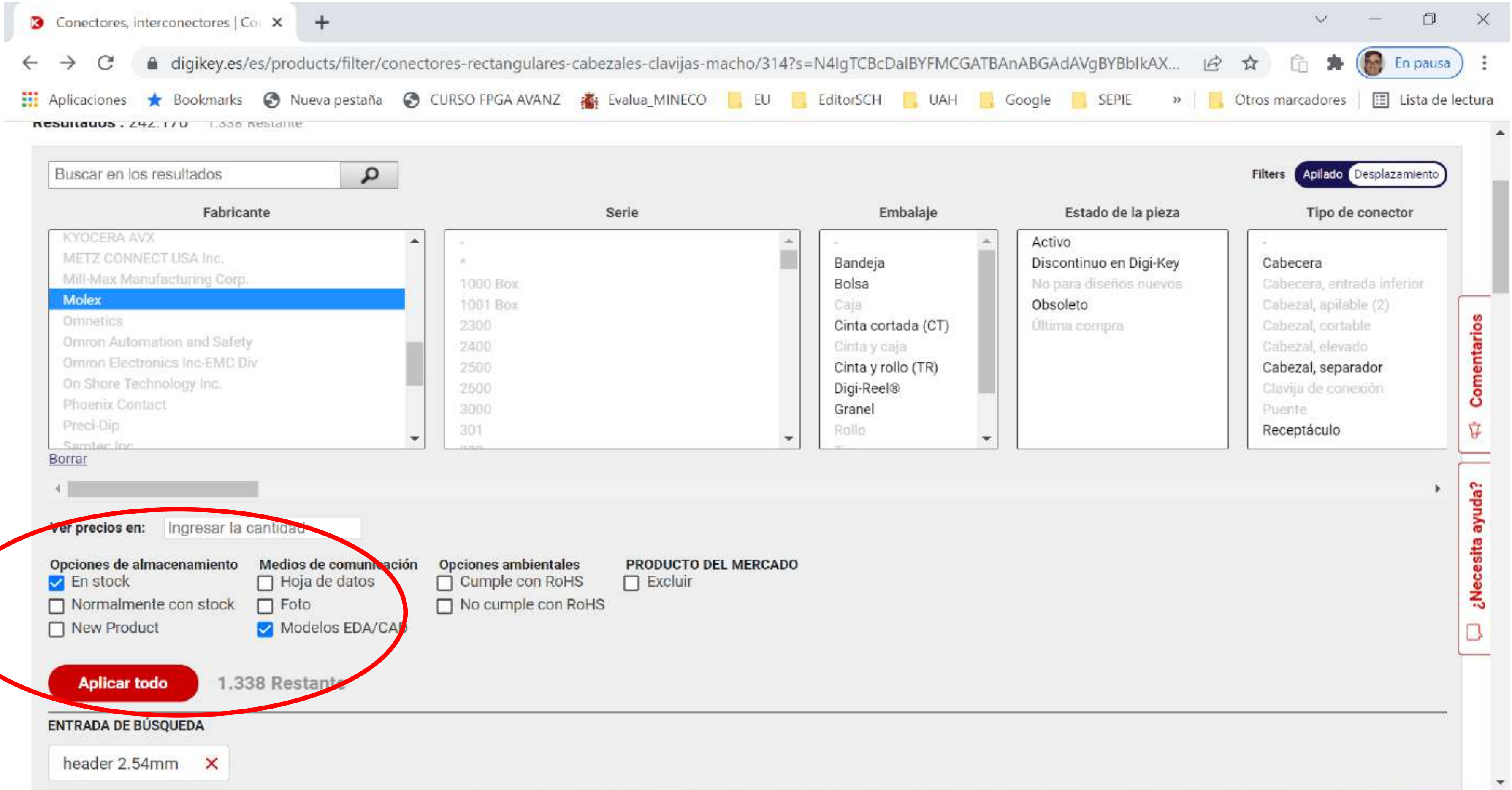
Header 1x2

- Searching the header component within Digikey



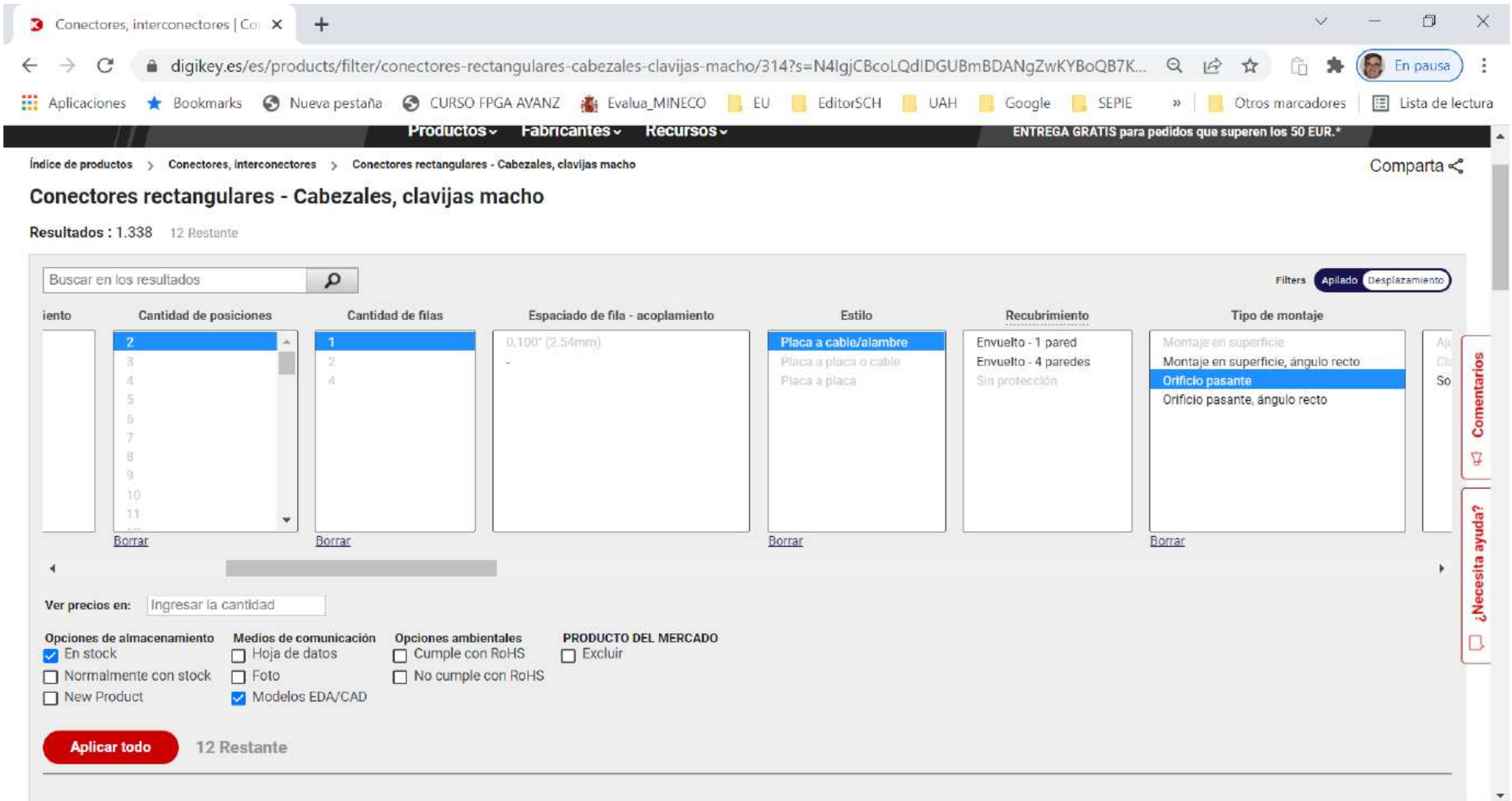
Selection of header from a provider

- Searching the header component within Digikey



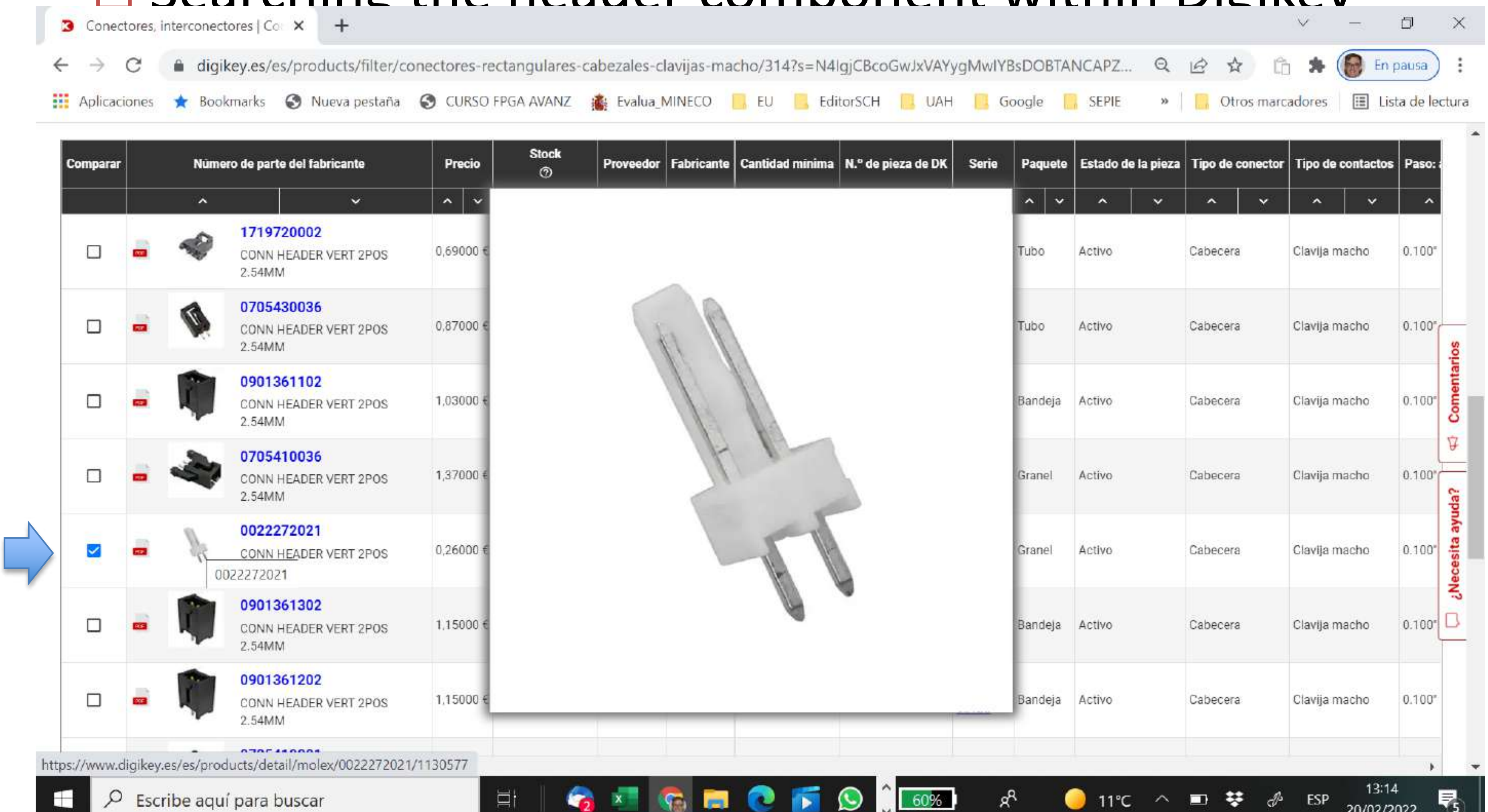
Selection of header from a provider

- Searching the header component within Digikey



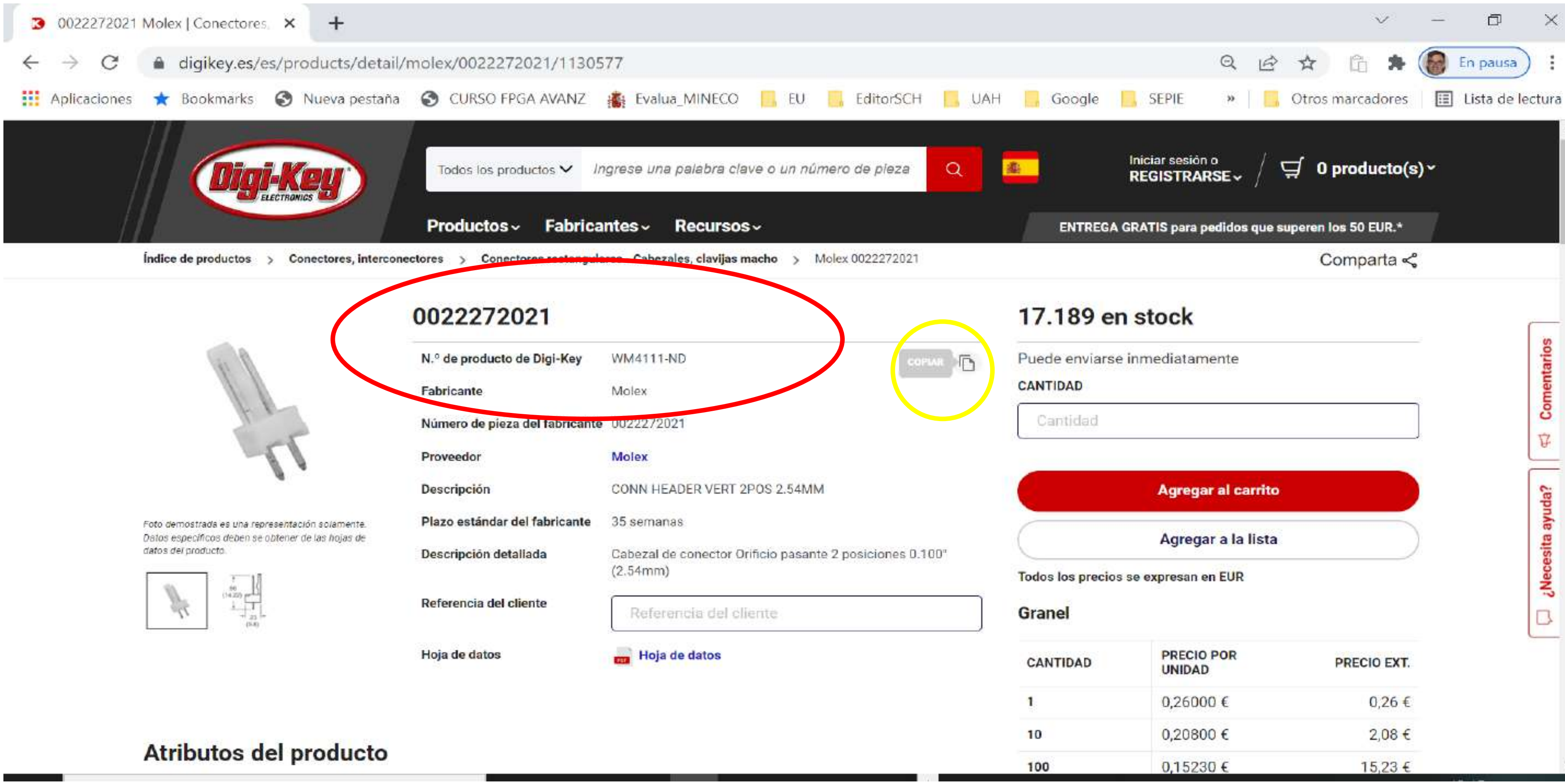
Selection of header from a provider

- Searching the header component within Digikey



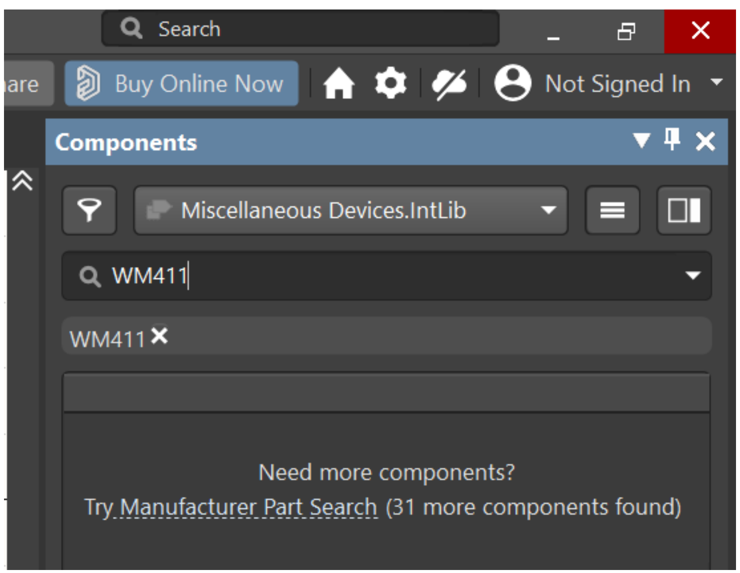
Selection of header from a provider

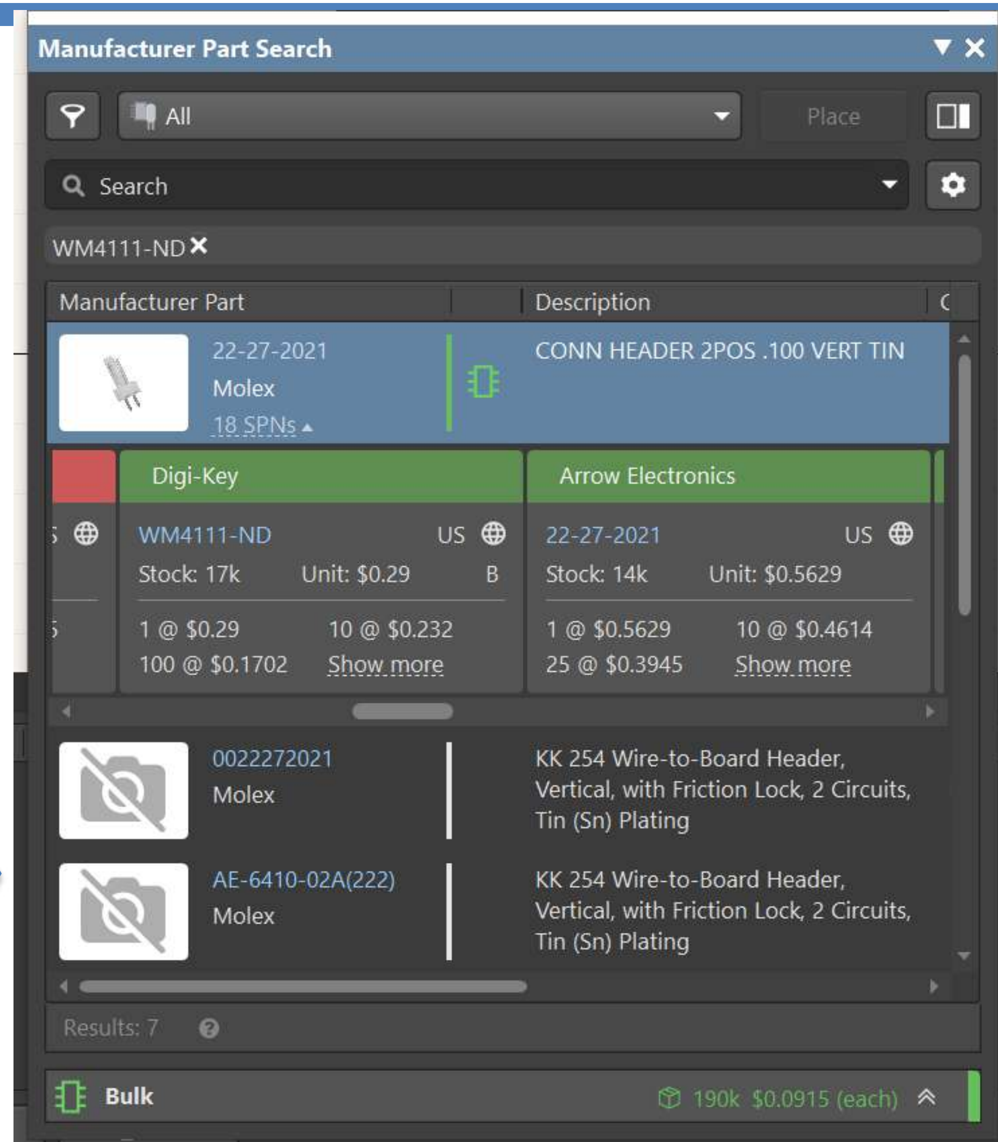
- Searching the header component within Digikey



Searching the component

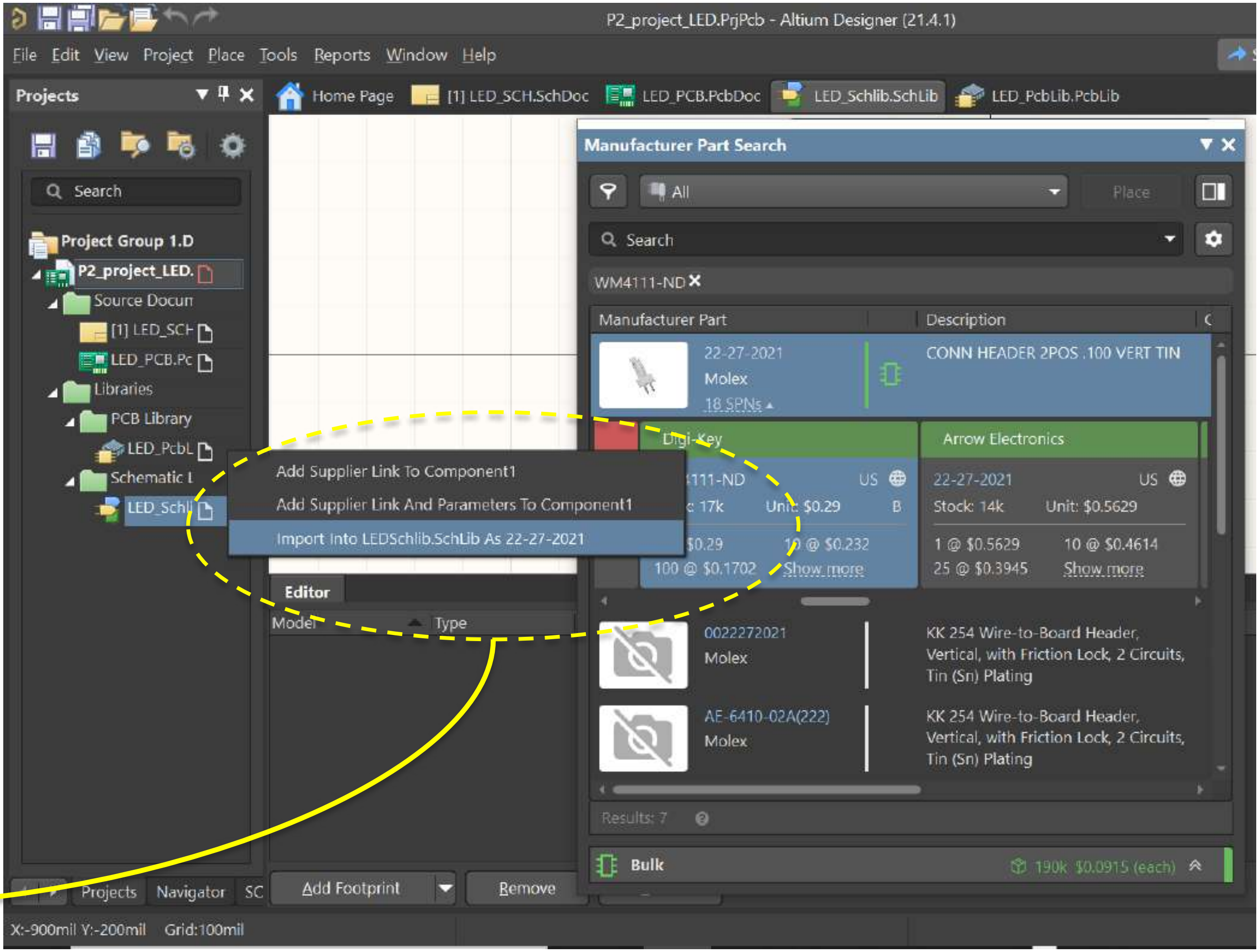
- Then, we will search the model inside of Altium Components (Manufacturer Research)





Importing the component properties

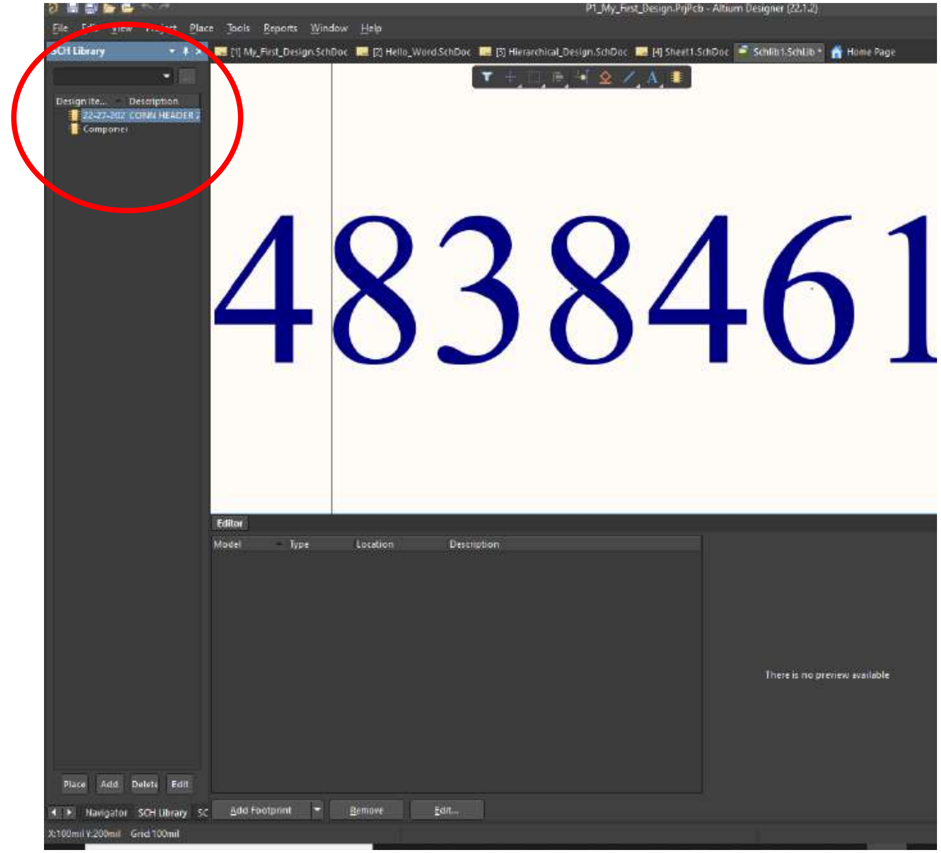
- Now, we should import the component to our Schematic Library (active window)



Right click within component description

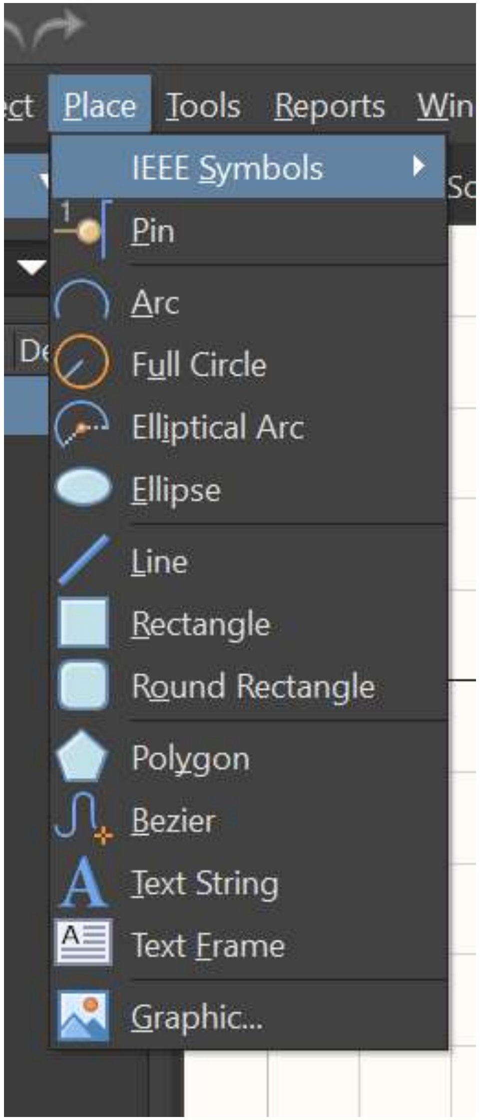
Importing the component properties

- Component properties are currently included in our SCH Library.
- However, the symbol should be created



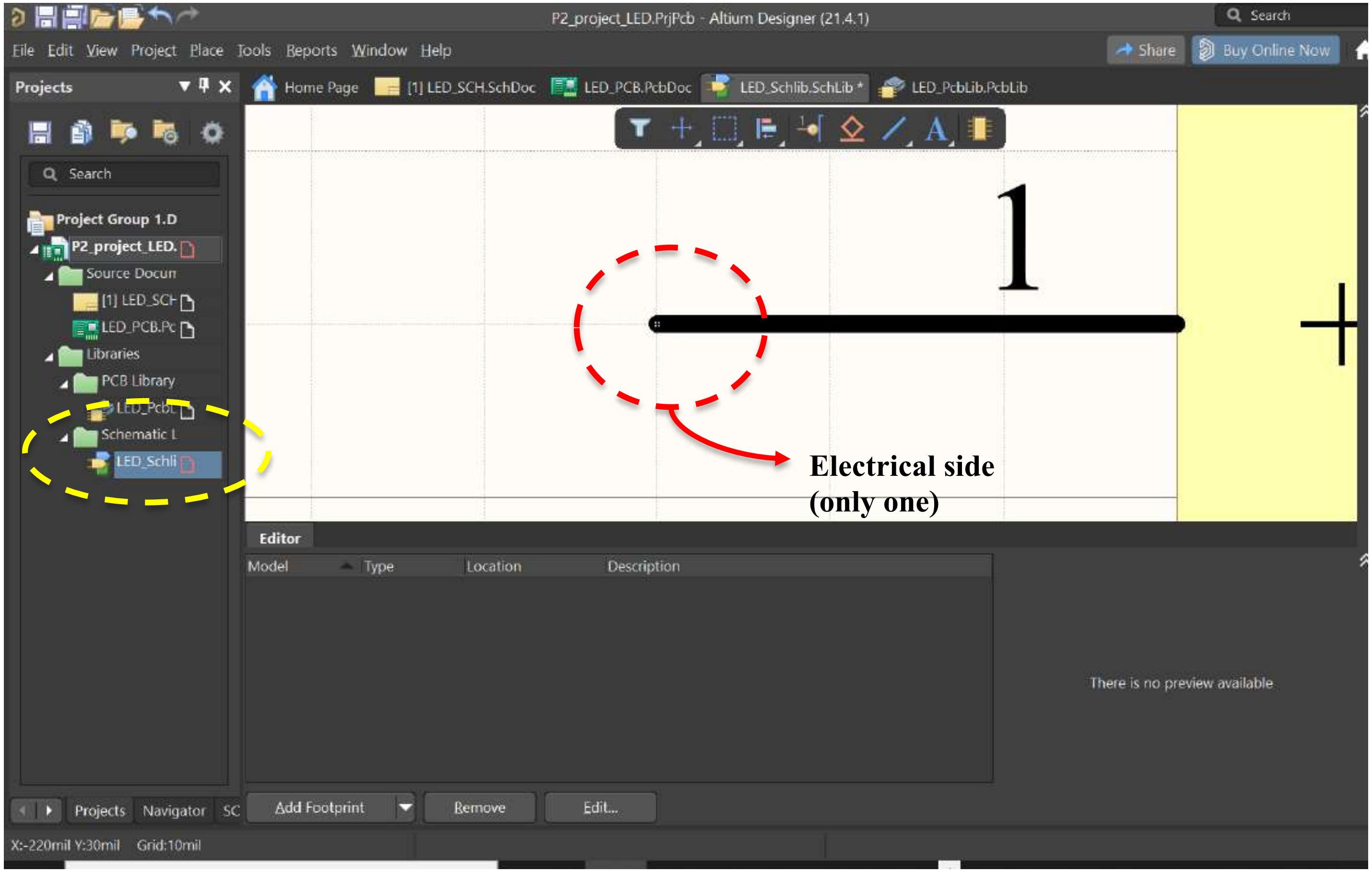
Drawing the component

- The composition of the component must be fulfilled with Pin and some shape (arc, polygon, etc.).
- Properties of the pin and shape must be checked.
- Pay attention to the pin. It must be include a name, number.

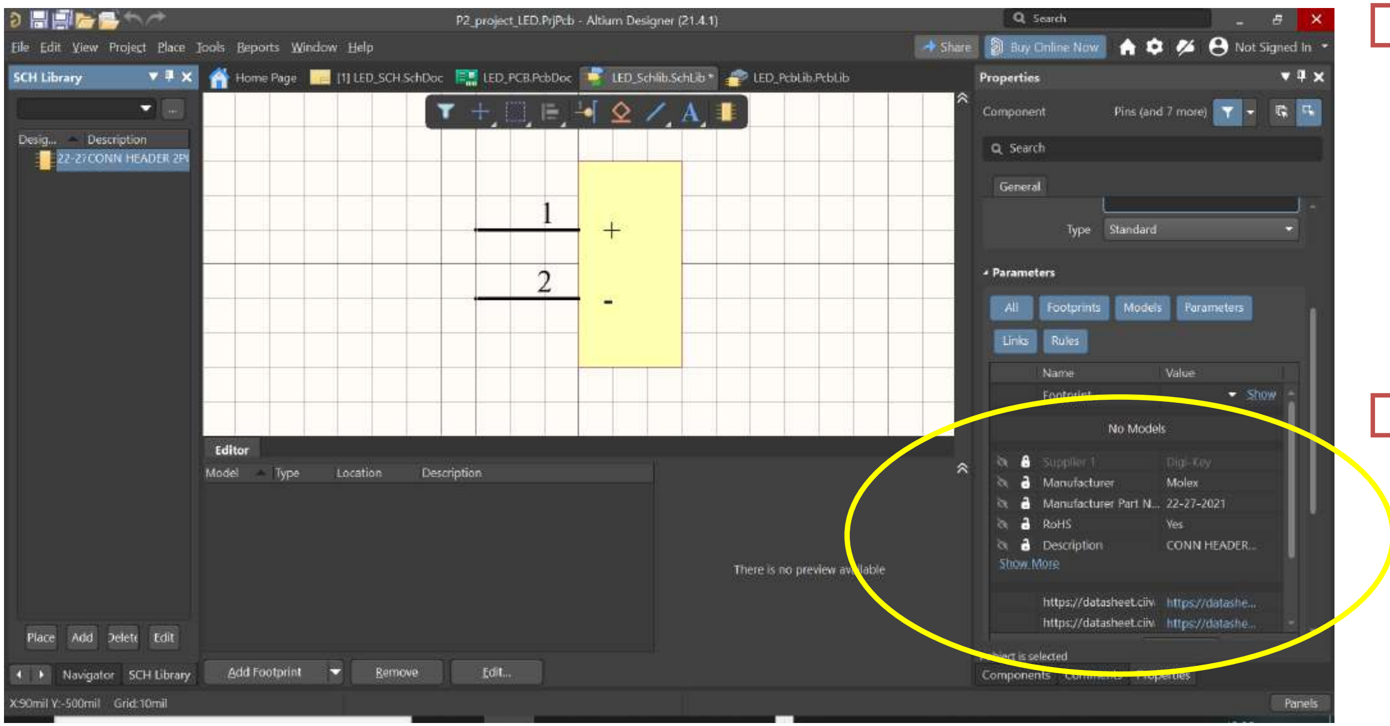


Drawing the component

- We can draw the new component (header) inside of SCH Lib environment



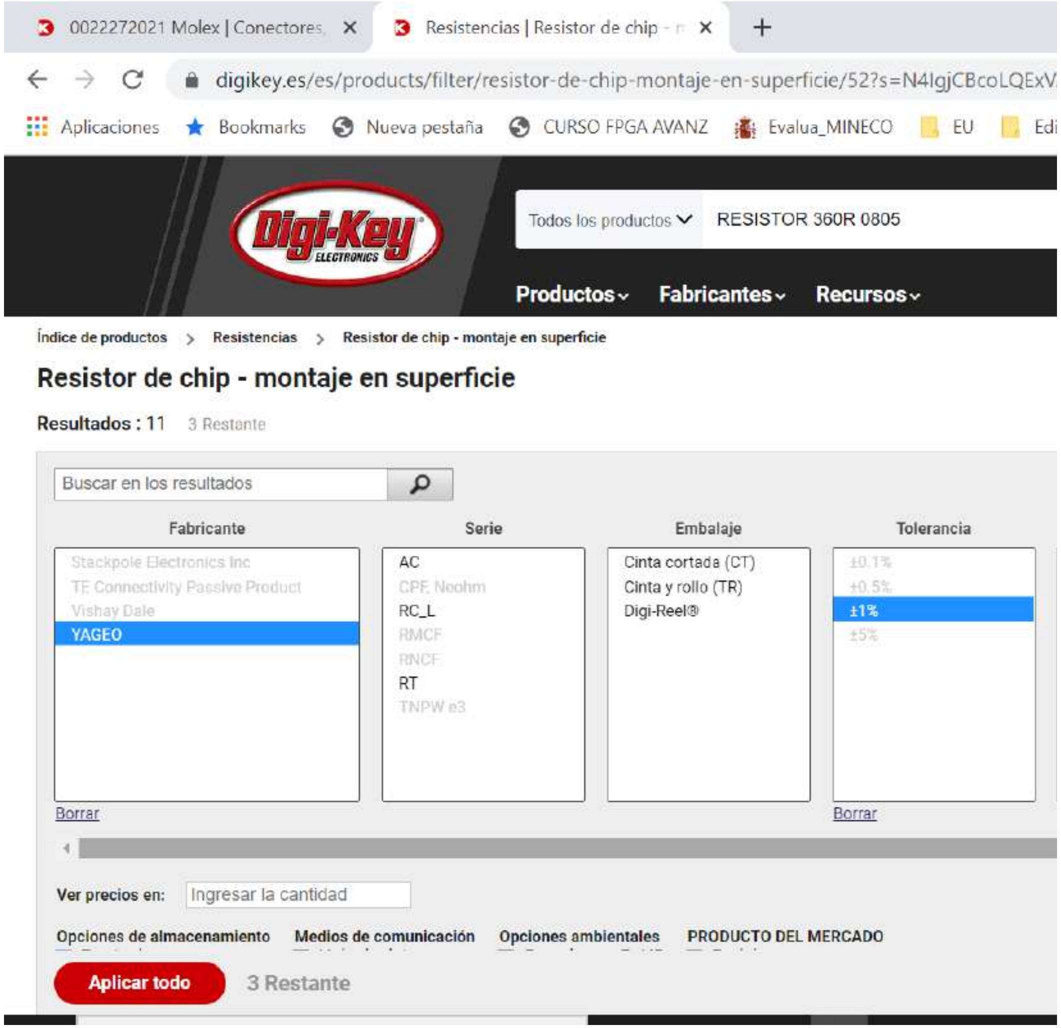
Drawing the component



- Here we can find some imported information.
- It contains relevant data regarding the device features.

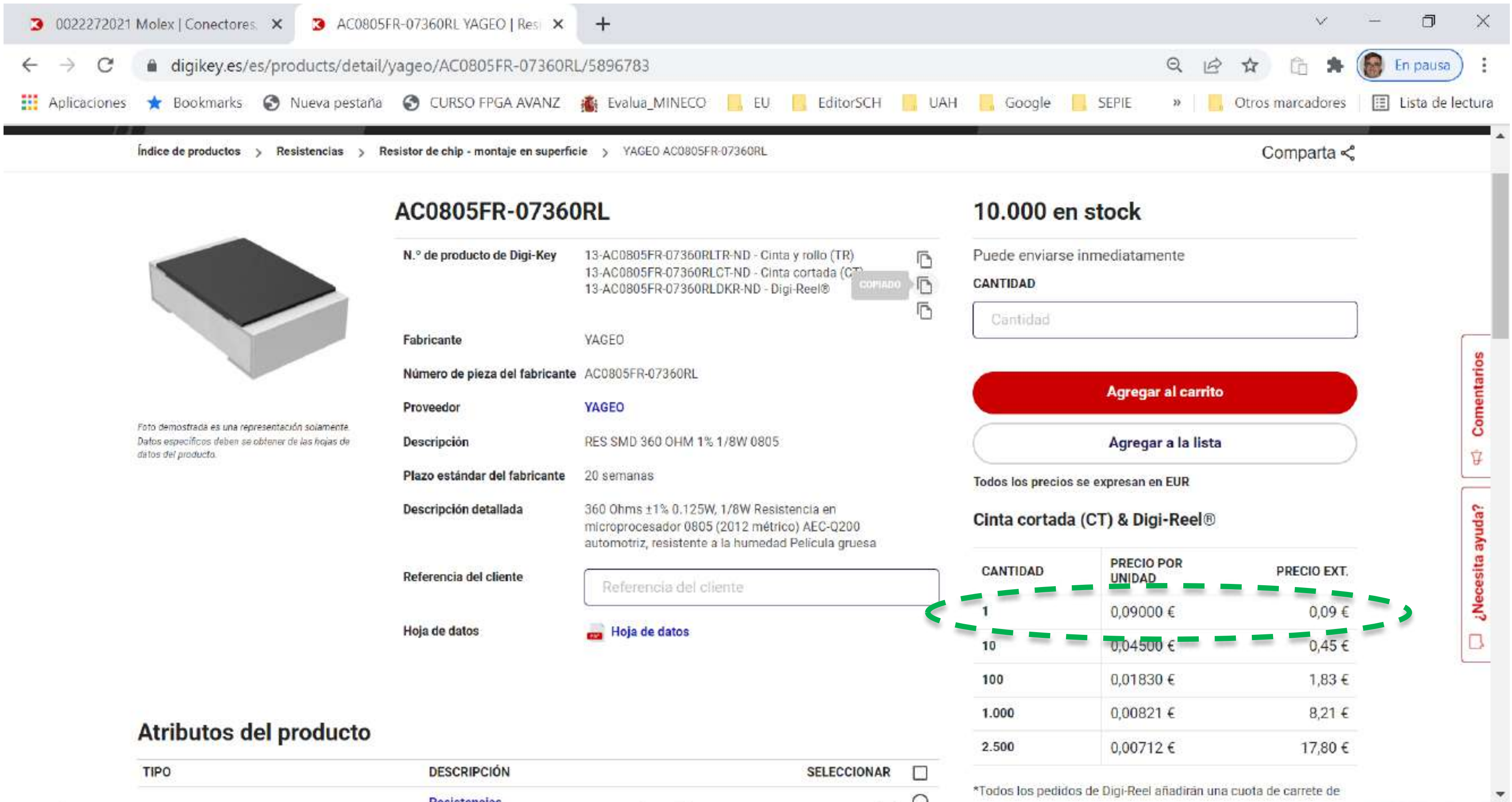
Selection of resistor

- We will deploy the same methodology than the diode.
- A SMD version will be selected.



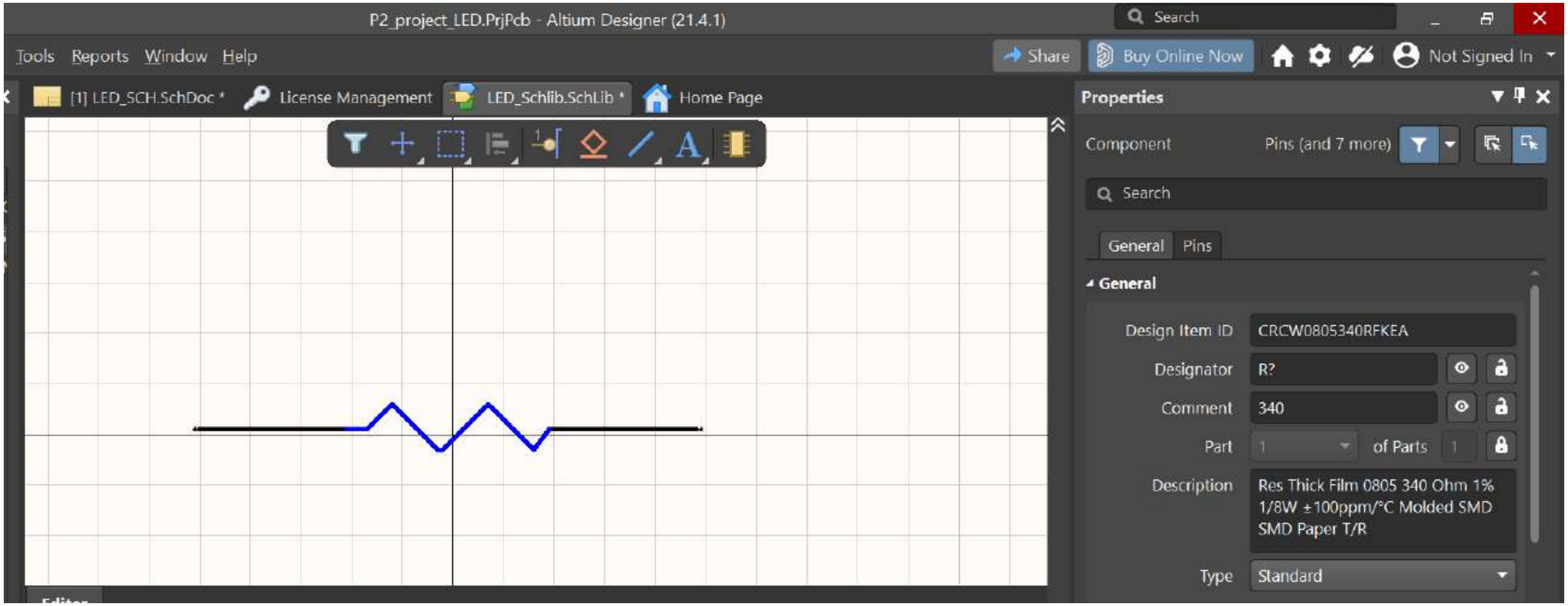
Selection of resistor

- We can choose several options. One criteria to choose can be the availability and quantity



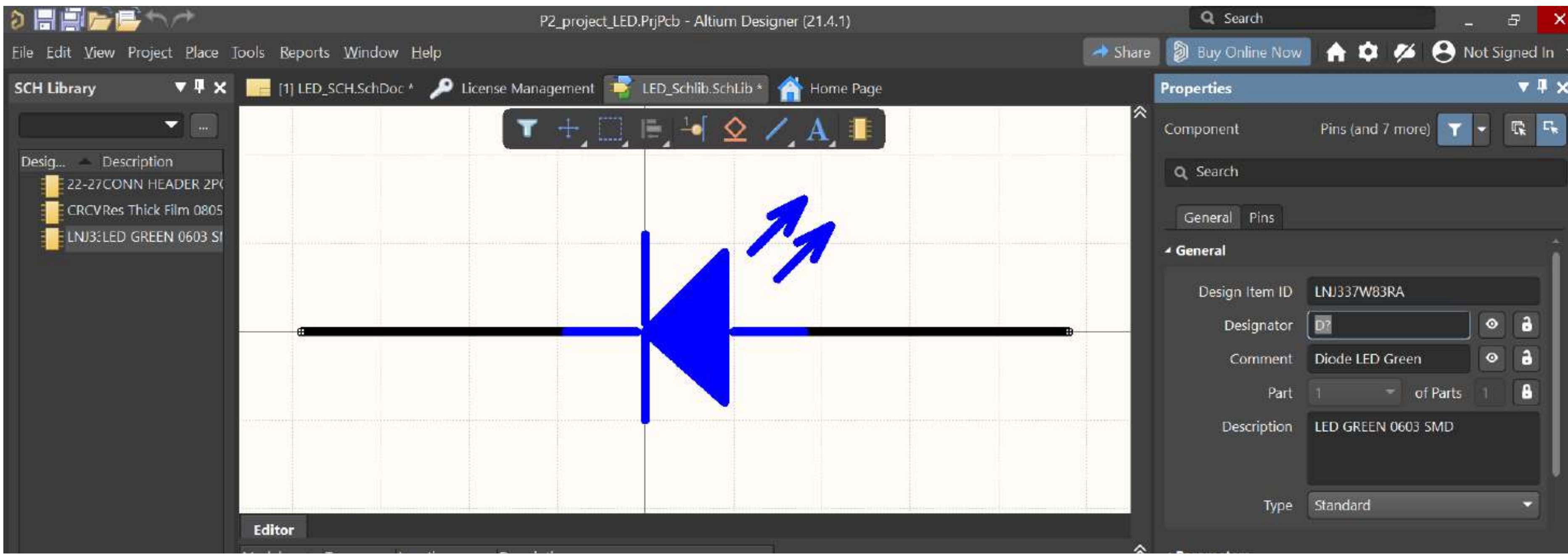
Resistor

- Remember to import the resistor model to the SCH Library.
- Then we should draw a resistor



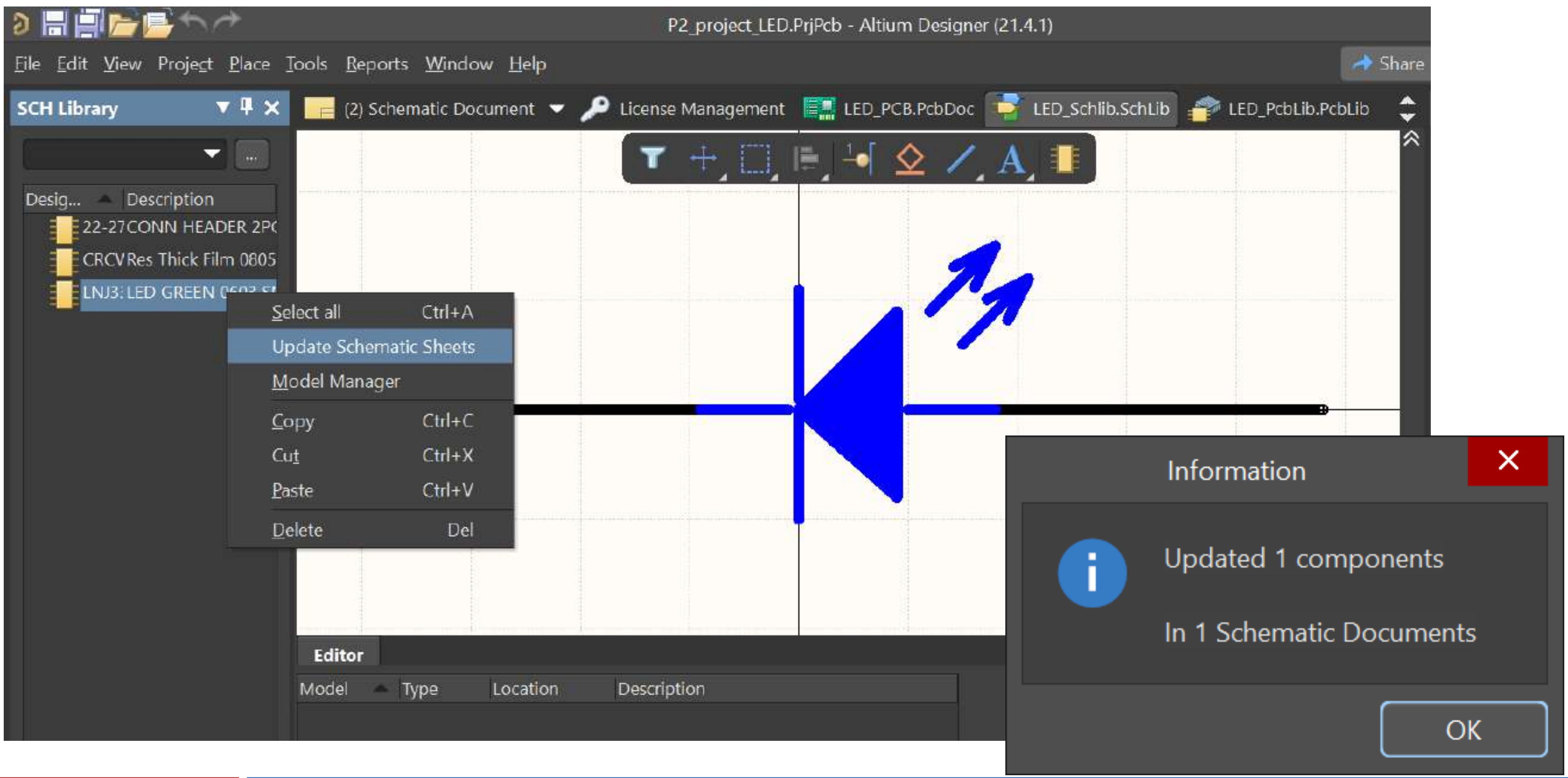
Diode

- We should do the same steps of header with the diode



Updating symbols

- If we wish updating the symbol, an automatic update could be used.

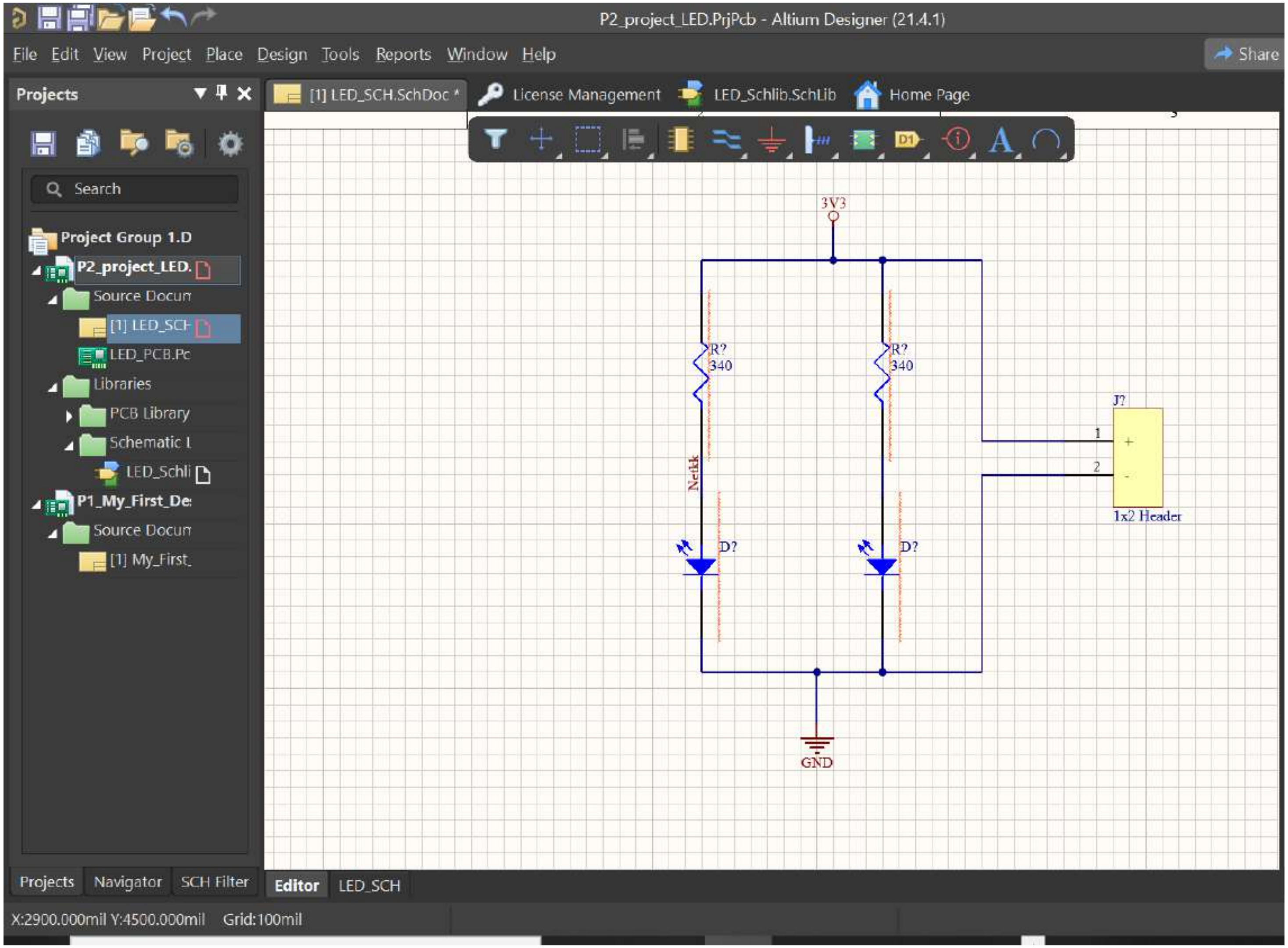


Drawing the schematic

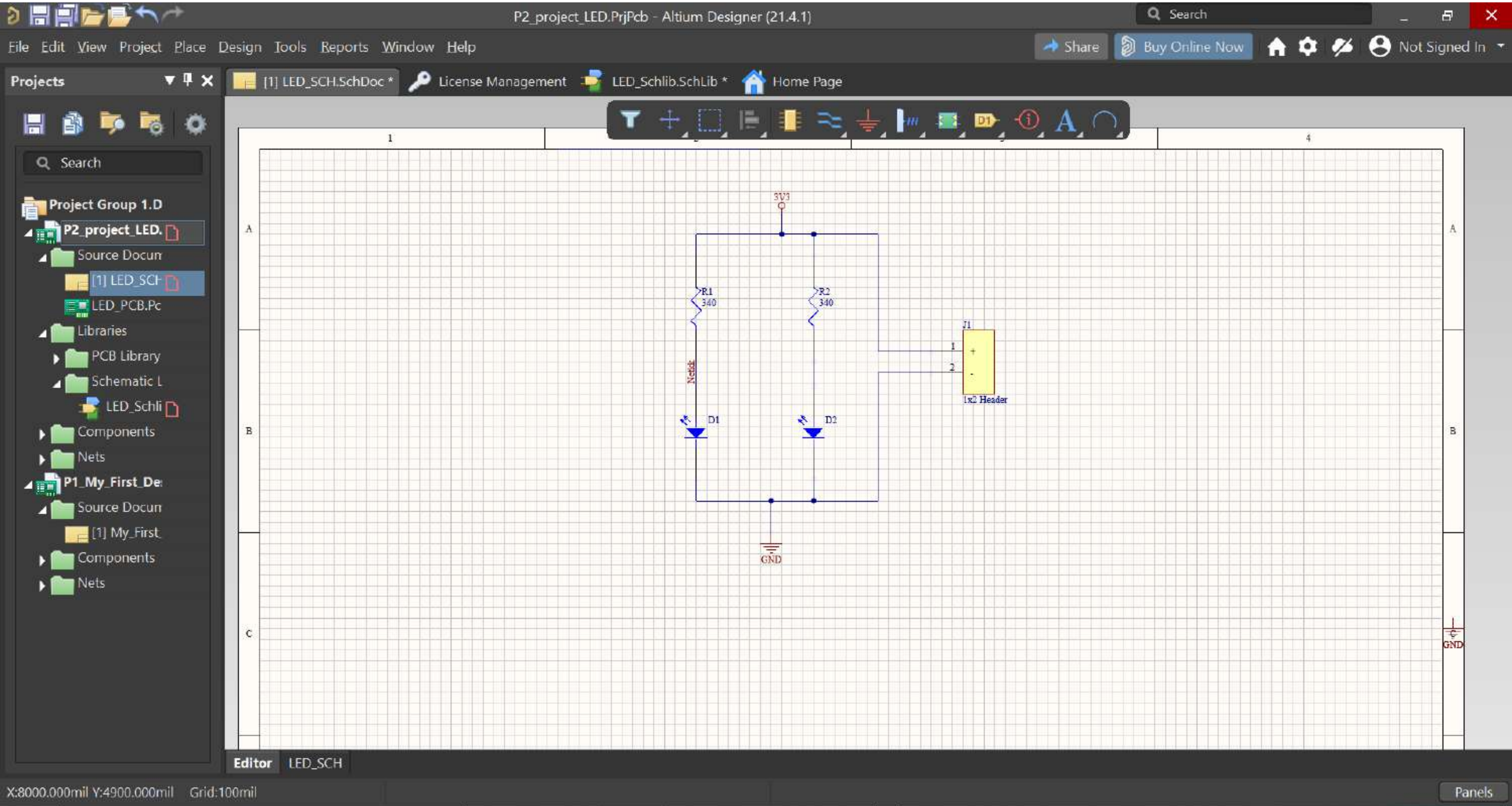
- When we have finished the composition of our schematic library, it is time for drawing of our design.
- Regarding power supply, we will include a 3,3V circle and a GND Power. NOTE: If you will need extra power supplies, it is suggested to copy and paste instead of to create a new one.
- Nets connected to a common point, share the net name.
- We can change the Net Name with Net Label (Place Menu)

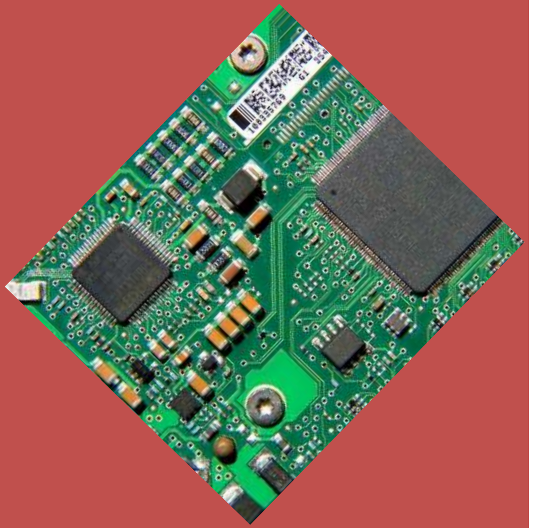
Before finishing ....

- Before annotating the design, errors (red colours) are showed in the sheet.



Final schematic view





Introduction to Printed Circuit Board (PCB) design with ALTIUM

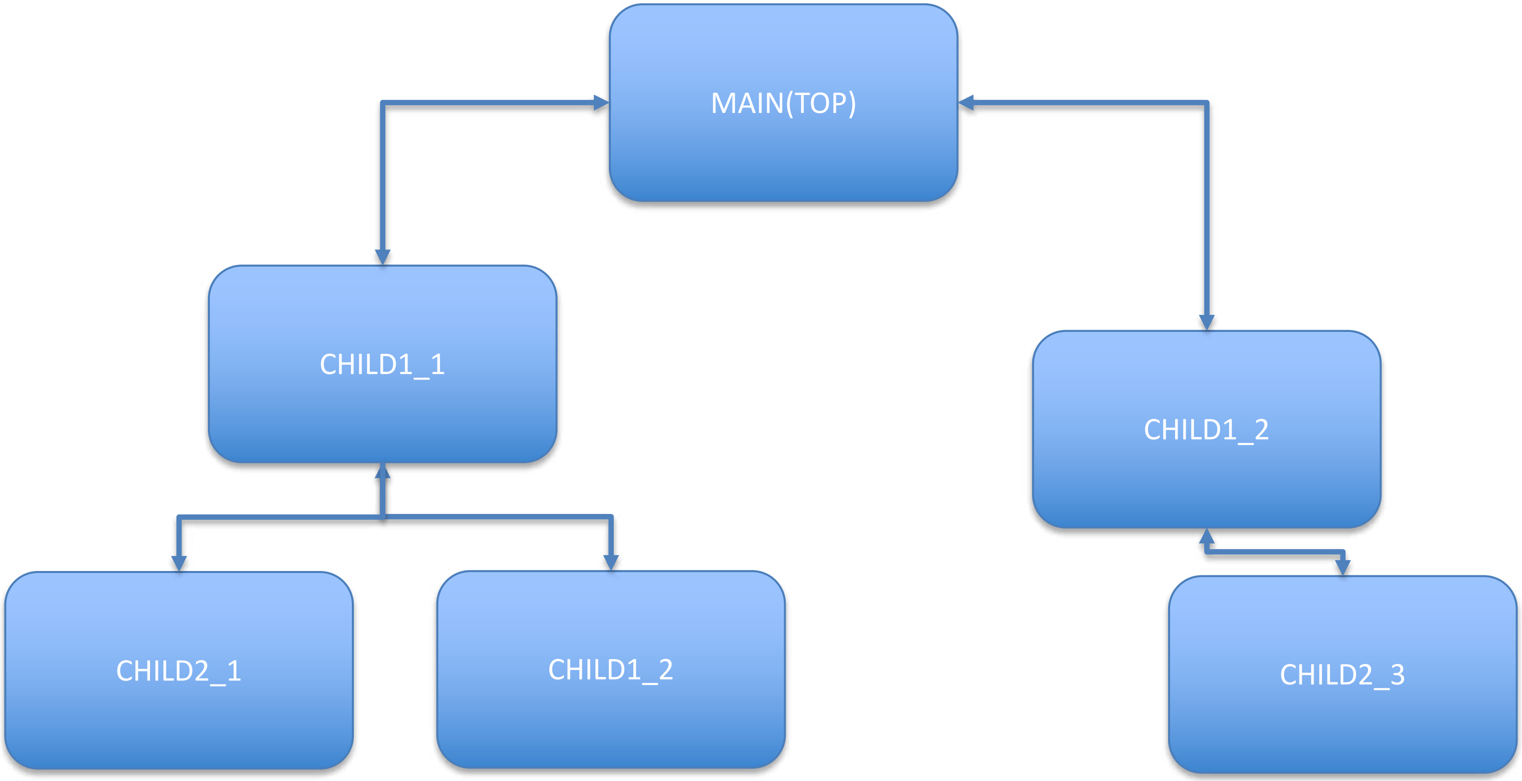
3.- Hierarchical Design

Hierarchical concept

- When a design is complex and big, it is usual to split in several sheets.
- Two approaches are possible:
  - Flat Design (horizontal connectivity)
  - Hierarchical Design (Top-Down approach).
- The second one is the most useful and used because has more flexibility and co-working approach is validated.

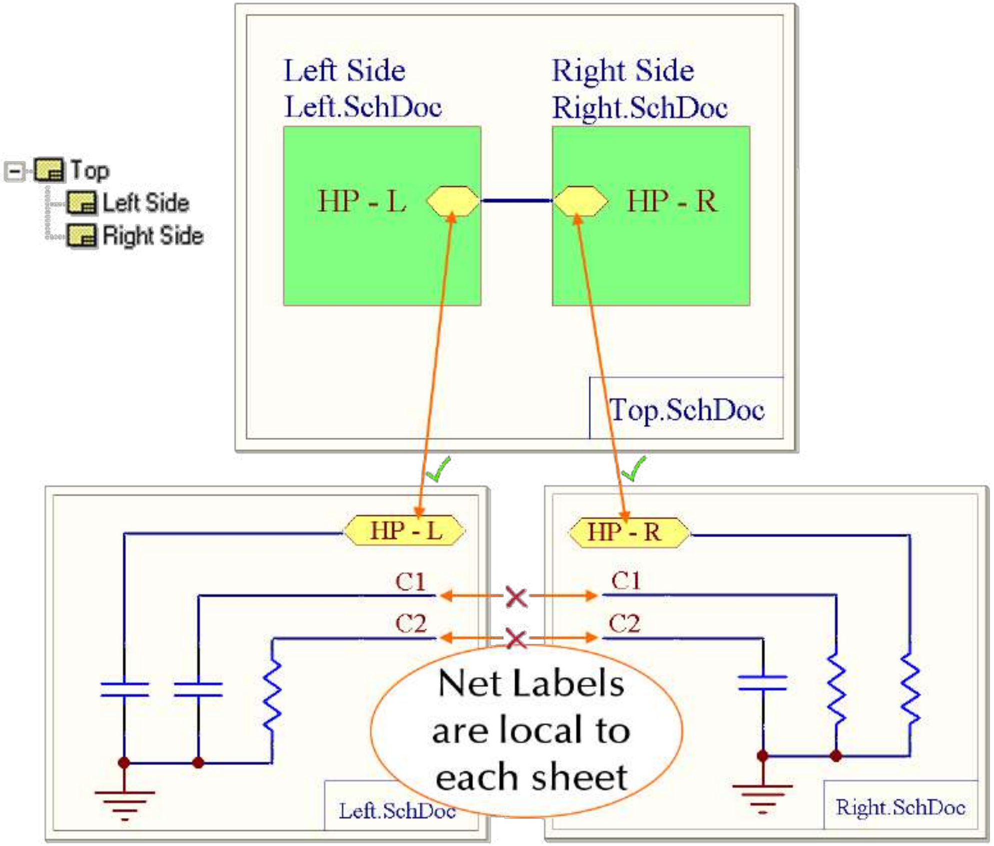
Hierarchical concept

- A top-down topology will have a view like this (maximum 3 levels):



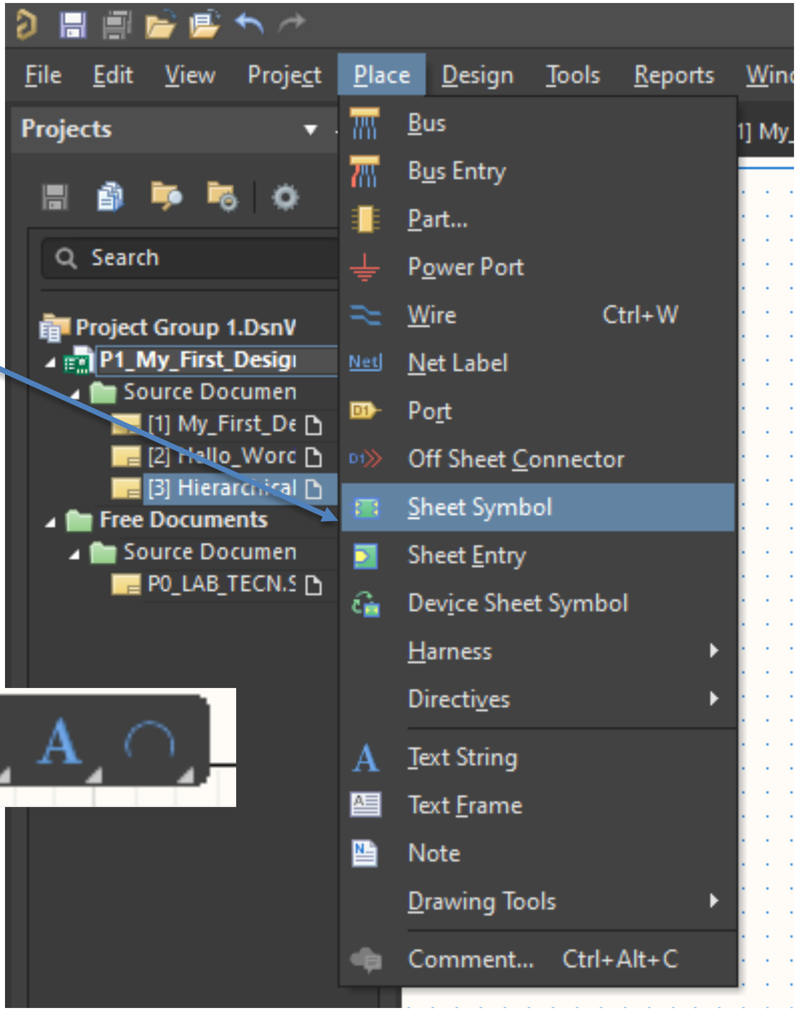
Hierarchical concept

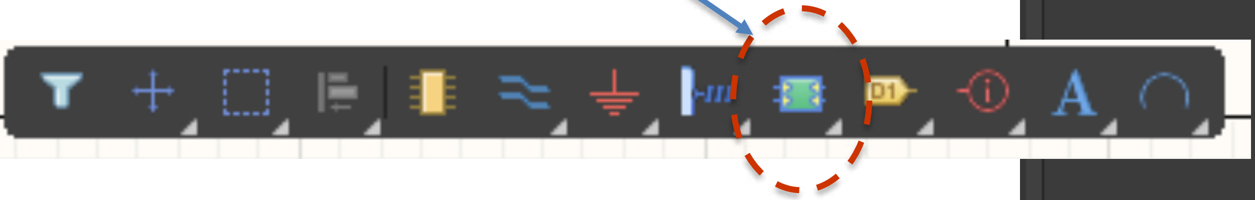
- The connexions between sheets will be done through Ports.
- Local nets are not connected with other sheets.



How to create a hierarchical design?

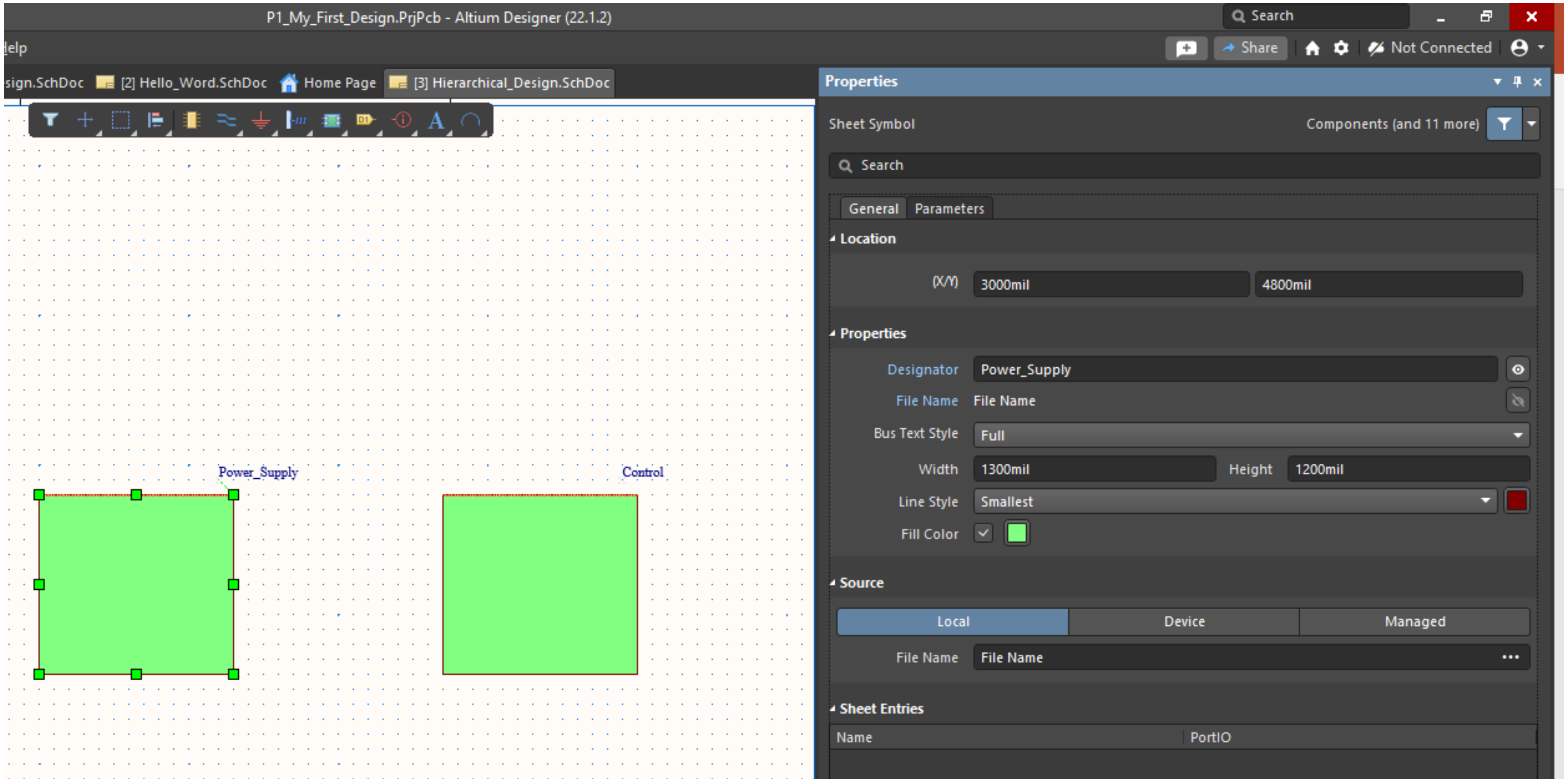
- The way to use this approach is through Sheet Symbol





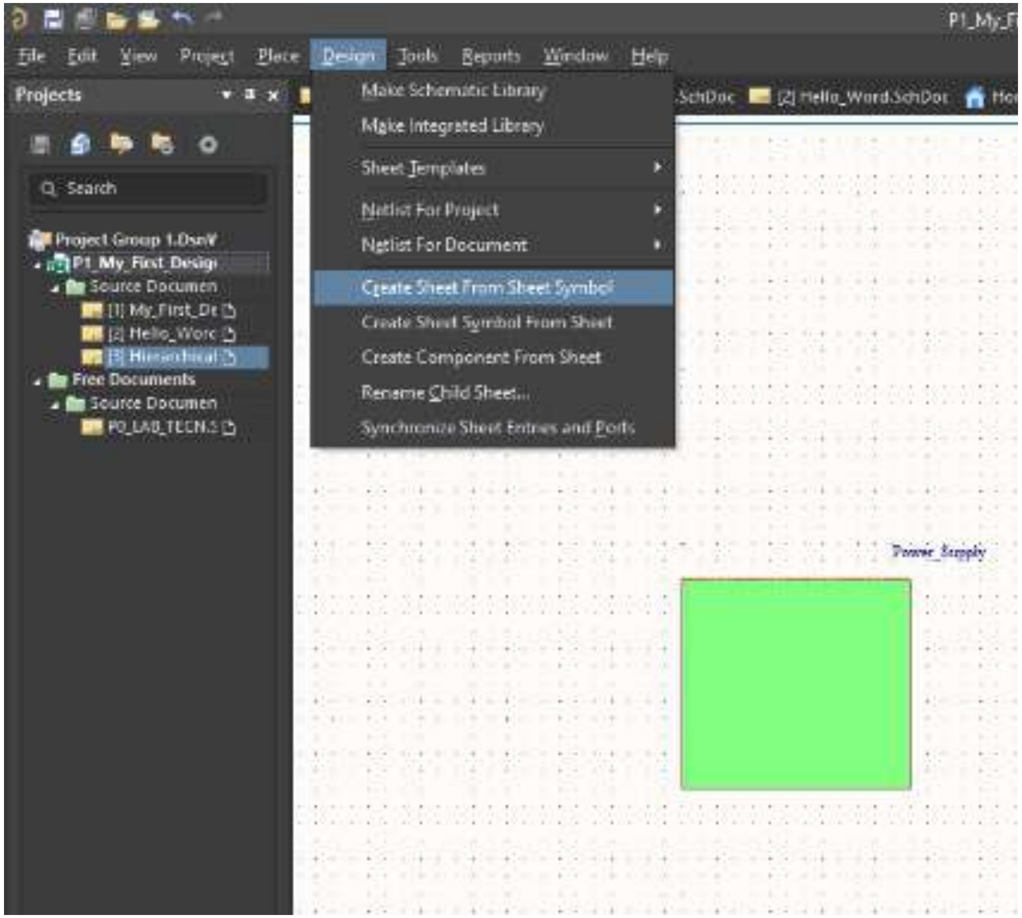
How to create a hierarchical design?

- We will change the sheet name in Properties Panel.



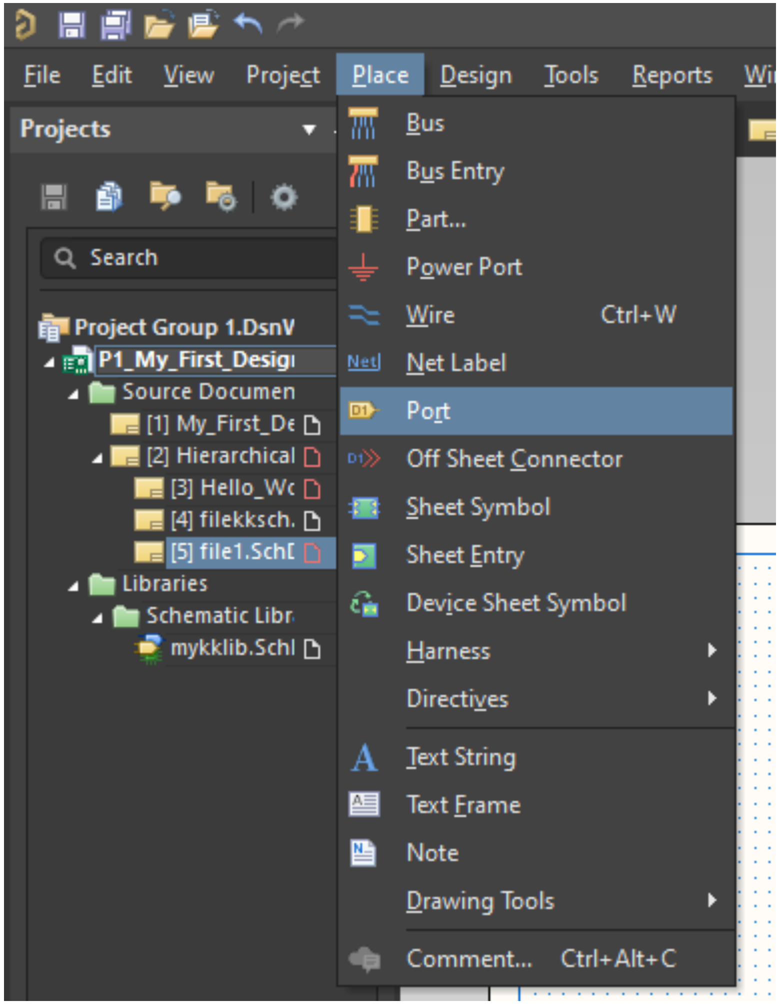
How to create a hierarchical design?

- Then, we need to link the symbols with new sheets



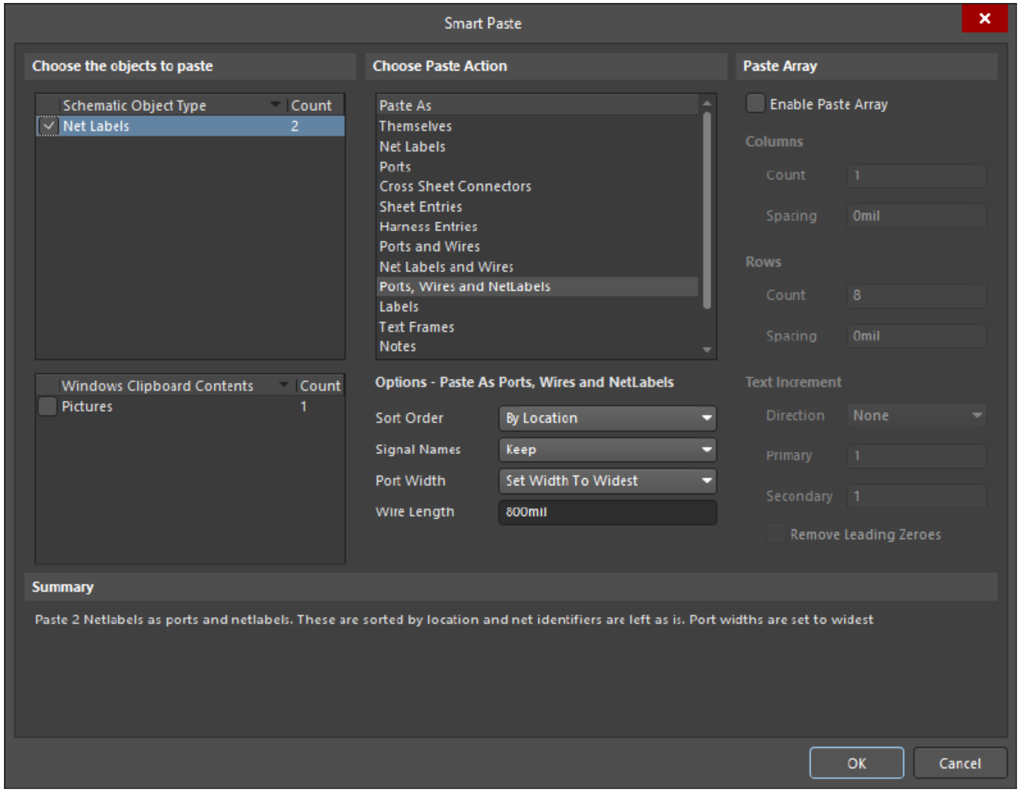
Creating the Ports

- When sheet file is finished, ports must be included.
- One by one can be done using **port** option inside menu Place.



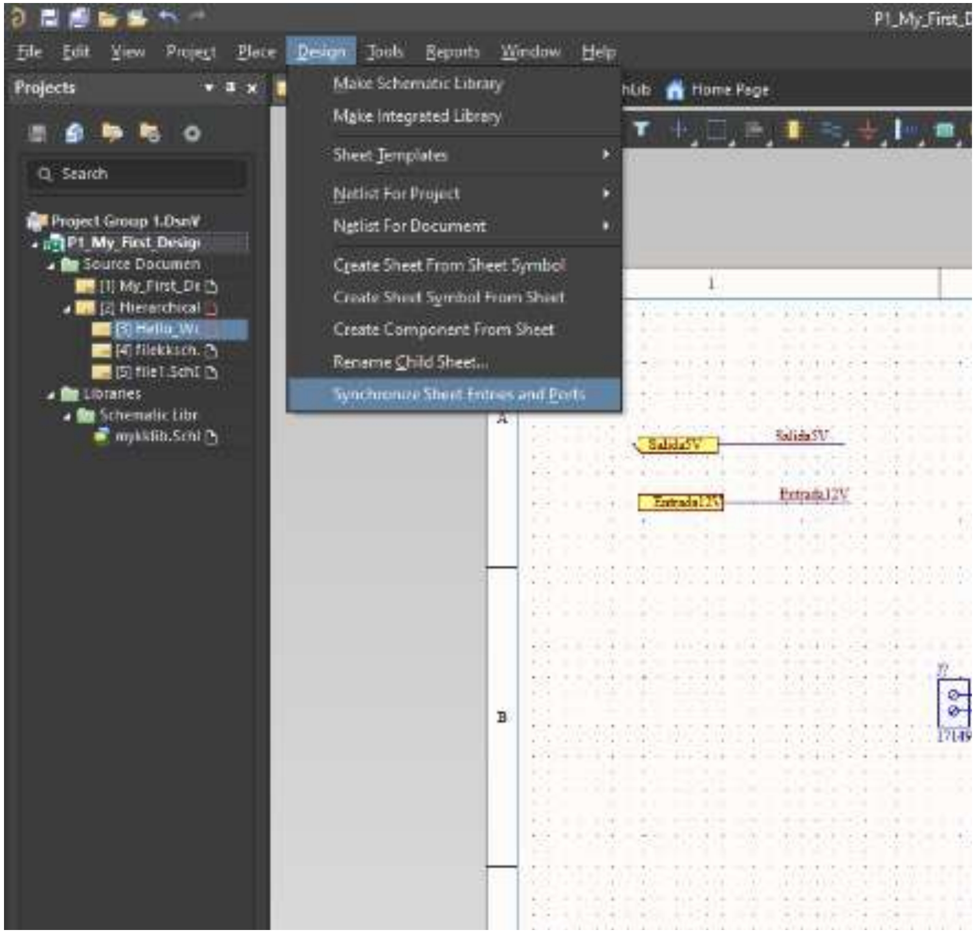
Creating the Ports

- Another way to create several ports in just one click: Smart Paste (Edit Menu),



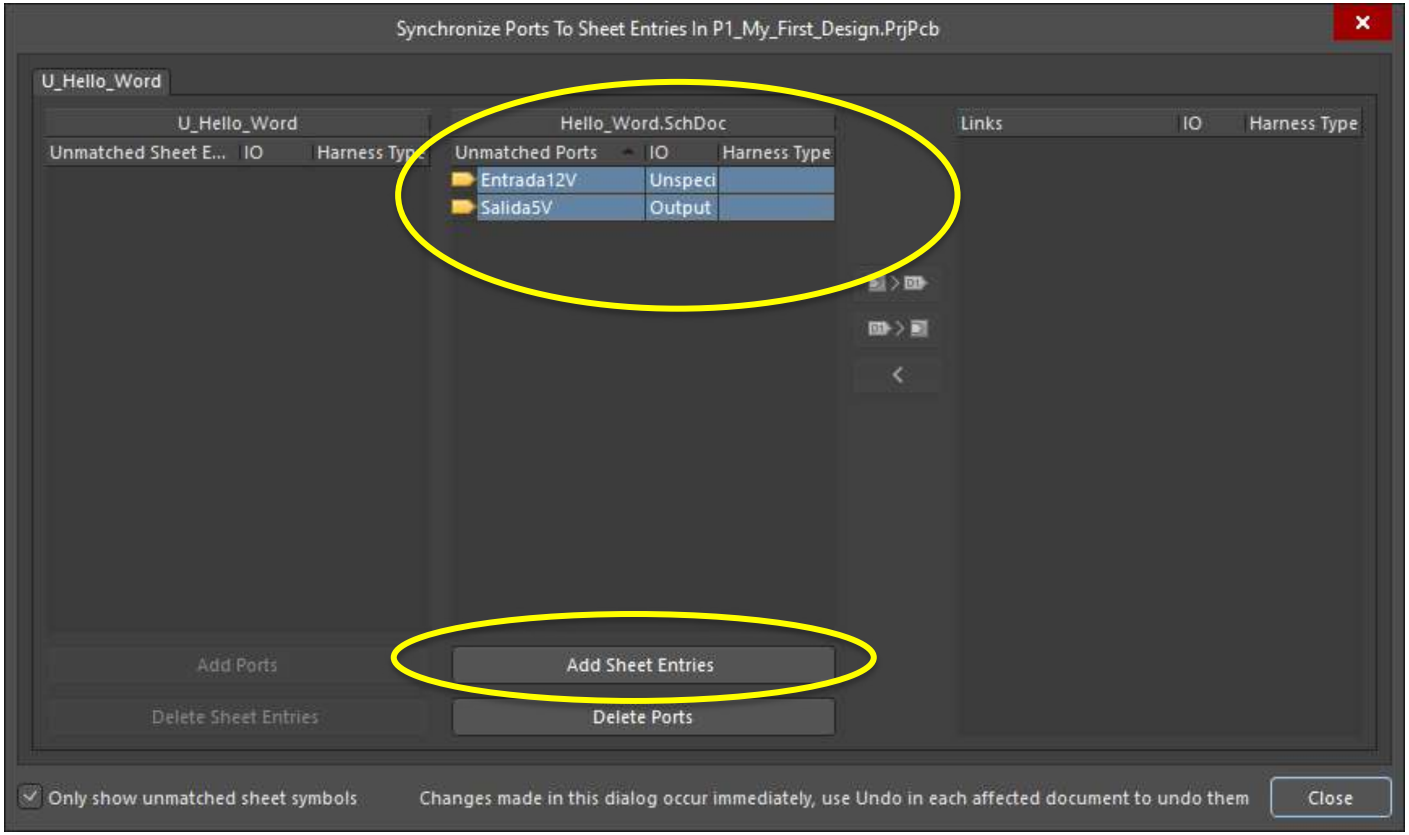
Creating the Ports

- Then, we should include the ports in the Sheet Symbol.
- This can be an automatic process (synchronizing).

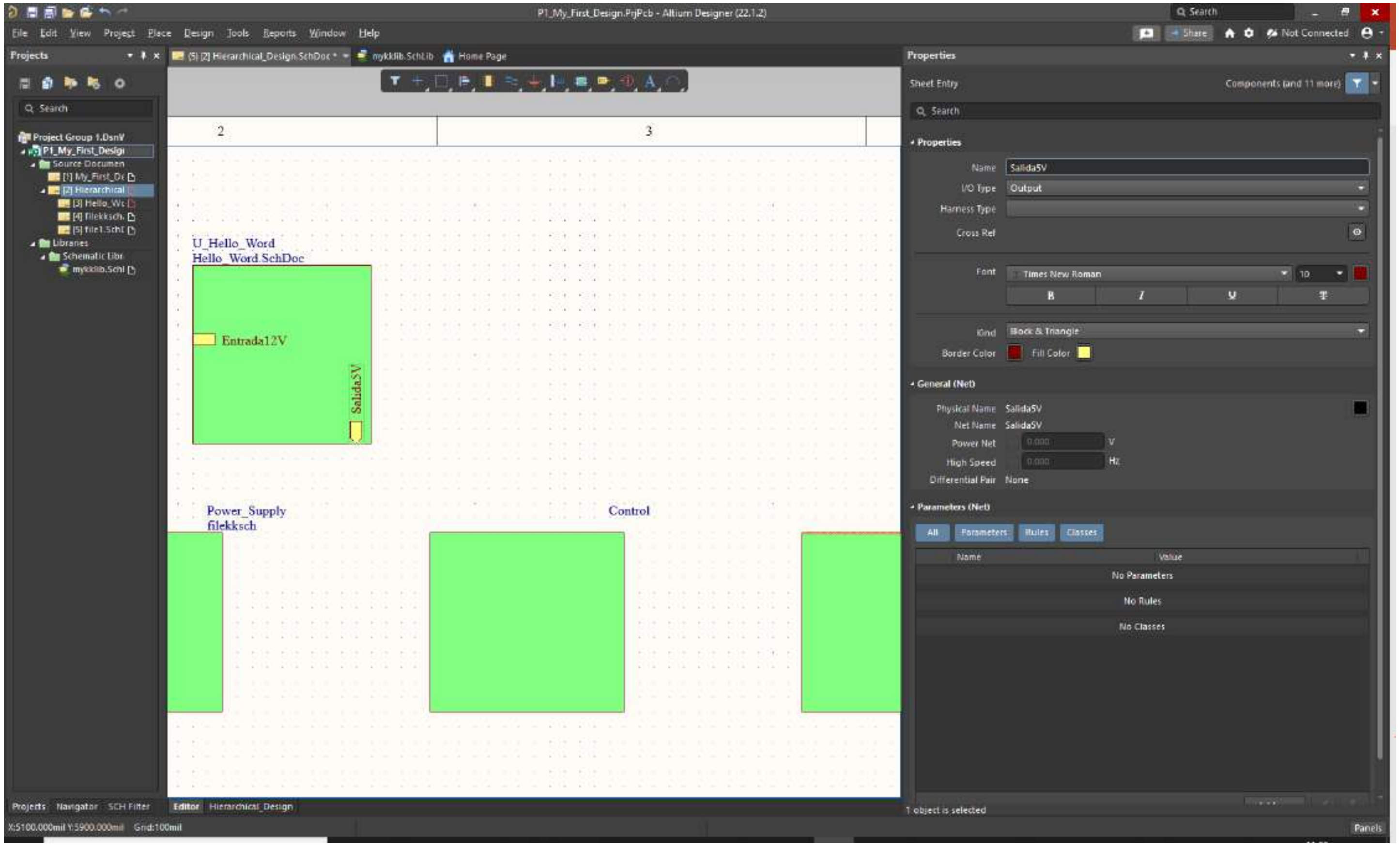


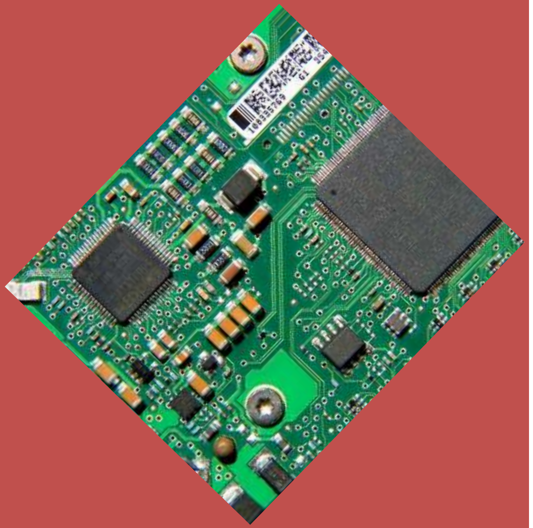
Creating the Ports

- Adding the ports to our symbol sheet



Creating the Ports





Introduction to Printed Circuit Board (PCB) design with ALTIUM

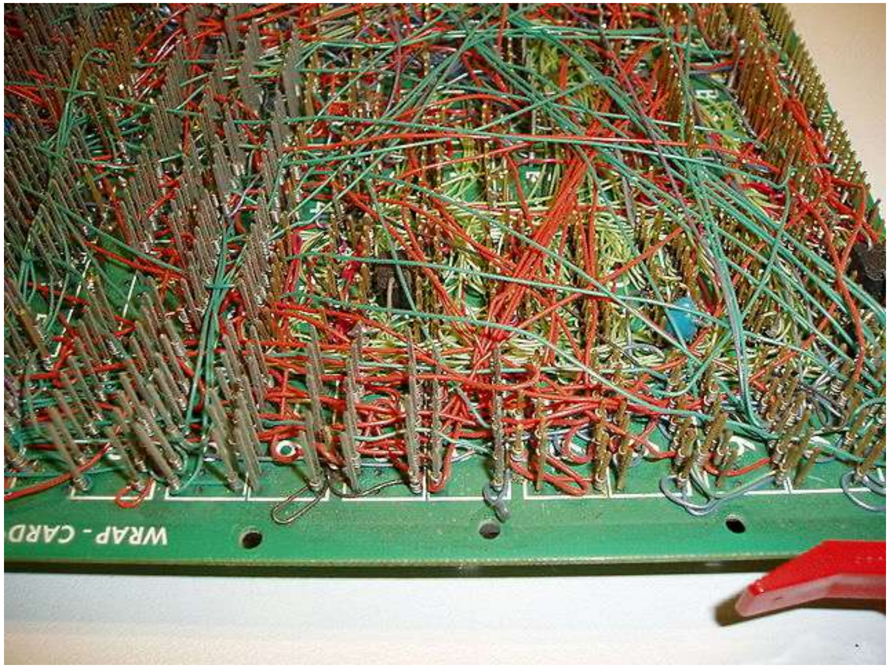
4.- Basic PCB concepts

Basic PCB concepts

- What is a printed circuit board (PCB)? A printed circuit board or PCB, is a plate or board used for placing the different elements that conform an electrical circuit that contains the electrical interconnections between them. [1]
- The different "sides" of a PCB is called layer. The simplest PCB is composed by 1 layer. The most common one uses at least 2 layers

PCB vs other options

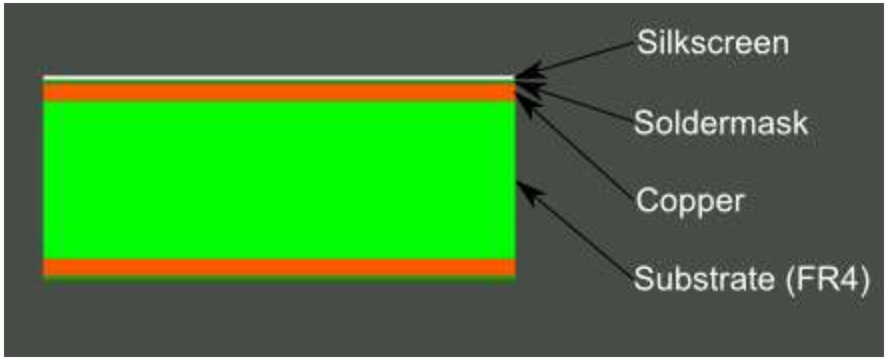
- Why a PCB? If we didn’t have it, we should make circuits as the figure → electrical, connection and thermal problems. [2]



PCB basic concepts

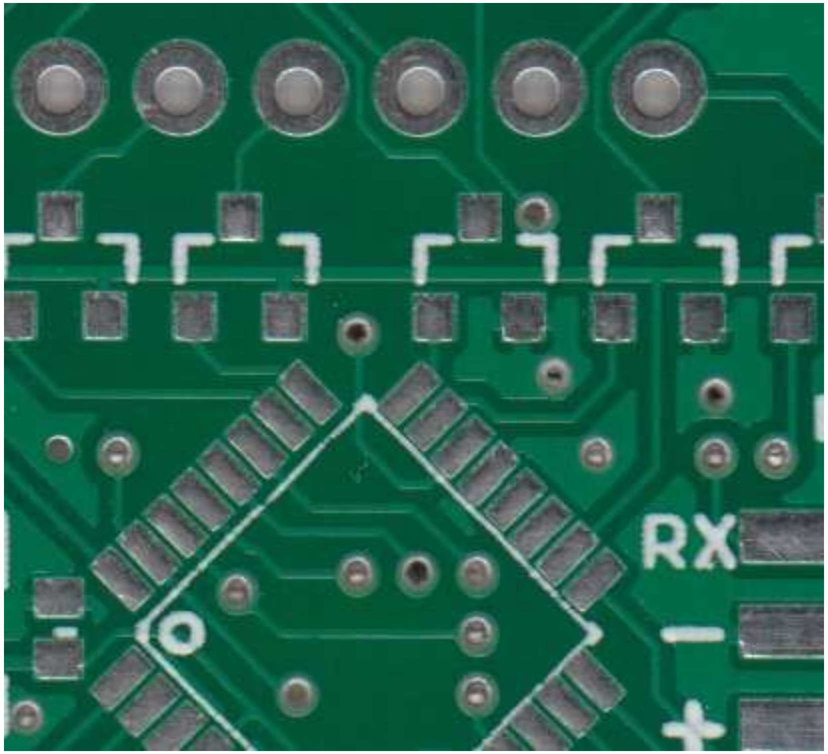
Layers of a PCB

- The figure shows a 2 layer PCB. [2]
- Bottom layer and top layer cannot be composed at the same level.
- FR4 is the substrate is usually fiberglass.
- Copper is the conductor element.



PCB basic concepts

- **Soldermask** is the layer on top of the copper.
- Usually the green colour is used.



PCB basic concepts

Silkscreen

- The white silkscreen layer is applied on top of the soldermask layer. The silkscreen adds letters, numbers, and symbols to the PCB that allow for easier assembly and indicators for humans to better understand the board. [2]

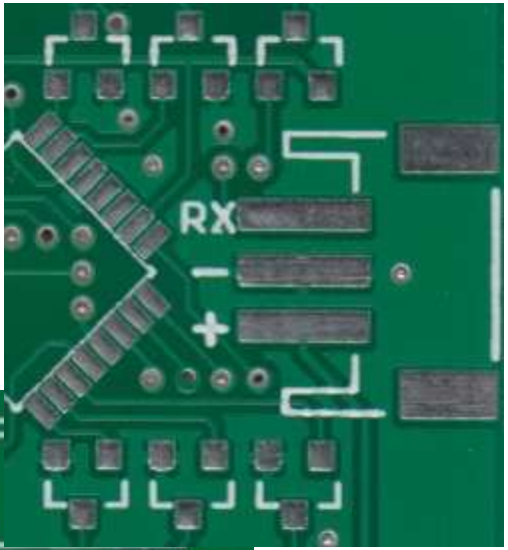
PCB basic concepts

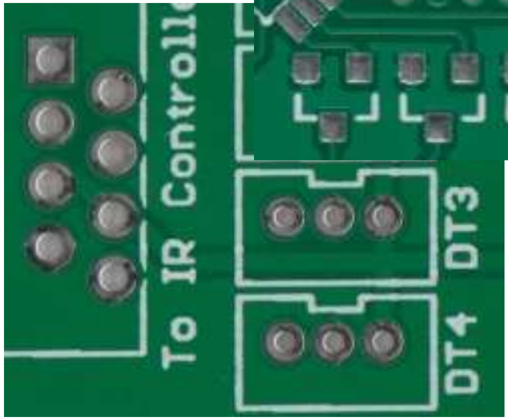
Pad concept.

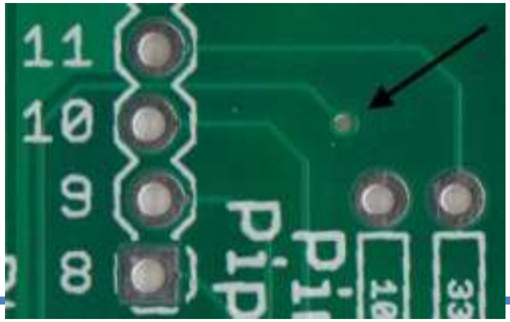
- A portion of exposed metal on the surface of a board to which a component is soldered. [2]

Via concept.

- A hole in a board used to pass a signal from one layer to another.







A PCB composed by multiple layers

- Layer Stackup

CAD Layer (conductive and nonconductive)	CAD Layer description
1	Top silkscreen/overlay (nonconductive)
2	Top soldermask (nonconductive)
3	Top paste mask (nonconductive)
4	Layer 1 (conductive)
5	Sustrate (nonconductive)
6	Layer 2 (conductive)
...	...
n-1	Sustrate (nonconductive)
n	Layer n (conductive)
n+1	Bottom paste mask (nonconductive)
n+2	Bottom solder mask (nonconductive)
n+3	Bottom silkscreen/overlay (nonconductive)

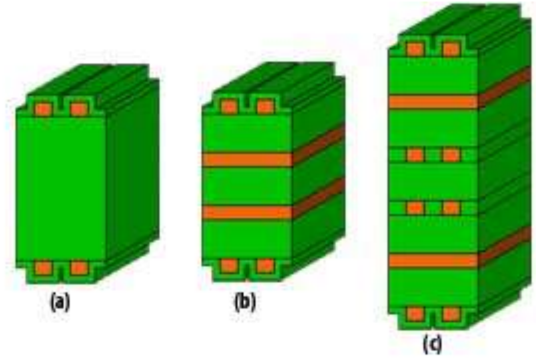


Fig 3. Example of 3 different PCB stackups: 2 layers (a), 4 layers (b) and 6 layers (c)

Types of footprint

Component packages (footprints)

- There are 3 main types of footprints:
- Thru Hole
- Surface Mounting
- BGA (Ball Grid Array)

Types of footprint

- SMDs are divided in several groups:
- SOIC (Small Outline Integrated Circuit)
- SOP (Small Outline Package)

- QFP (Quad Flat pack)

Types of footprint

- BGA

Types of footprint

- A summary of footprints (I/II)

IC Chip Packages	
CBGA - Ceramic Ball Grid Array	PQFP - Plastic Quad Flat Pack
CCGA - Ceramic Column Grid Array	PSOP - Power Small Outline Package
CerDIP - Ceramic Dual-in-Line Package	QFN - Quad Flat No Leads Package
CerPack - Ceramic Package	QSOP - Quarter Size Outline Package
CLCC - Ceramic Leadless Chip Carrier	SBDIP - Sidebraze Dual-in-Line Package
CPGA - Ceramic Pin Grid Array	SC-70 - Small Outline Transistor
CQFP - Ceramic Quad Flat Pack	SIP - Single-In-Line Package
D2PAK or DDPAK - Dou-ble Decawatt Package	SOIC - Small Outline IC Package
D3PAK - Decawatt Package 3	SOJ - Small Outline J-Lead Package
DFN - Dual Flat No Leads Package	SOT-23 - Small Outline Transistor
DPAK - Decawatt Package	SPDIP - Shrink Plastic Dual-in-Line Package
FBGA - Fine-Pitch Ball Grid Array	SSOP - Shrink Small Outline Package

Types of footprint

- A summary of footprints (II/II)

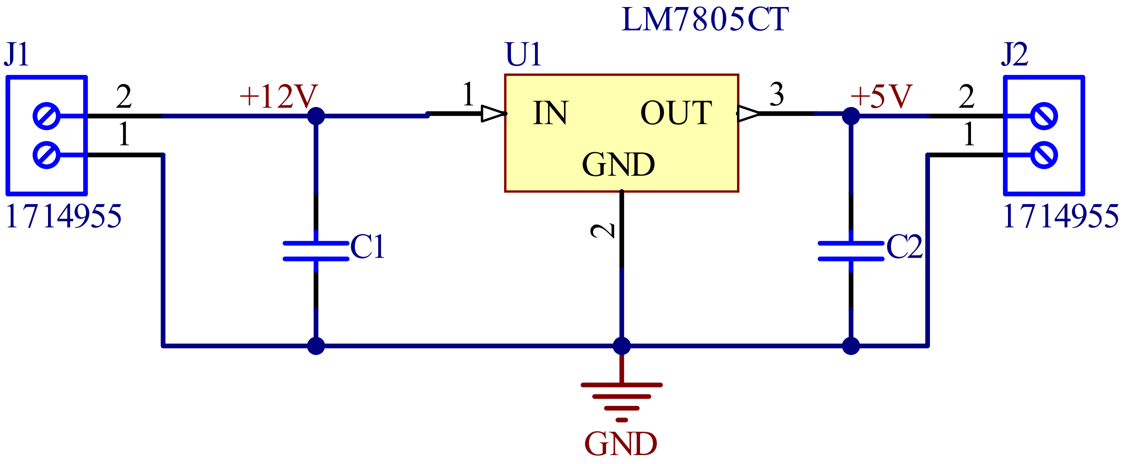
JLCC - J-Leaded Ceramic Chip Carrier	TDFN - Thin Dual Flat No Leads Package
LFBGA - Low Profile Fine-Pitch Ball Grid Array	TFBGA - Thin Fine-Pitch Ball Grid Array
LGA - Land Grid Array	TQFN - Thin Quad Flat No Leads Package
LQFP - Low-Profile Quad Flat Package	TQFP - Thin Quad Flat Pack
MLP - Micro Leadframe Package	TSOP - Thin Small Outline Package
MQFP - Metric Quad Flat Package	TSSOP - Thin Shrink Small Outline Package
MSOP - Micro Small Outline Package	UTDFN - Ultra Thin Dual Flat No Leads Package
PBGA - Plastic Ball Grid Array	UTQFN - Ultra Thin Quad Flat No Leads Package
PDIP - Plastic Dual-in-Line Package	VFBGA - Very Thin Fine-Pitch Ball Grid Array
PLCC - Plastic Leaded Chip Carrier	VSOP - Very Small Outline Package
PPGA - Plastic Pin Grid Array	VSOP - Very Small Outline Package
PQFN - Power Quad Flat No Leads Package	XQFN - Extreme Thin Quad Flat No Leads Package

Several references to know more

- [1] Electrosoft Engineering. “Concepts and terminology used in Printed Circuit Boards (PCB)” http://www.pcb.electrosoft-engineering.com/04-articles-custom-system-design-and-pcb/01-printed-circuit-board-concepts/printed-circuit-board-pcb-concepts.html
- [2] Sparkfun Start Something“PCB Basics”. https://learn.sparkfun.com/tutorials/pcb-basics/all
- [3] MadPCB “Chip Package” https://madpcb.com/glossary/chip-package/

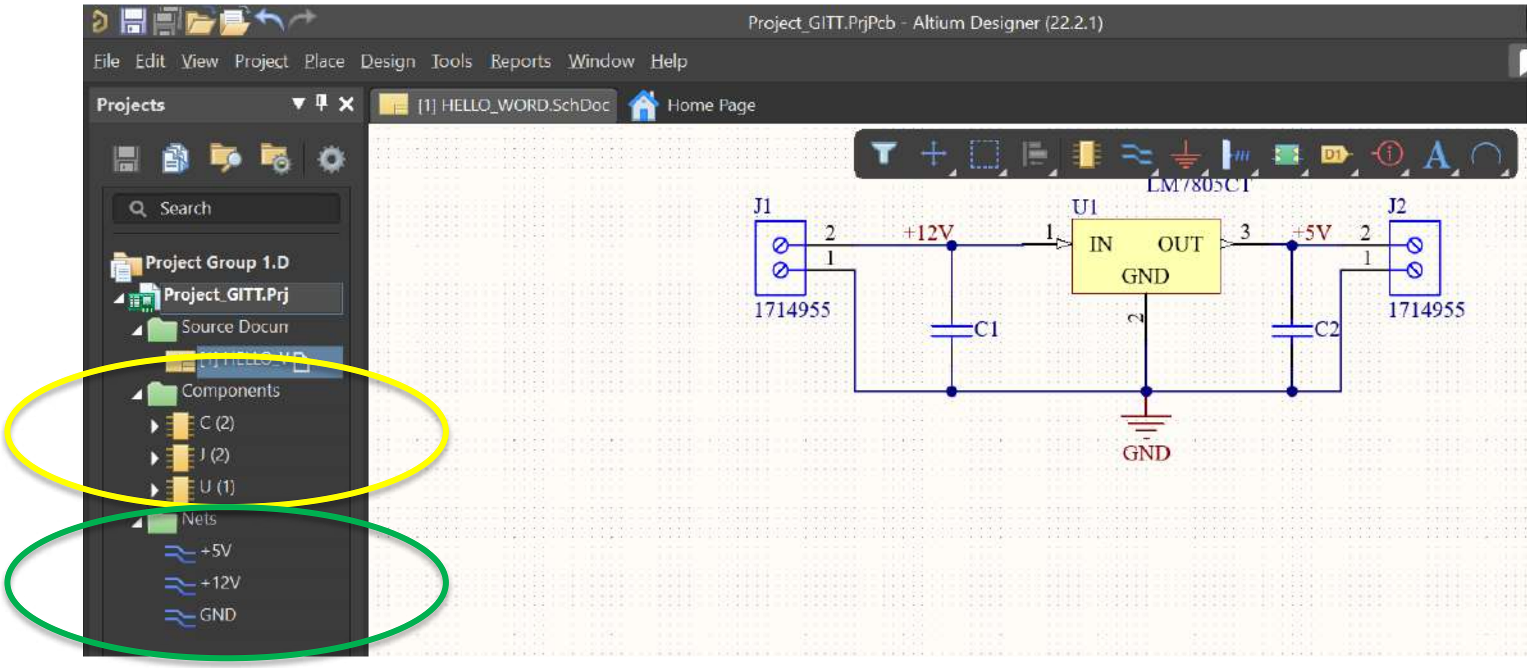
My first Layout

- Based on a simple schematic, the plan is to implement a basic layout design using commercial footprints and devices.



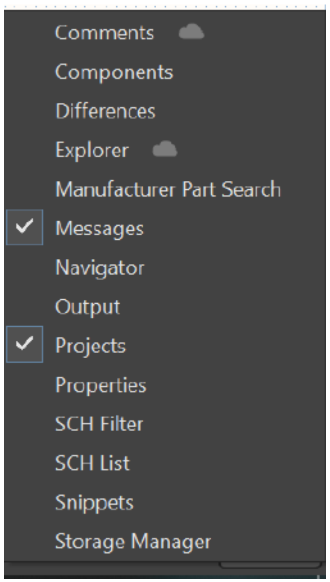
My first Layout

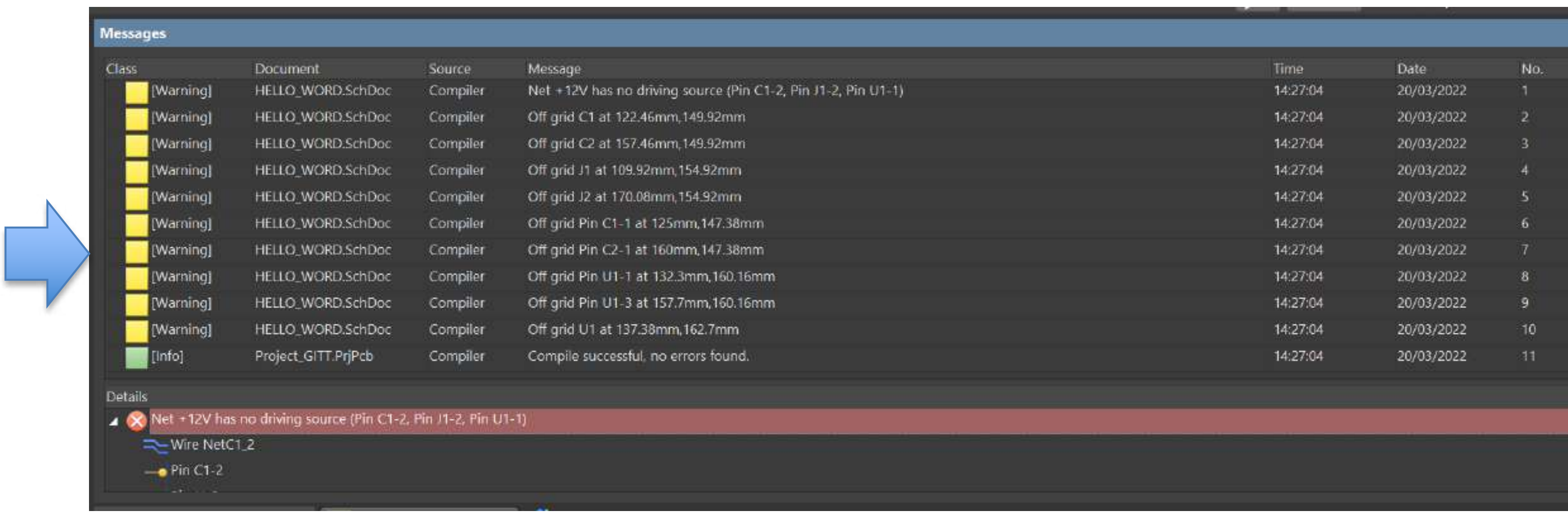
- Before doing anything else we can check the components and nets used in the Project.



My first Layout

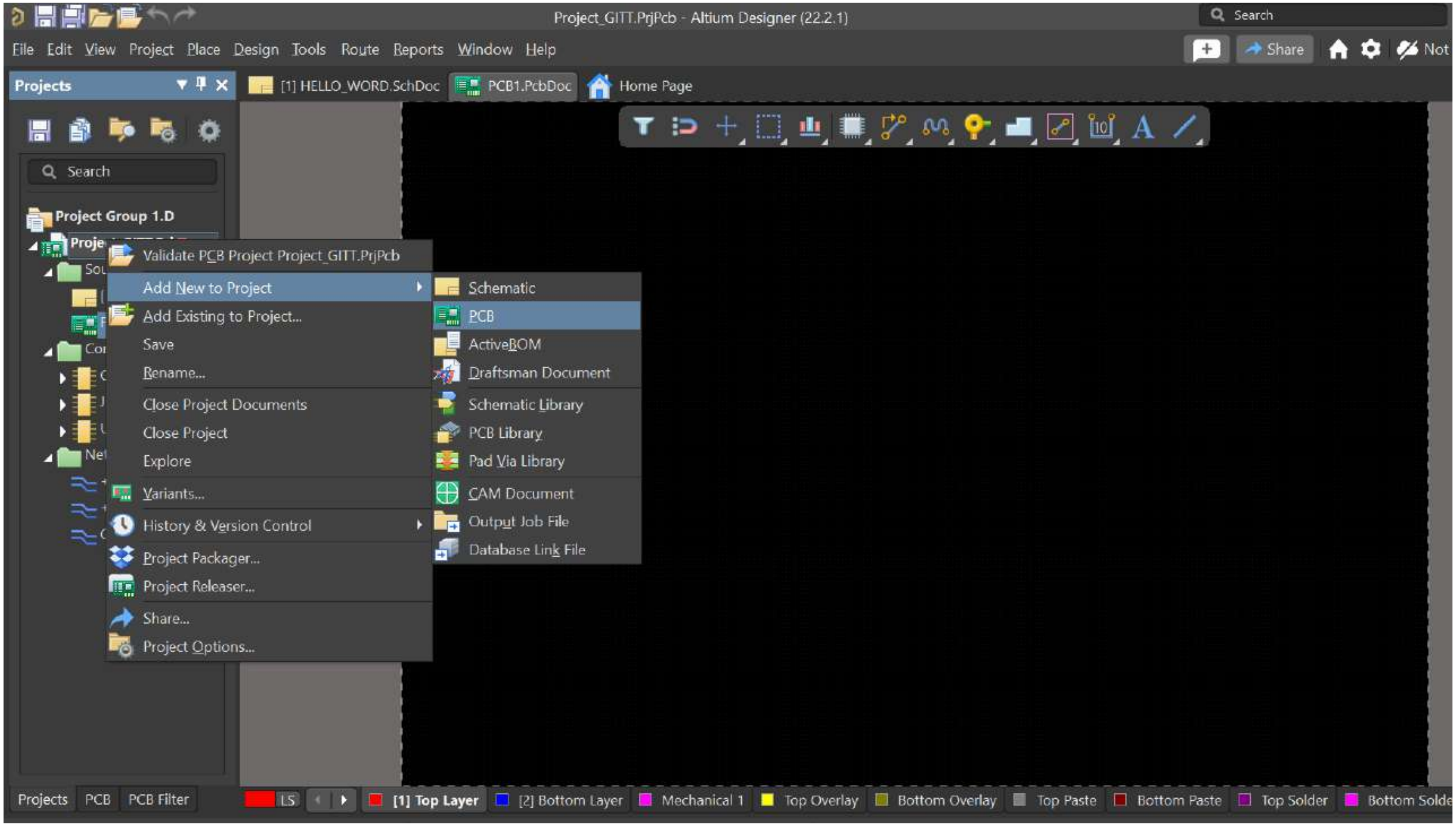
- Additionally, we will check any type of electrical error (Project → Validate PCB Project).
- Messages can be displayed through panel tab.





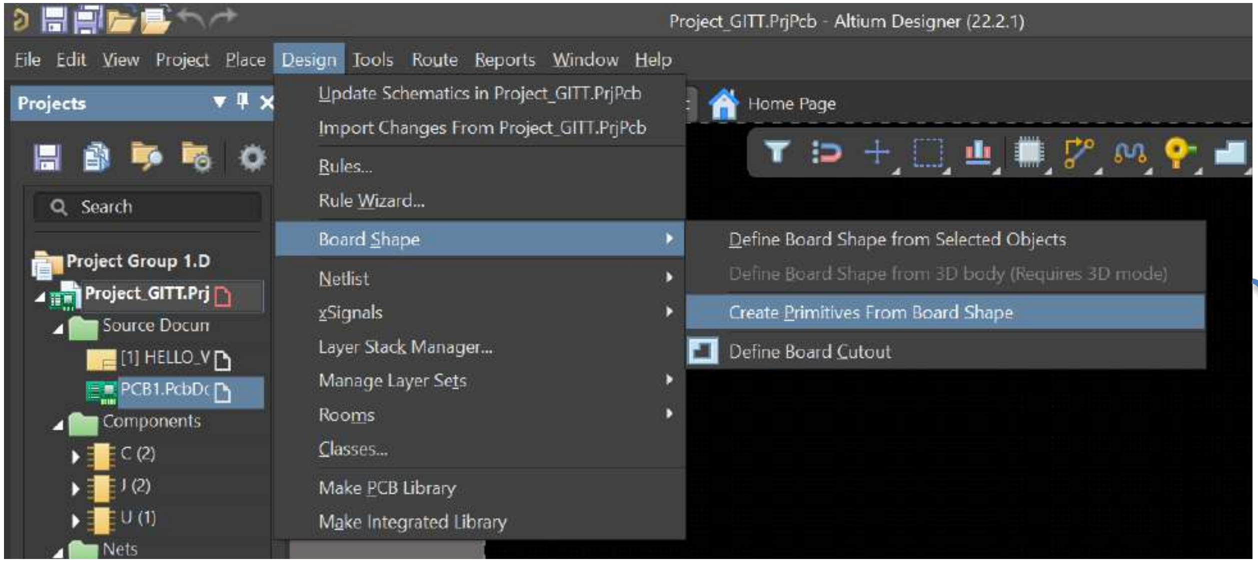
My first Layout

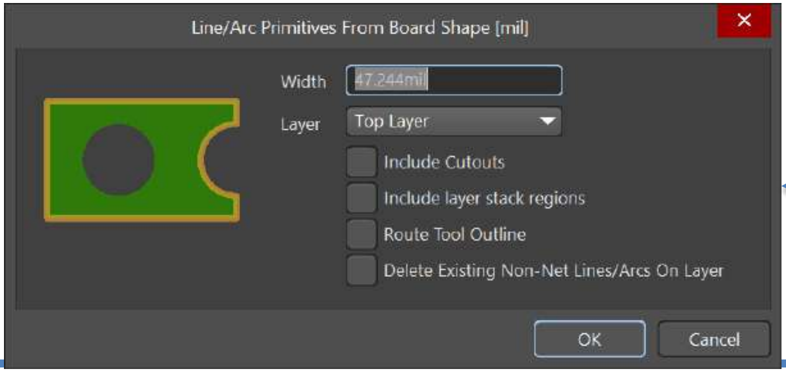
- Creating a PCB "Sheet"



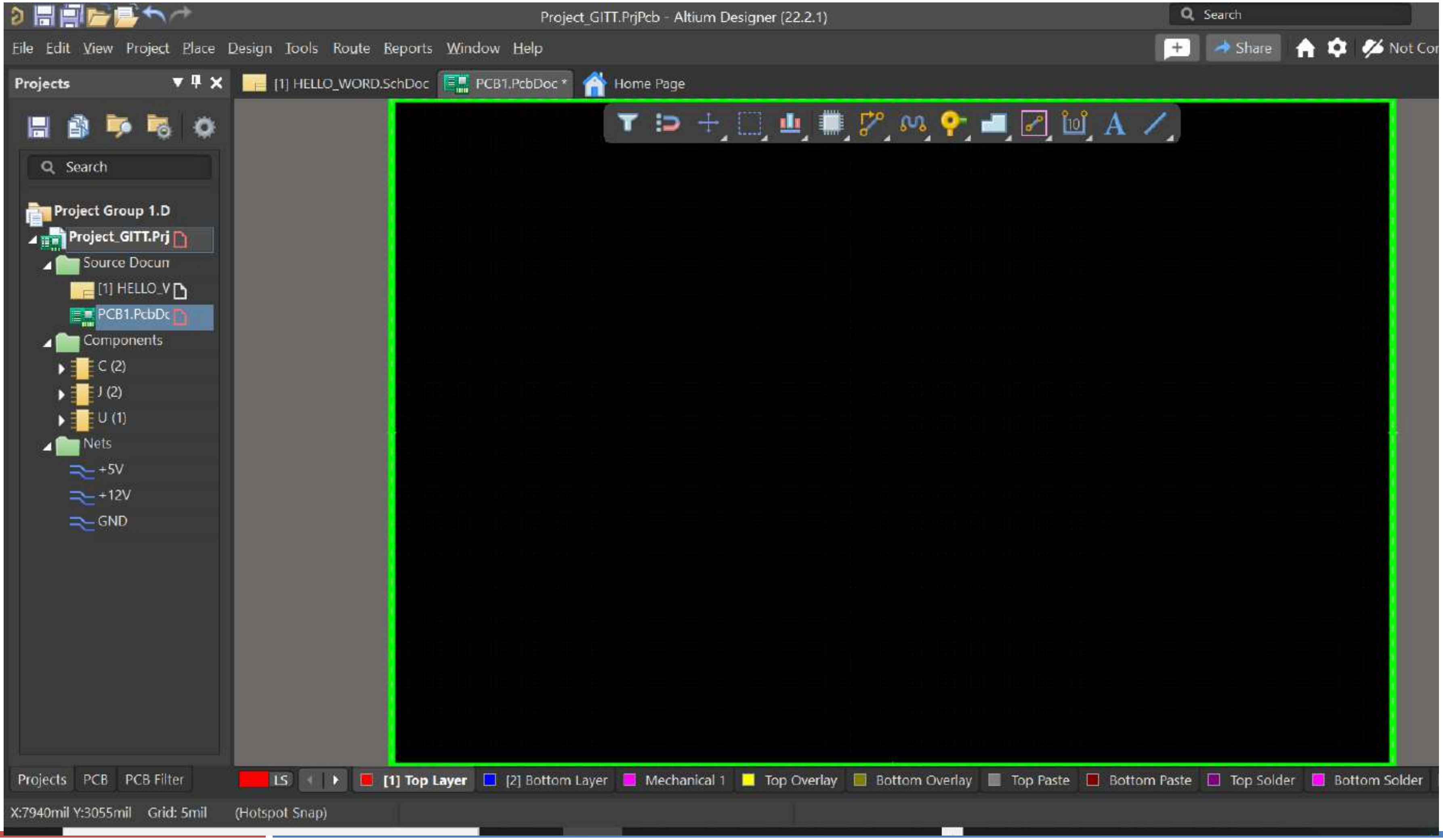
My first Layout

- Defining the size of the PCB



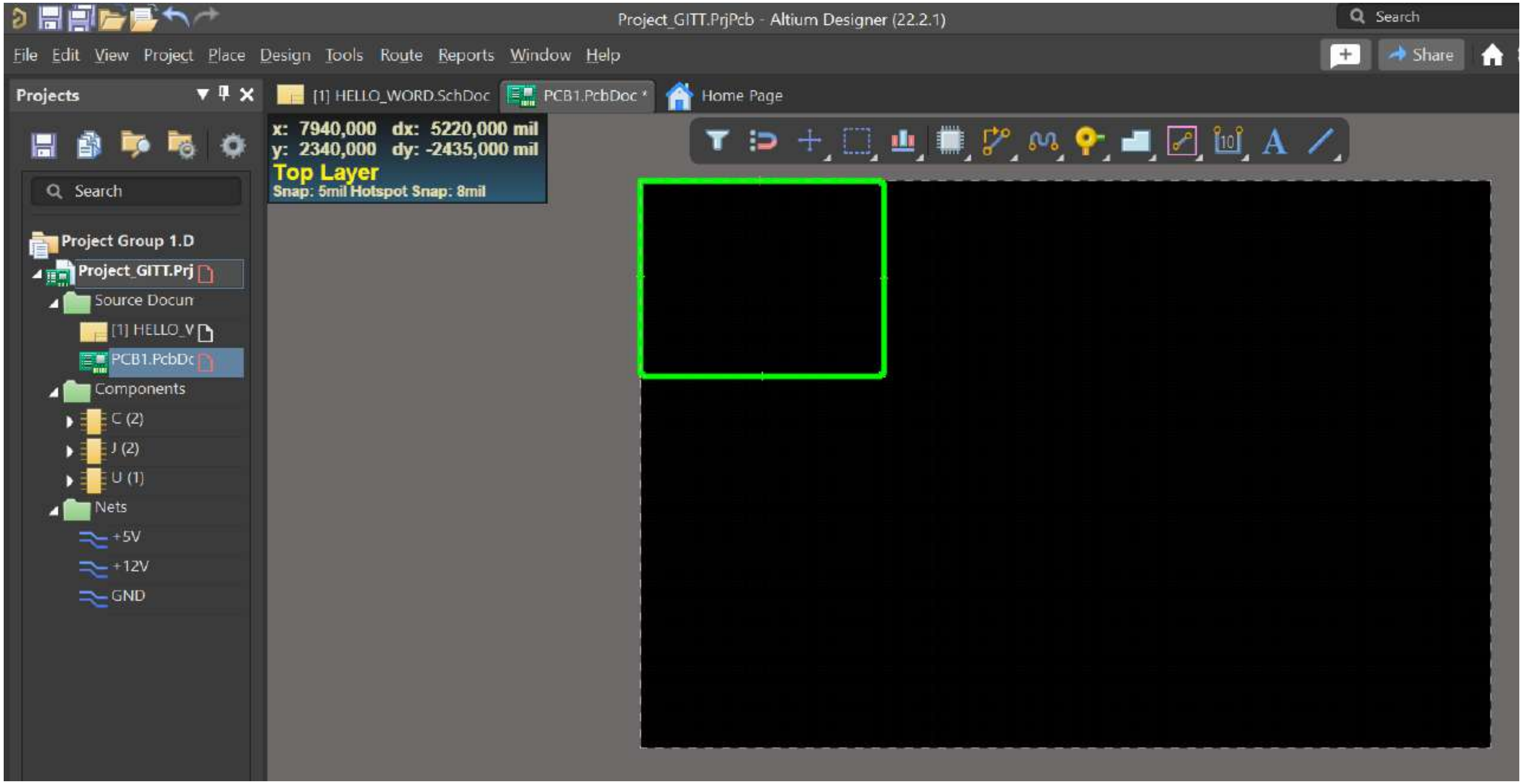


My first Layout

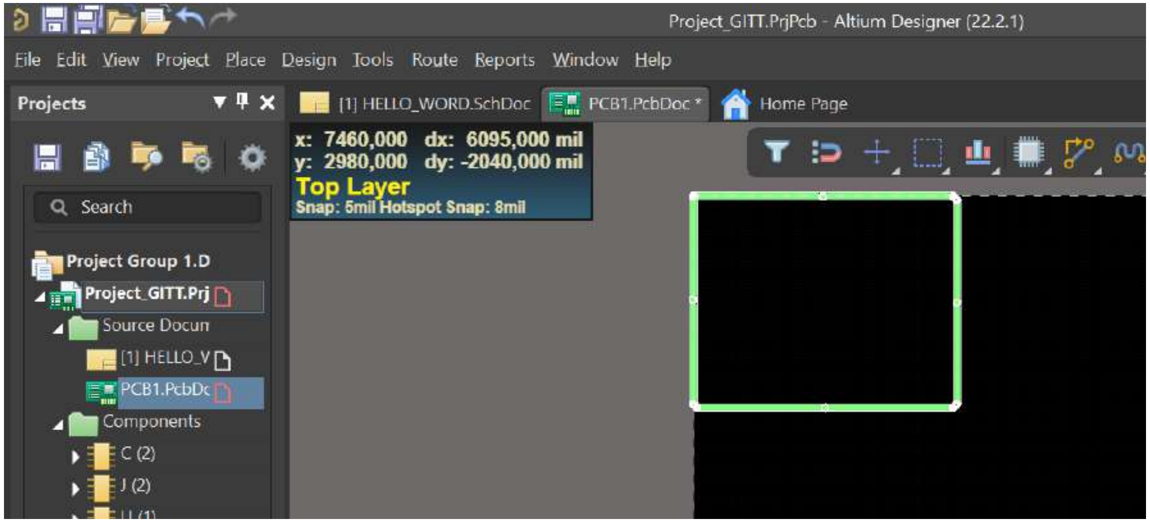


My first Layout

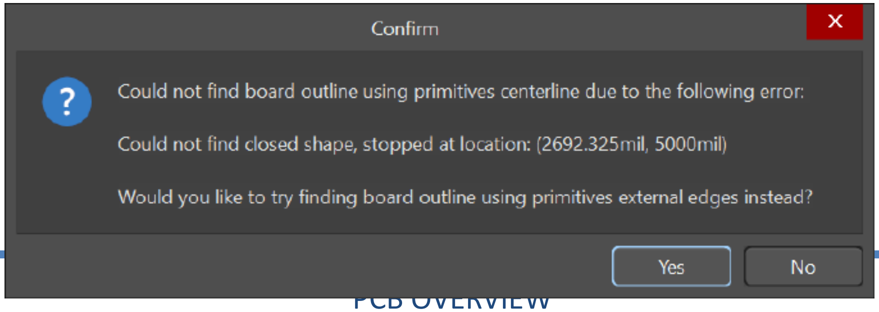
- The default are for our PCB is too big. Then, we should reduce the size moving the green rectangle.



- So, we should click each border to hold on….

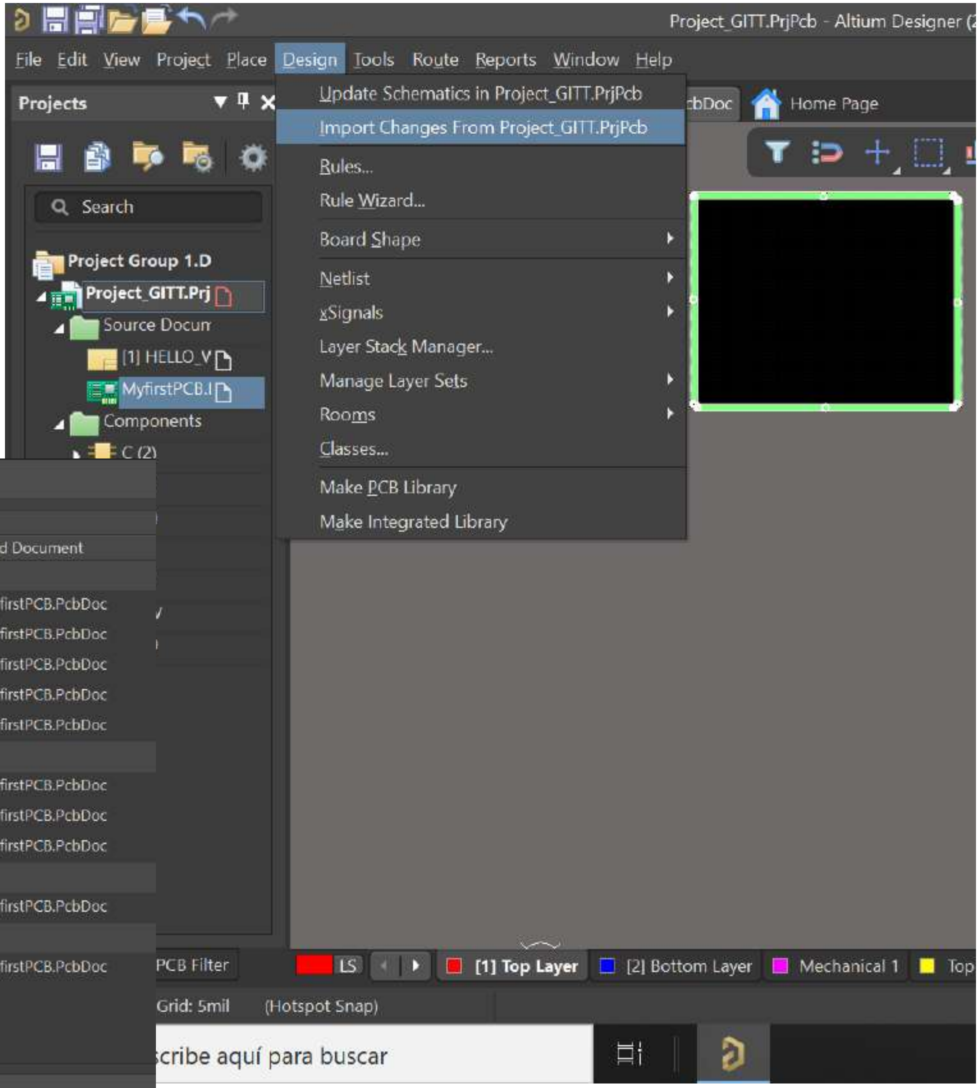


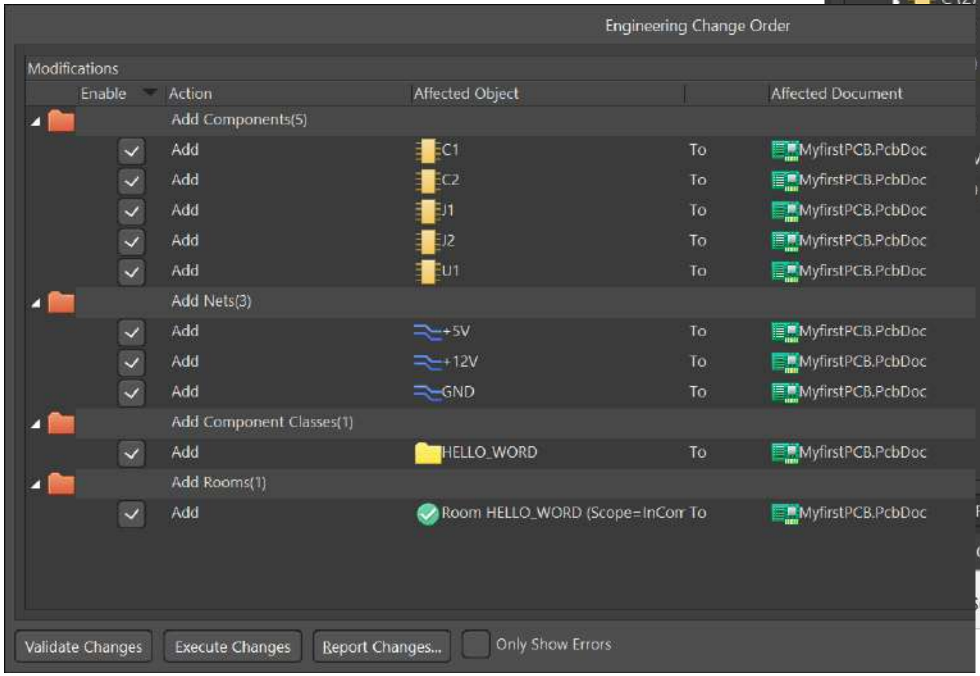
- …and create a new size for our board. (Design → Board Shape → Define Board Shape from Selected obj.



My first Layout

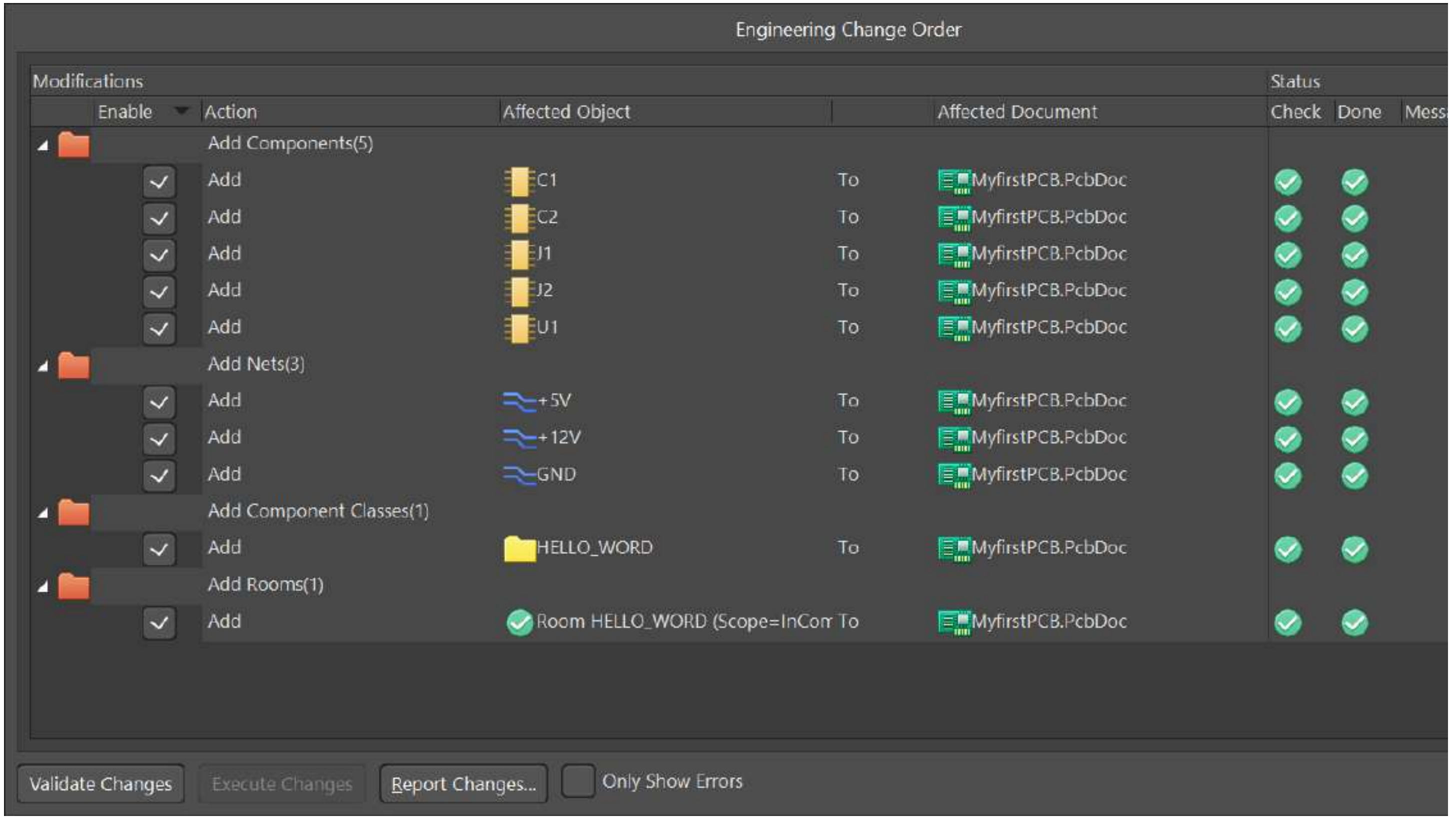
- Then we should import the Schematic symbols to our PCB Layout.





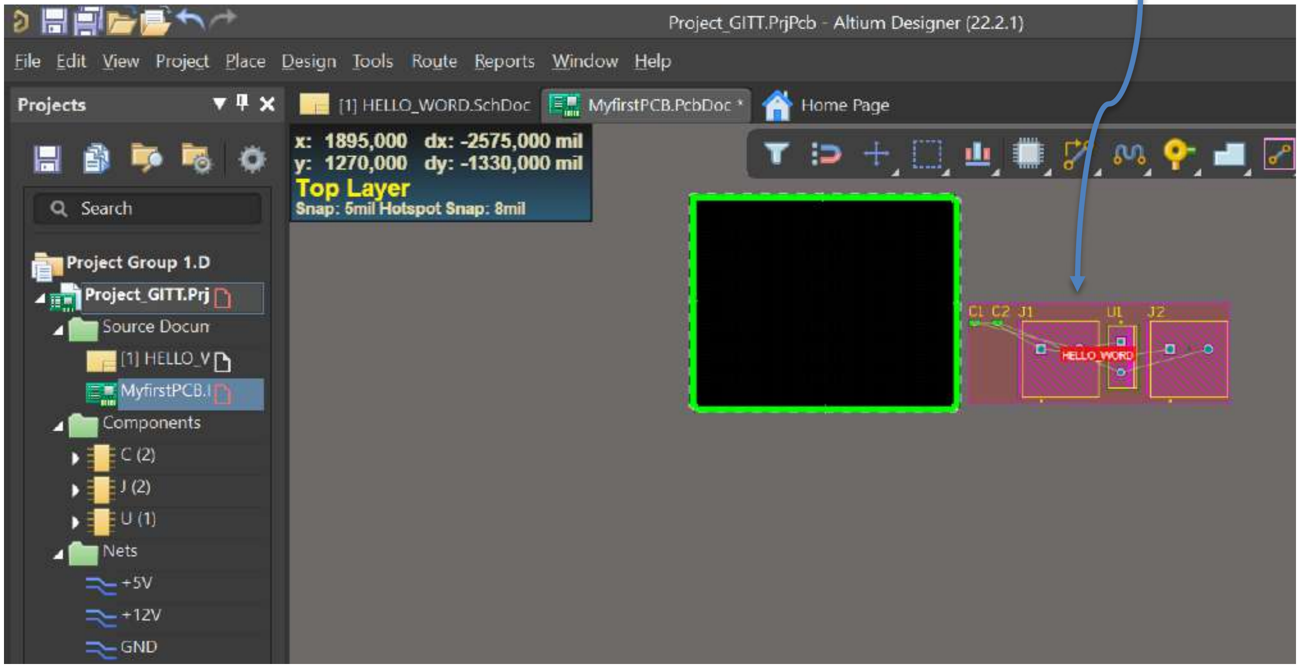
My first Layout

- Validate changes and our nets, components and romos are imported to layout.



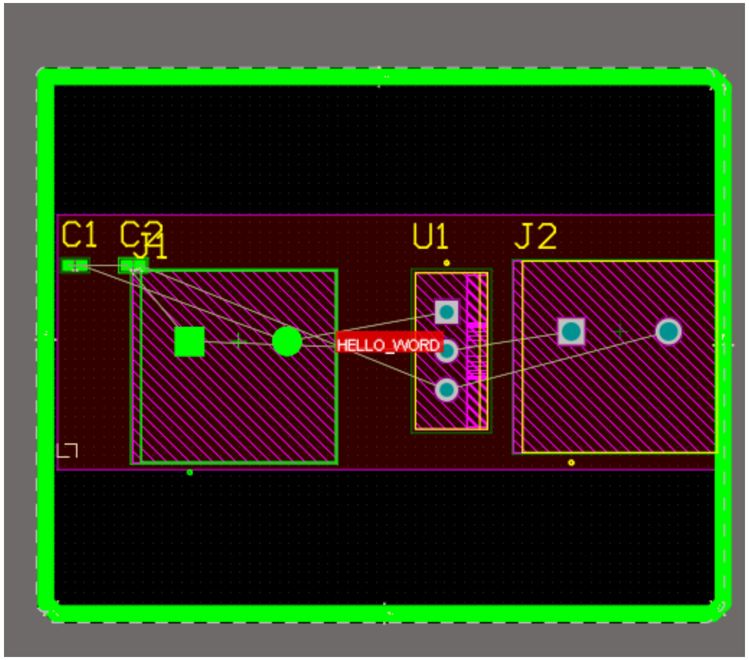
My first Layout

Imported elements from Schematic Tool



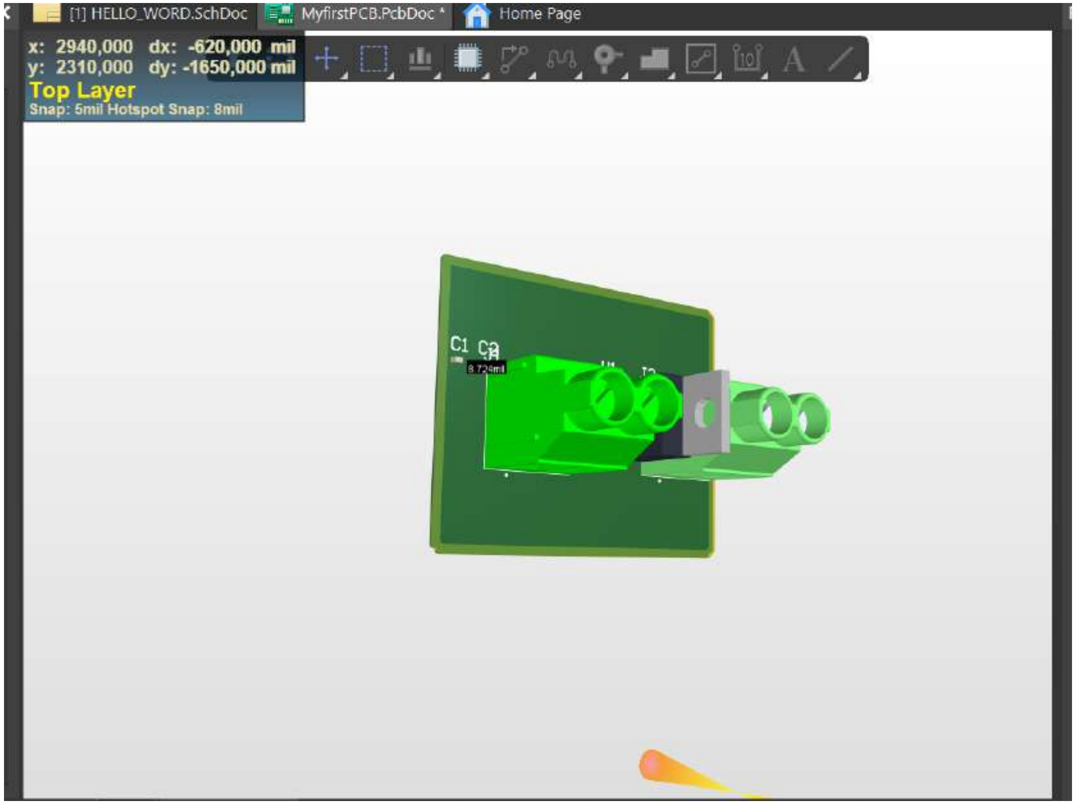
My first Layout

- We should move in the room in the board area.



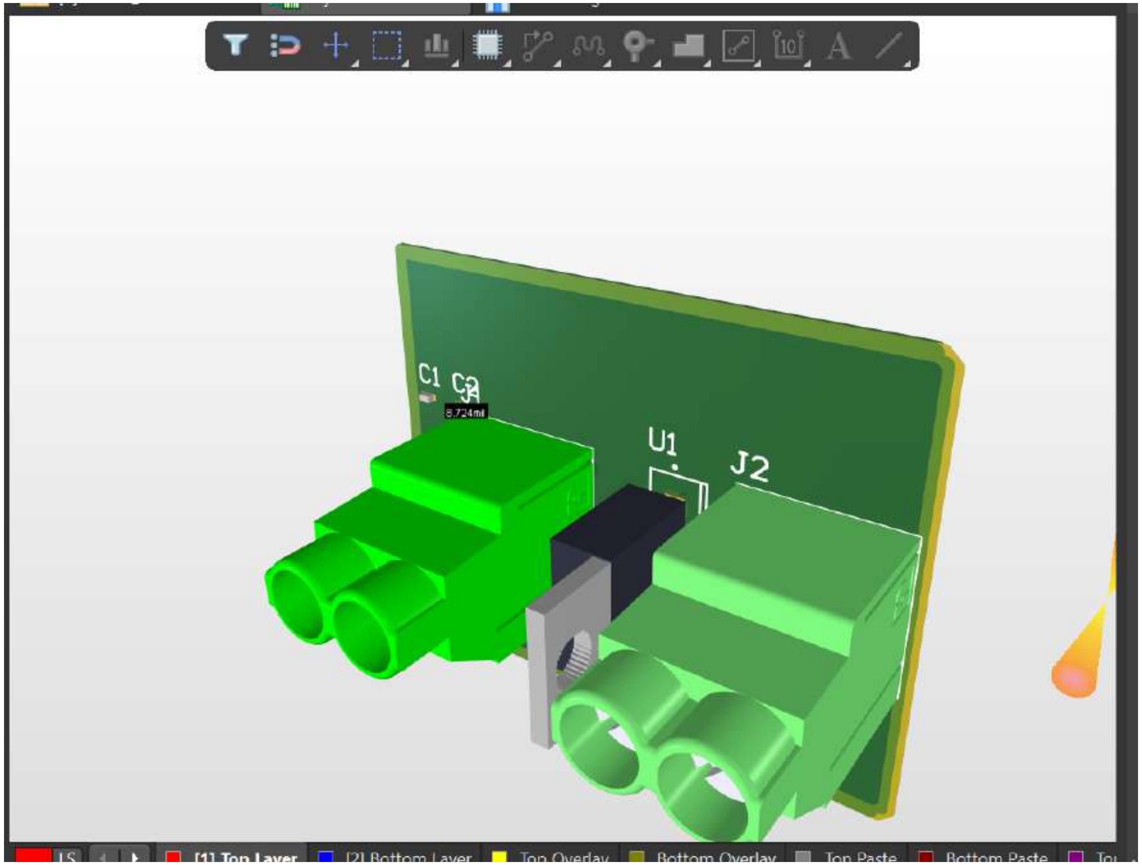
My first Layout

- We can switch between 2D and 3D view with 2 or 3 keys.

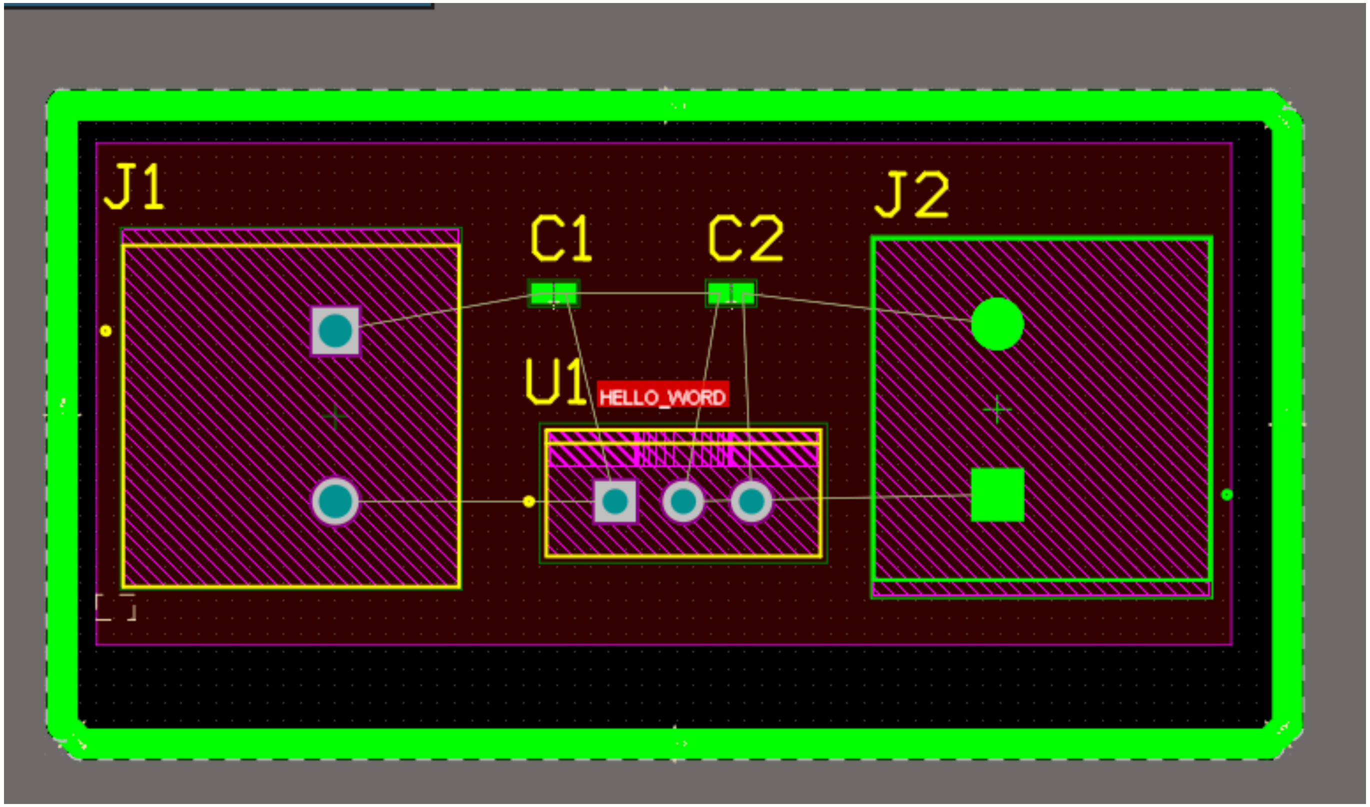


My first Layout

- To move the 3D view in several ways, press SHIFT + Right click

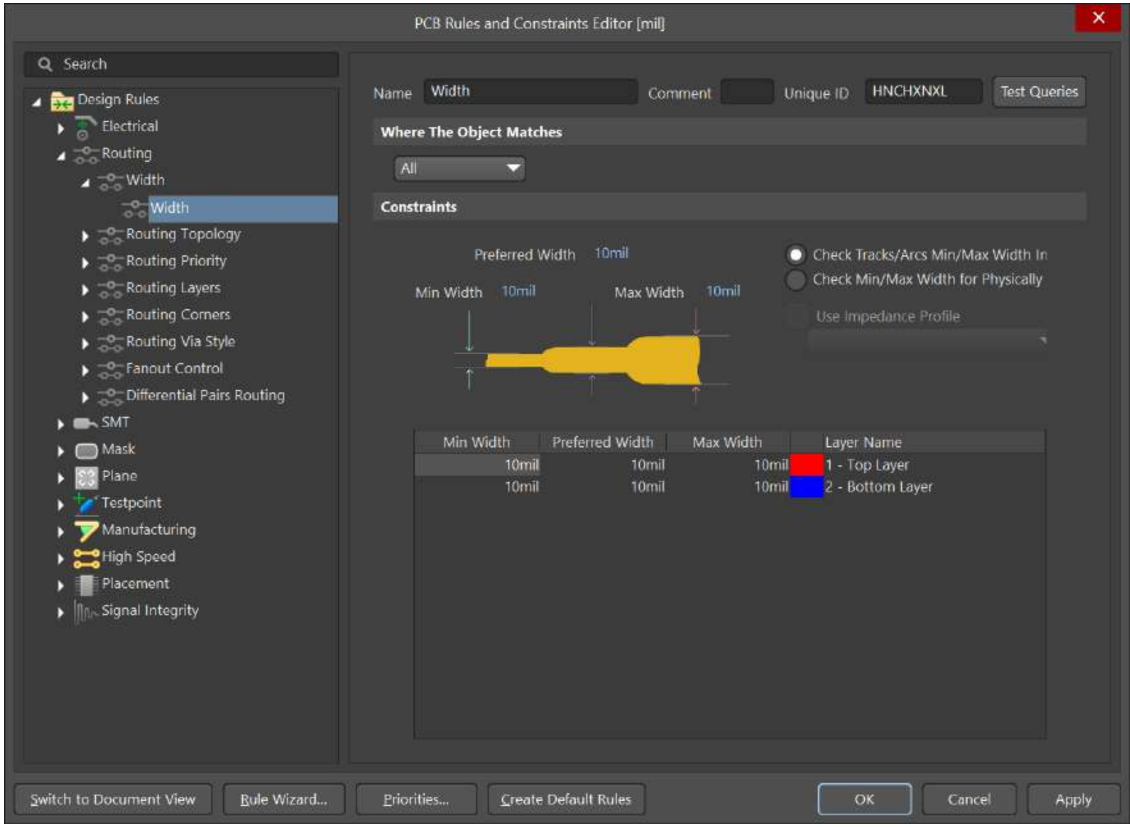


My first Layout



My first Layout

- Now it is time to start the routing process. First of all, we should define the track widths.



My first Layout

- What is the right width for the tracks? It depends on the maximum current through them
- Digital signals rarely exceed a maximum current of 500mA (examples: Arduino output pins maximum current of 500mA, Arduino output pins: maximum current 40mA, raspberry Pi: 16mA, DC motor of 9volt DC motor, max. current consumption 500mA)

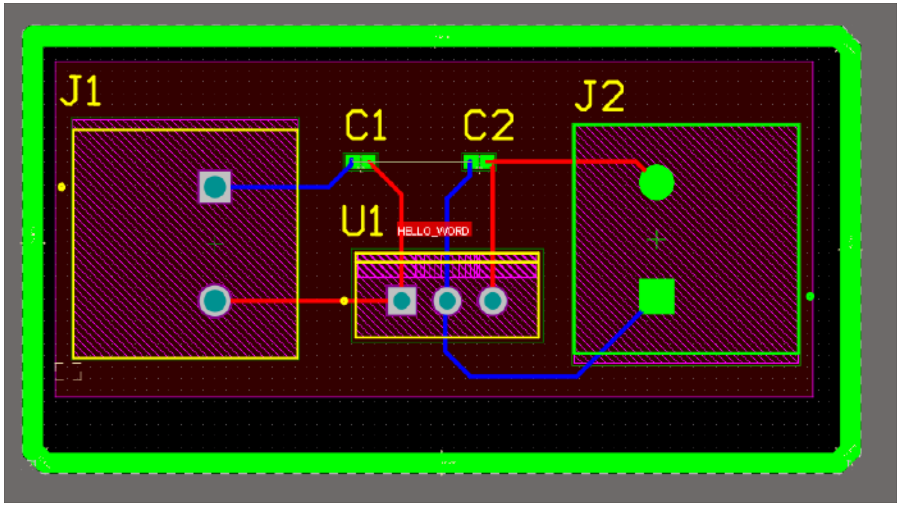
Current	Inches	Millimetres
10 A	0,15748	4 mm
5 A	0,0787402	2 mm
4 A	0,0590551	1.5 mm
3 A	0,0393701	1 mm
2 A	0,019685	0,5 mm
0.5 A	0,00787402	0,2 mm

Other recommendations:

- Distance between tracks: It depends on the voltage. For digital signals 0,3mm is enough.
- Routing with 45º avoiding 90º.
- Choose the right width track according to the type of via.
- Take care when you route tracks for SMD components. Are you in the right layer?
- Colour the priority tracks (GND, VCC and others). Further details: https://www.youtube.com/watch?v=AhJEJWTKUVI

My first Layout

- Then … we are ready to route (CTRL + W or Route → Interactive Routing).

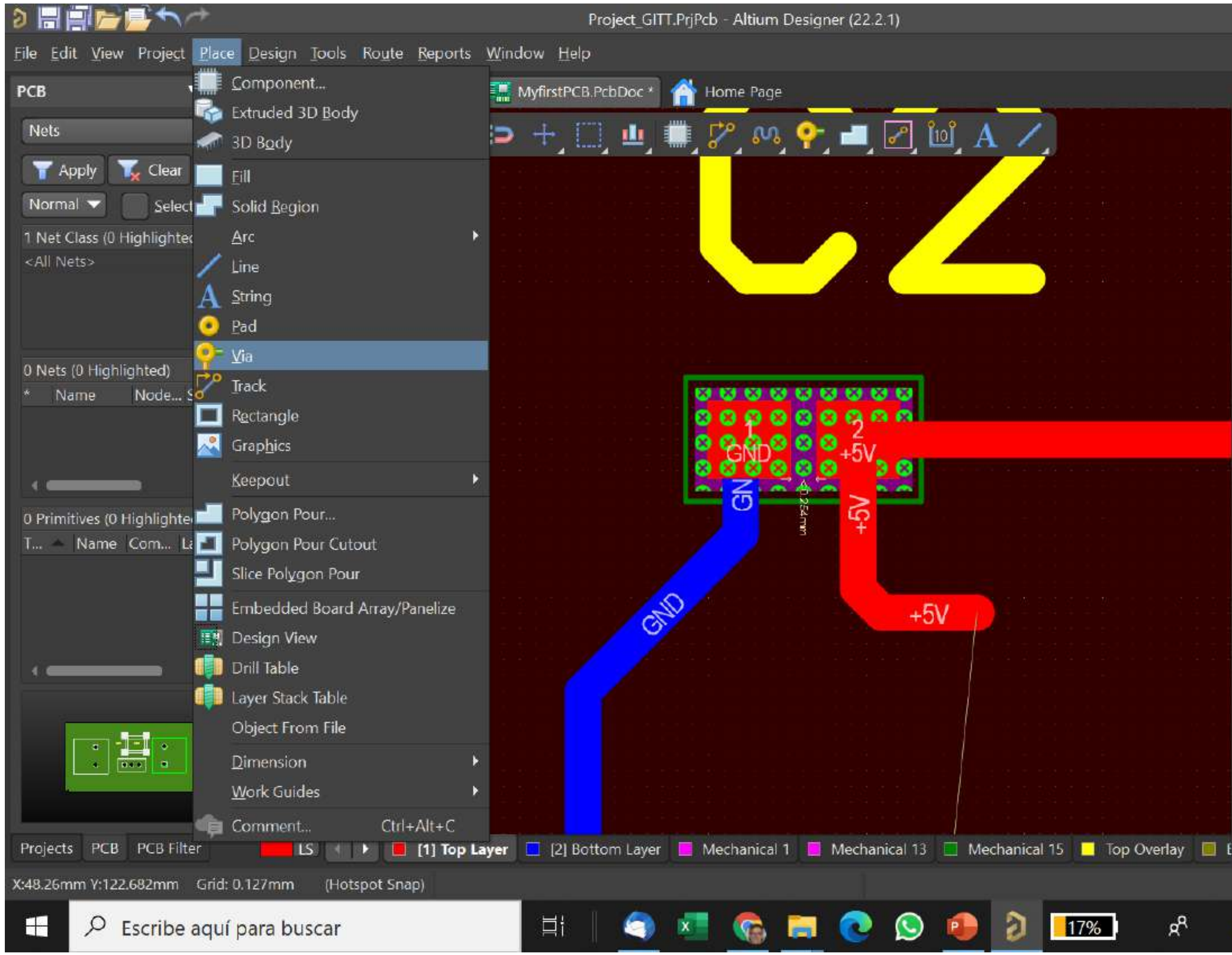


- We can change the layer to be routed.
- Track width can be dynamically changed pressing 2 key.

My first Layout

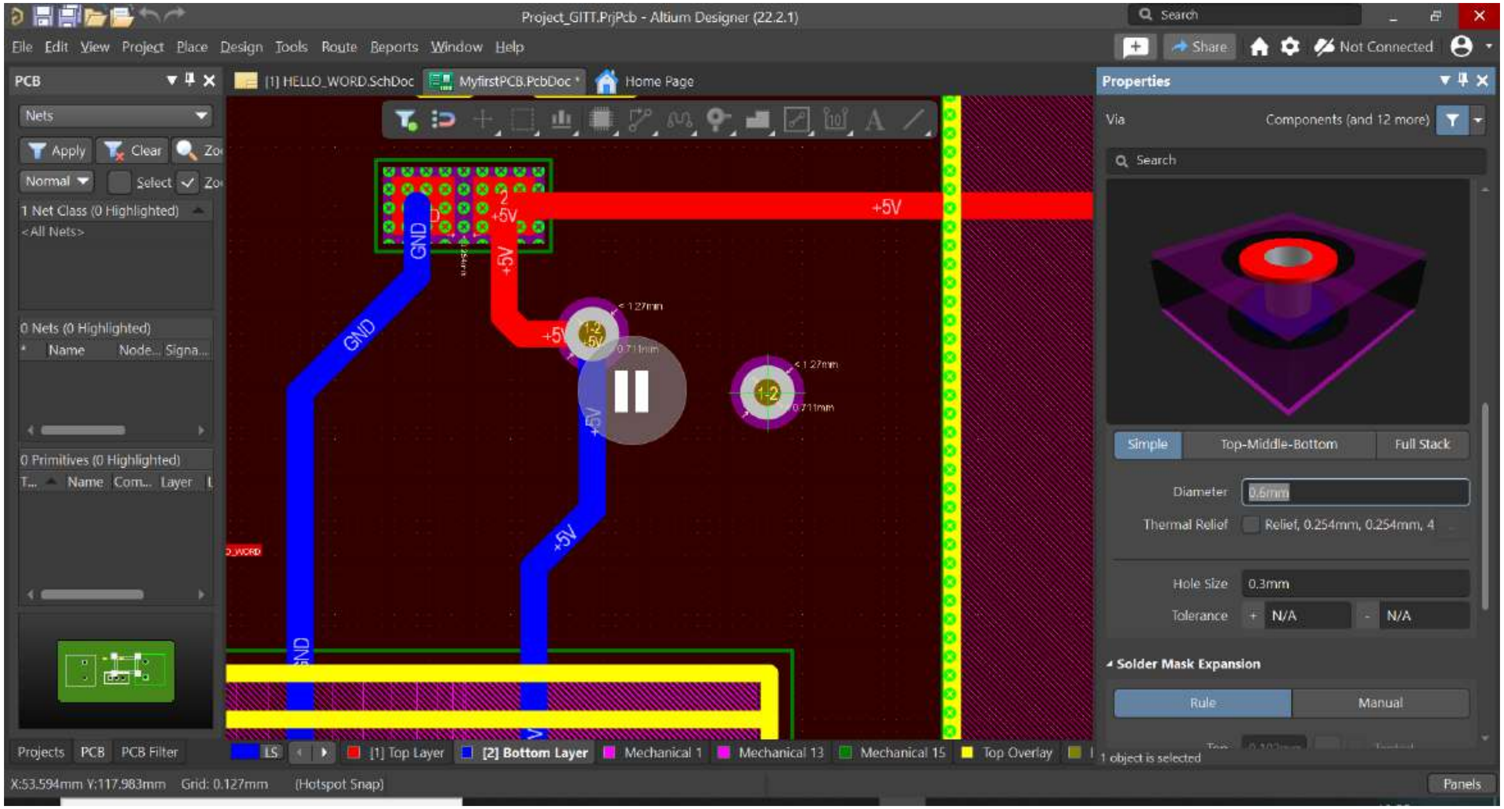
Placing vias

- Via is a hole to communicate a same track through several layers.



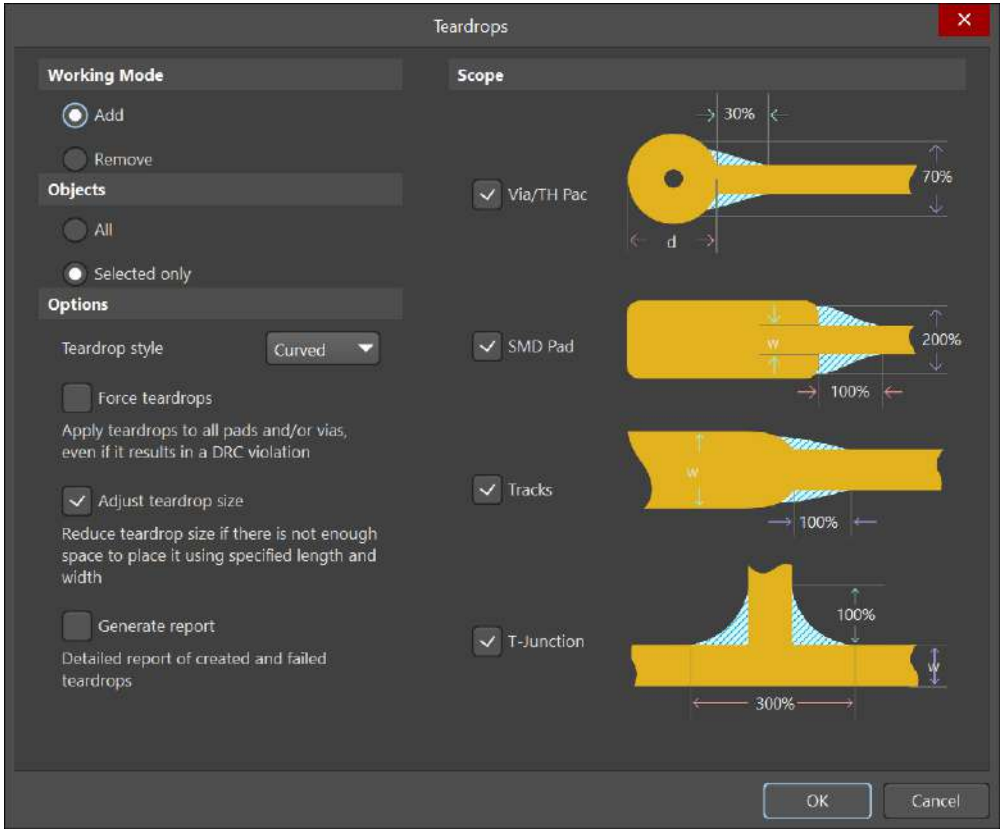
My first Layout

- The via width and size must be equivalent to the track width.



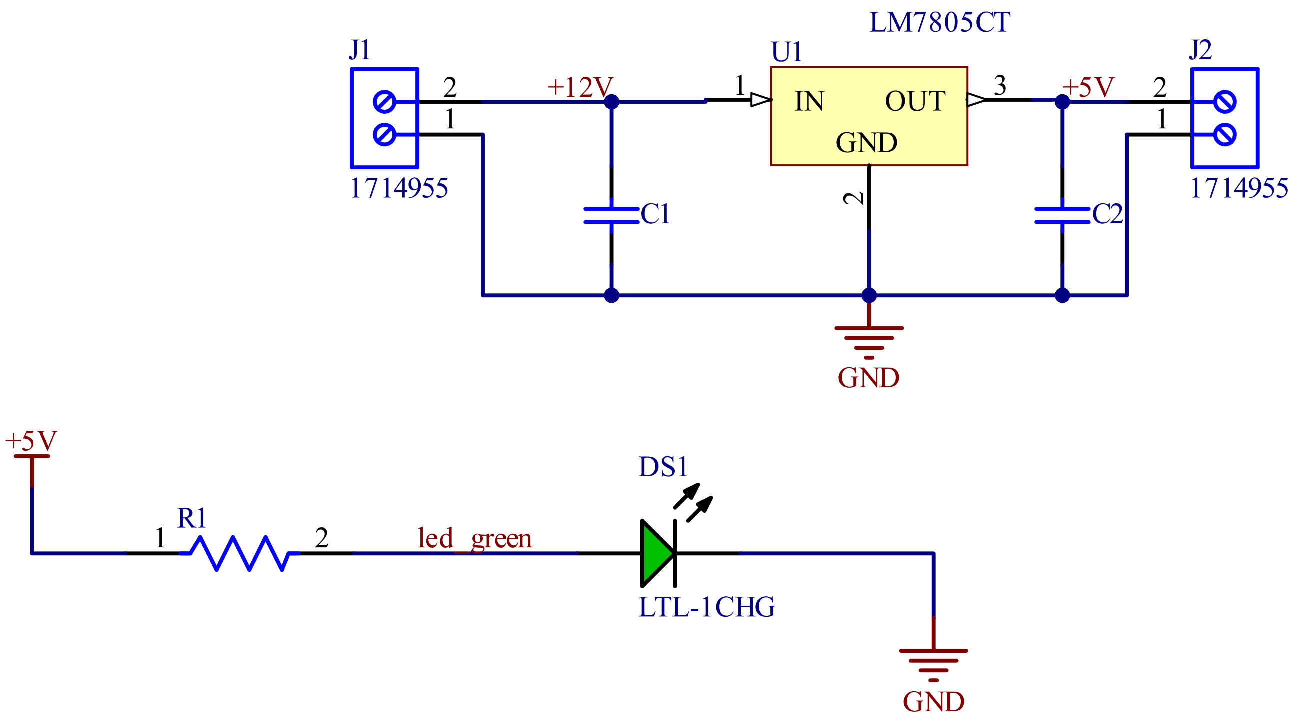
My first Layout

- When routing stage has finished we could do a “track smooth” using Tools → Teardrops. Default options can be validated to improve the track connexions.



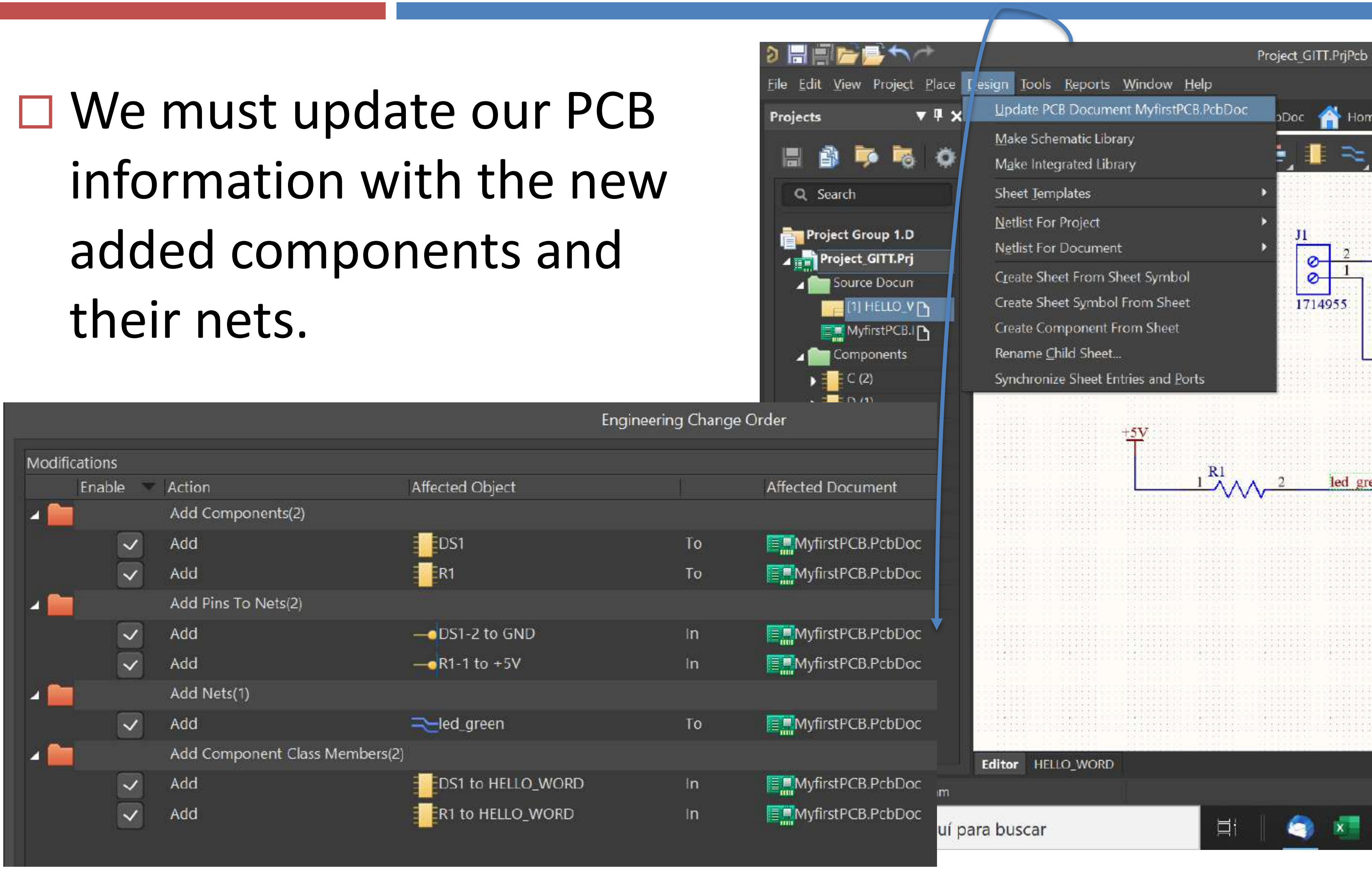
My first Layout

- Eco feature is still active between Schematic and Layout Tool.
- Now, we are going to add a LED and a resistor.



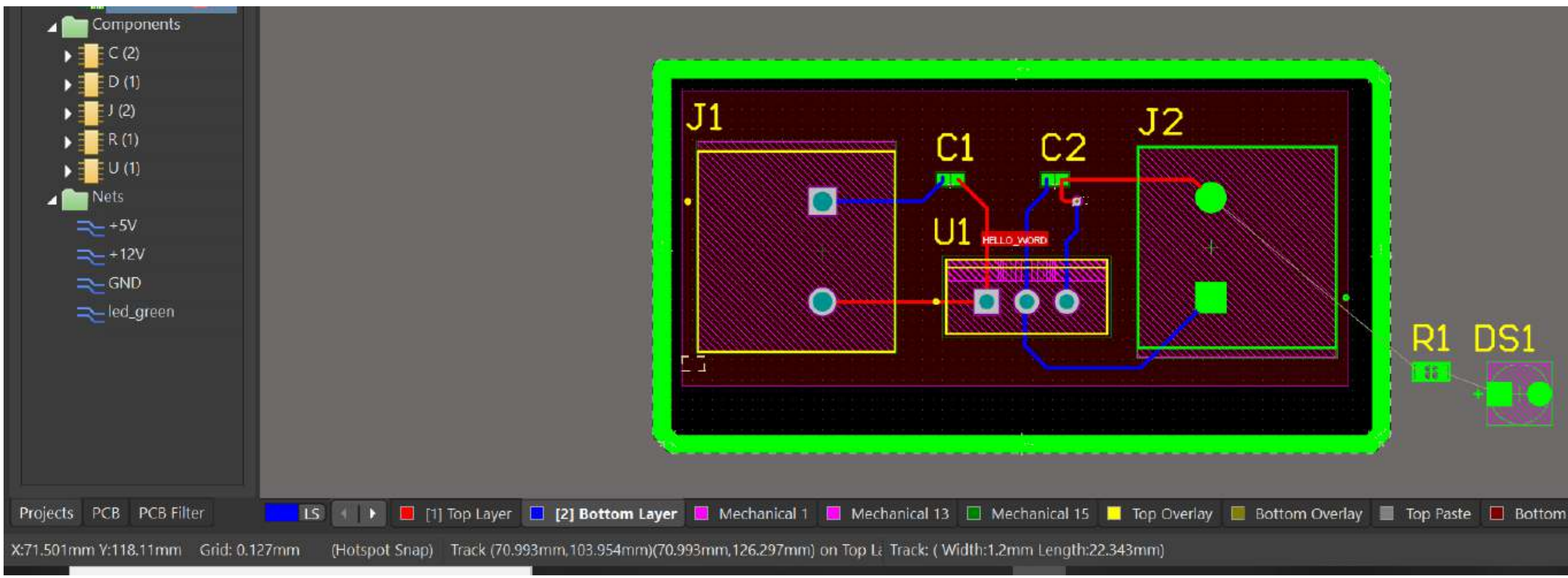
My first Layout

- We must update our PCB information with the new added components and their nets.



My first Layout

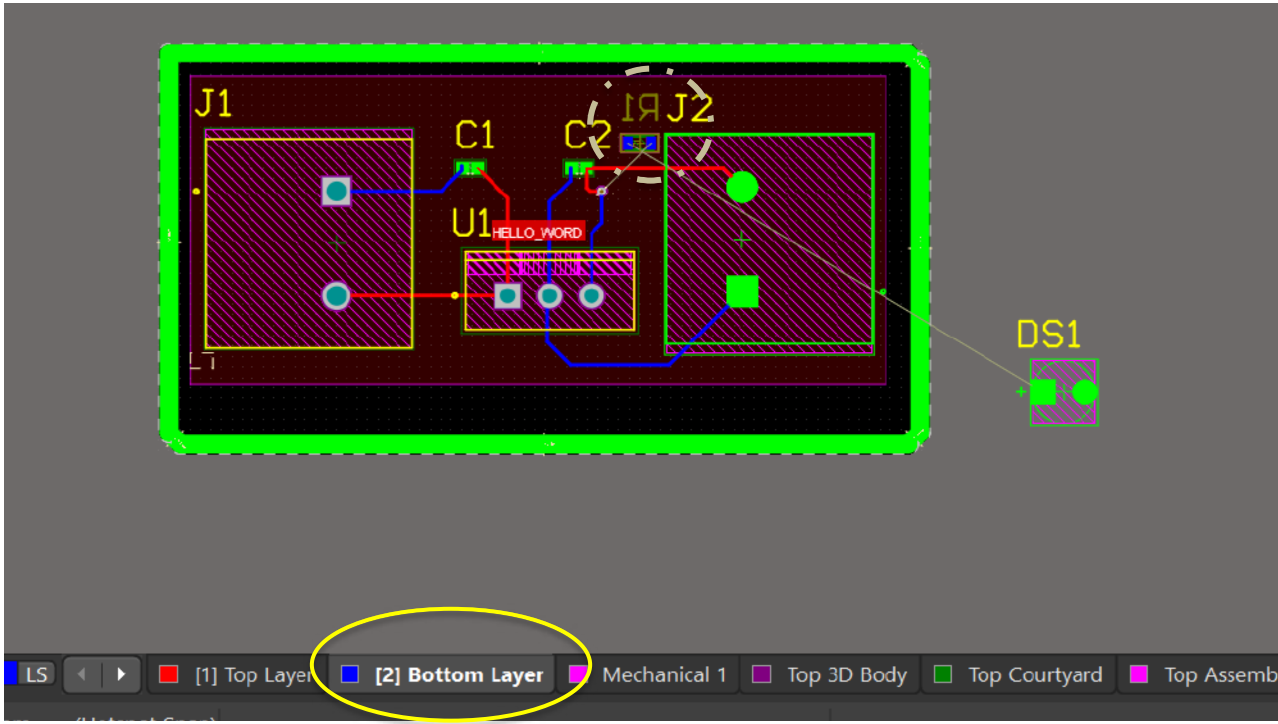
- New components have been imported to our PCB Layout.



My first Layout

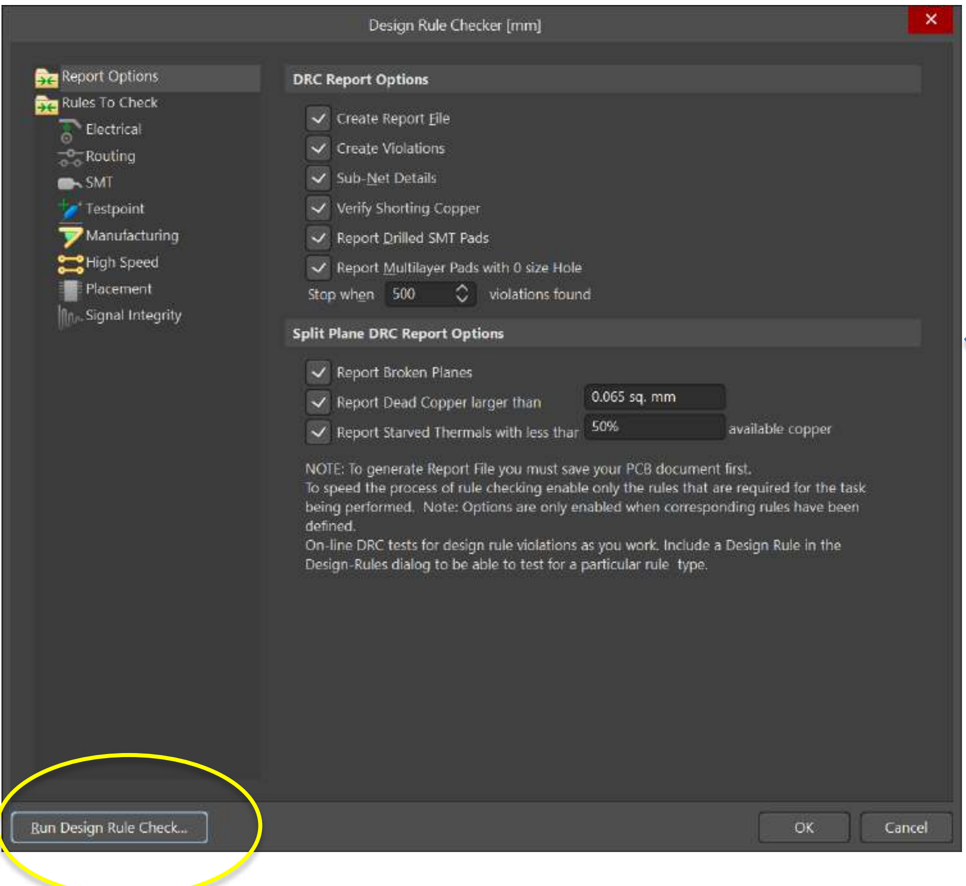
- To place a component in the Bottom Layer:

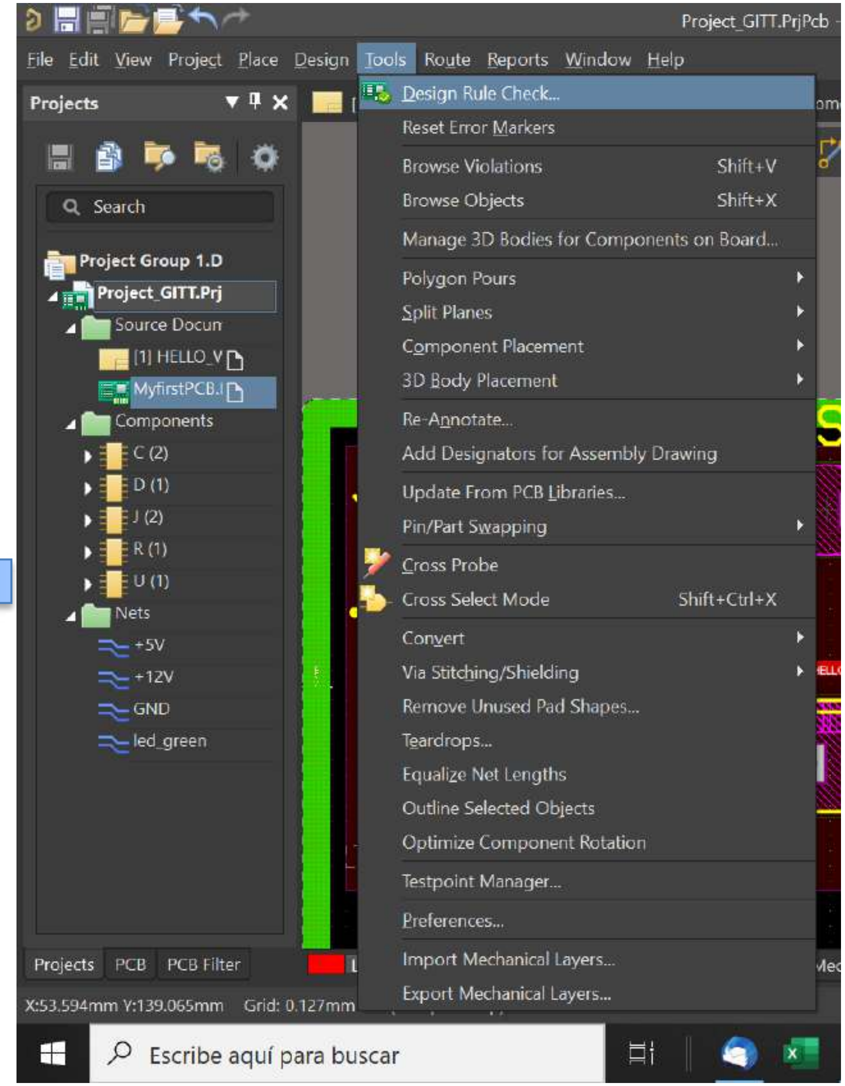
1. Choose Bottom Layer
2. Select the component
3. Press ‘L’ Key.



My first Layout

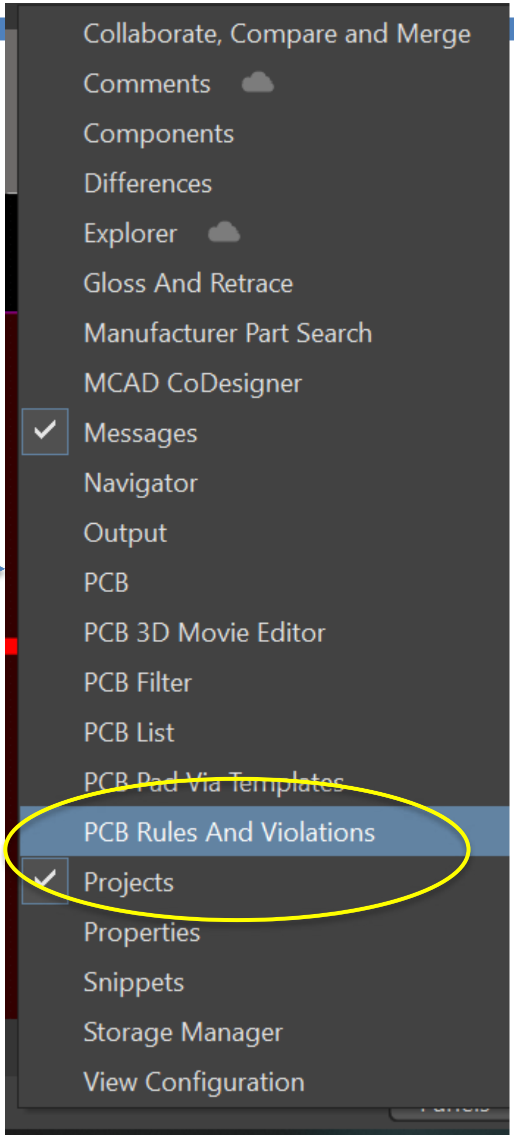
- Checking the PCB → Detecting violations



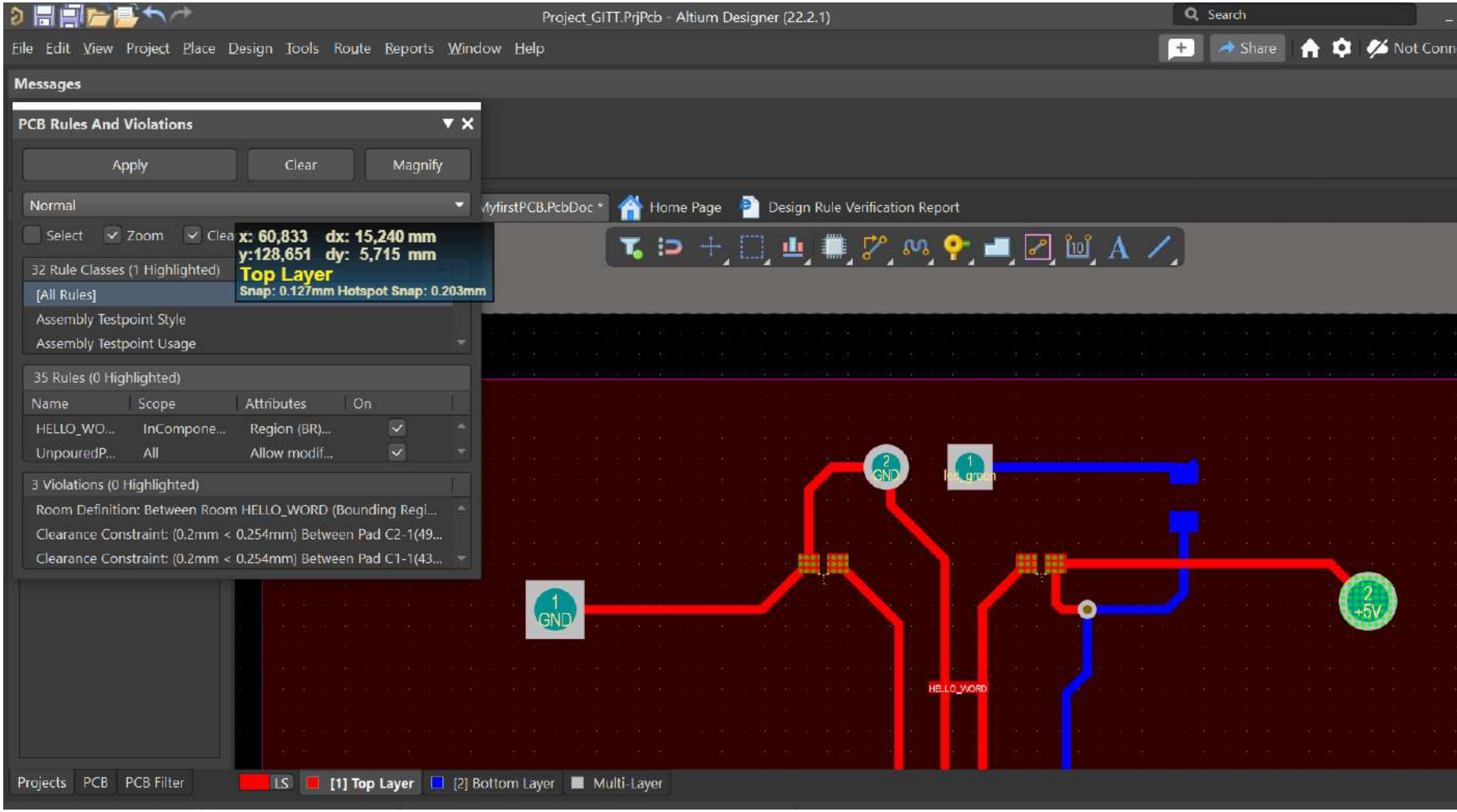


My first Layout

- To display *the violations* in a suitable mode:

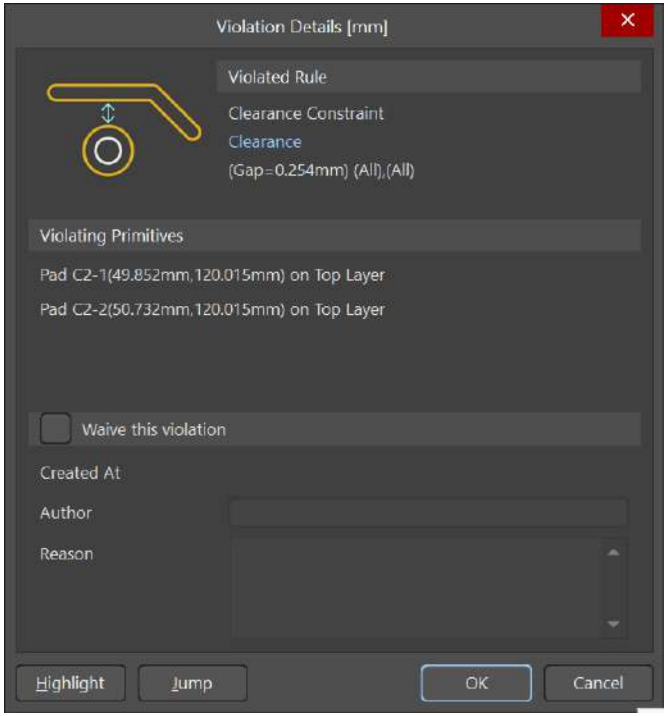


My first Layout



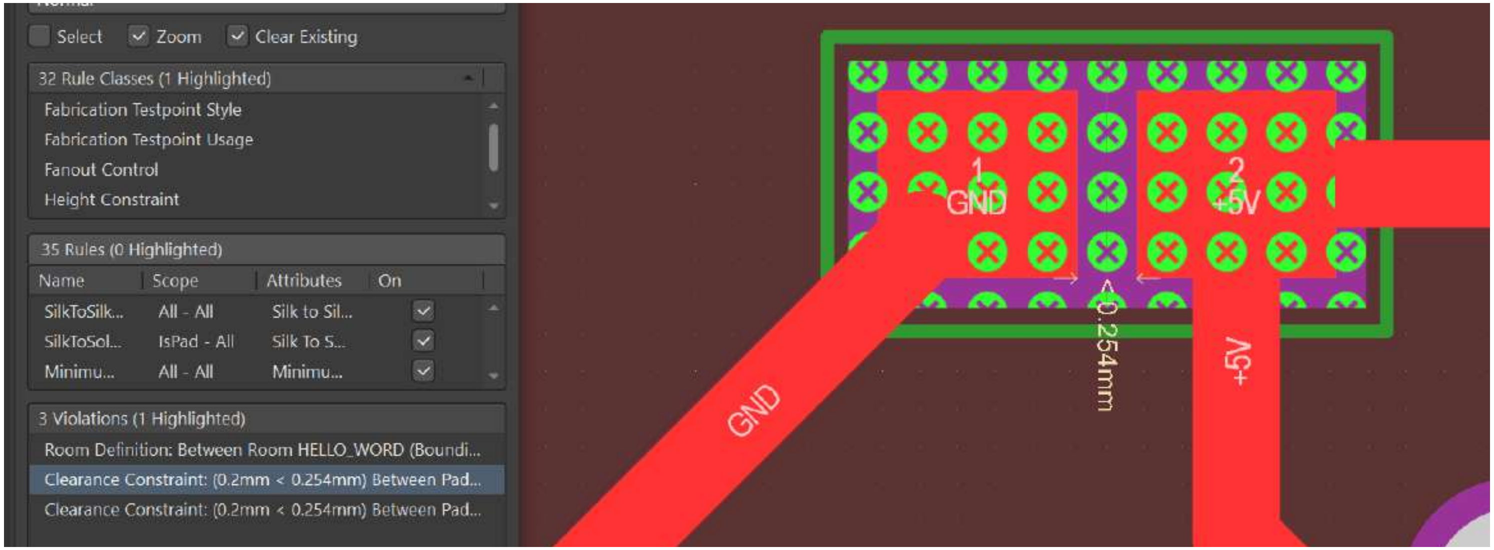
My first Layout

- Remember that there is a set of electrical rules defined by default.
- Some of them could not adjust to standard footprints



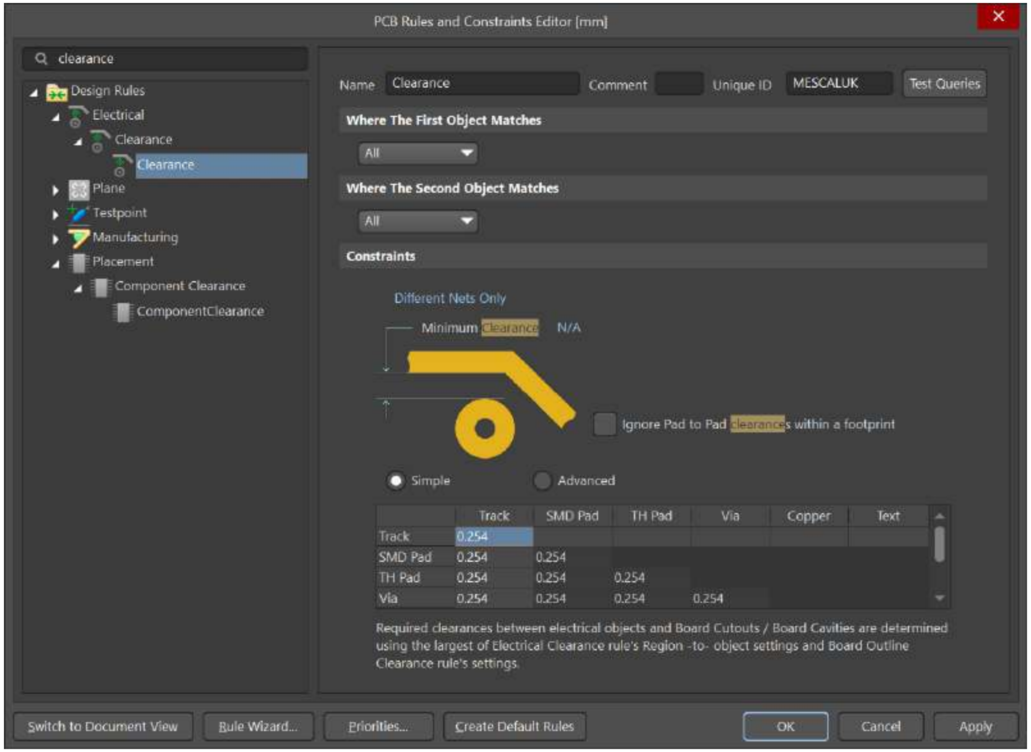
My first Layout

- This example displays how minimum distance is not fulfilled. However, the footprint is taken from manufacturer.



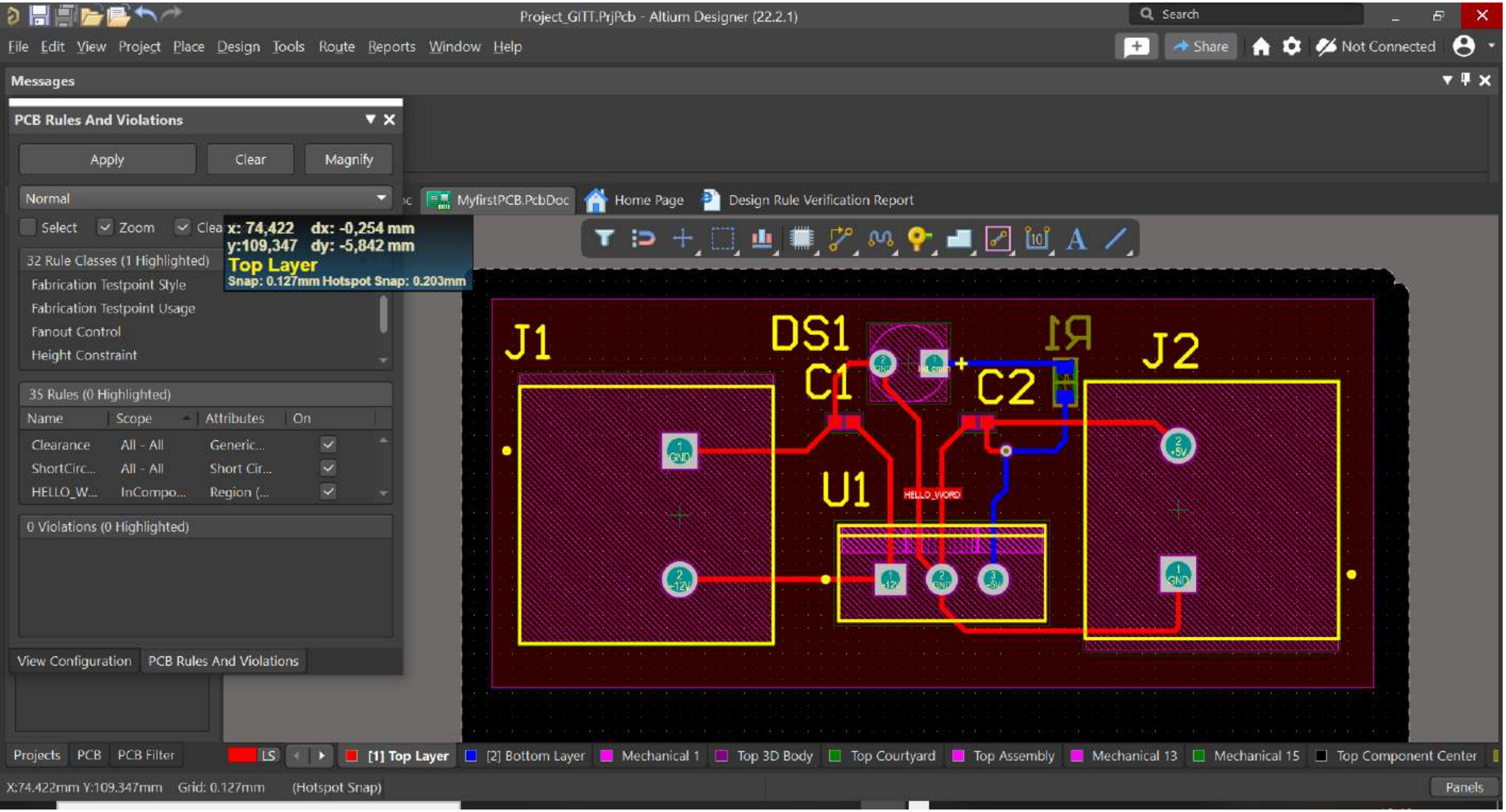
My first Layout

- To avoid any footprint modification, we are going to modify the electrical rule.
- Design → Rules → Search Clearance.
- Now, change it!!



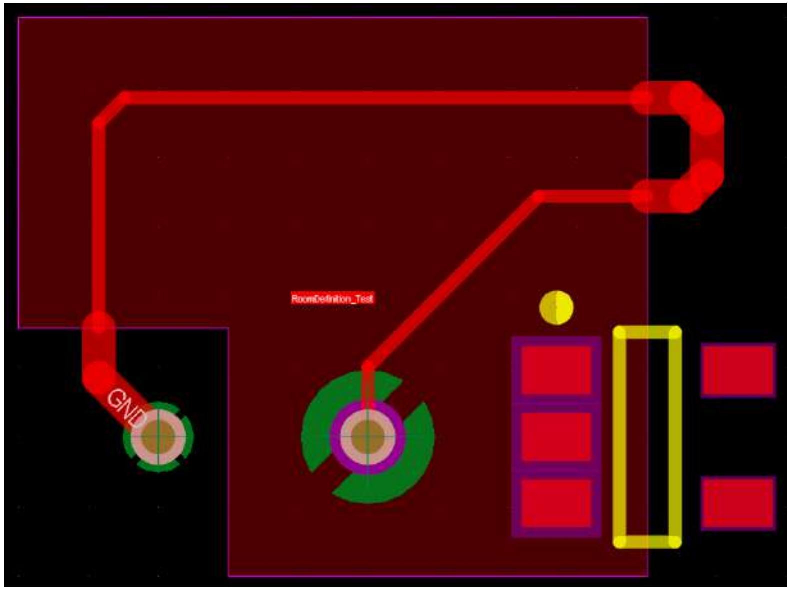
My first Layout

- Final View



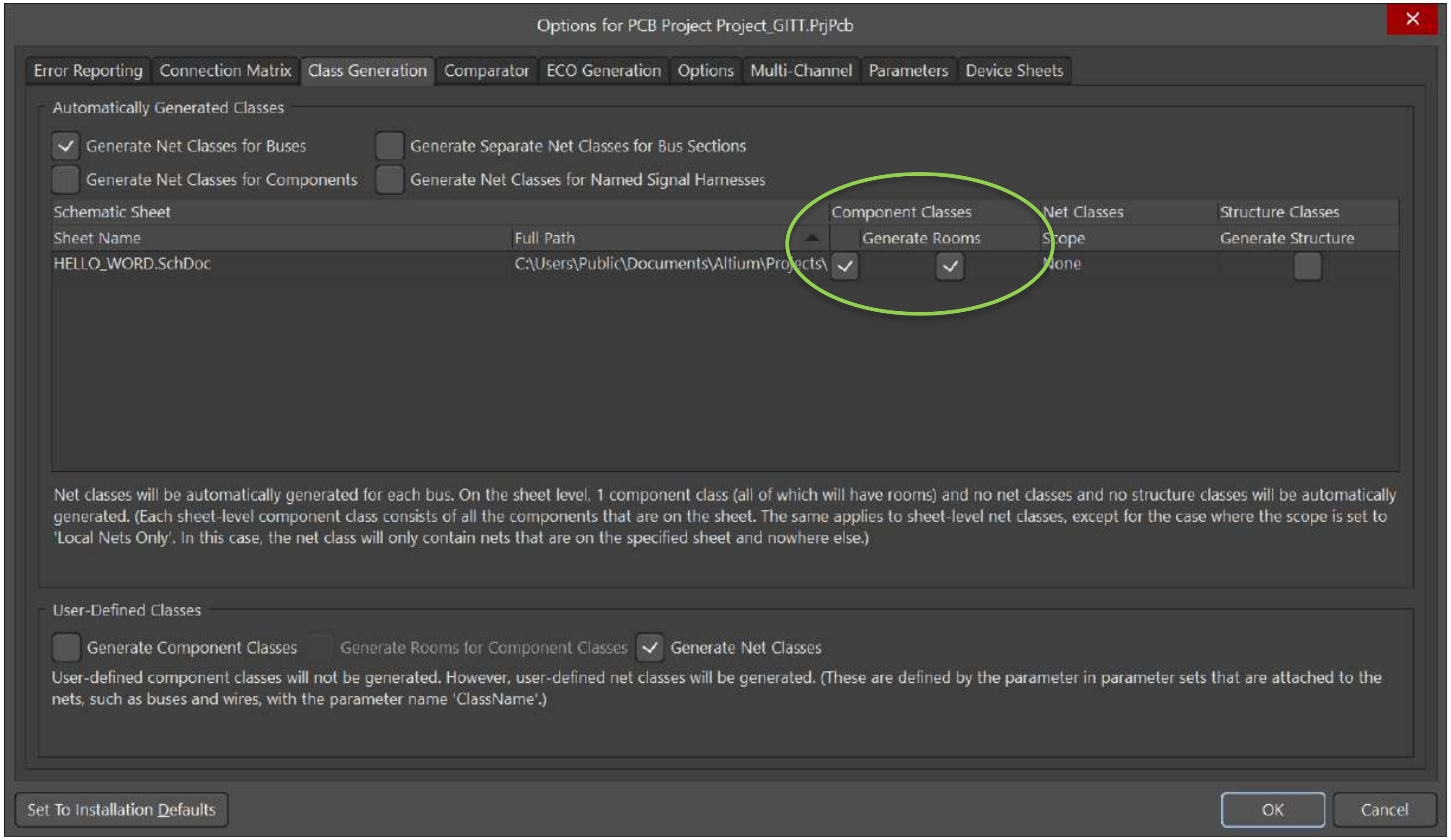
Room

- Room Concept: It can be defined as a virtual area for PCB Layout where the components belonging to that region inherit a common set of specifications (rule designs, track width, same region, etc.)
- By default a room area is set by the system per sheet. It means that a hierarchical design contains as rooms as symbol sheets are defined.

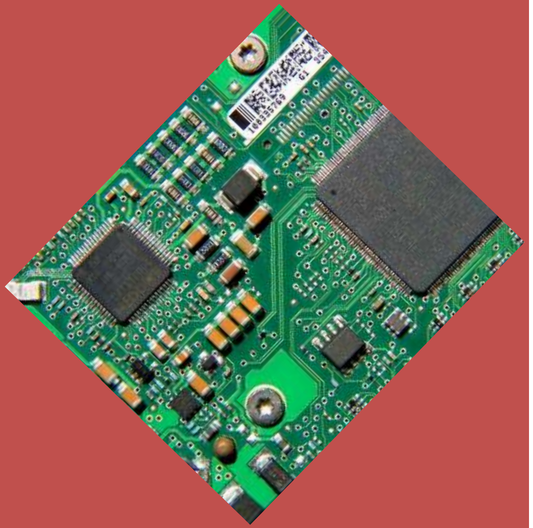


Room

- Unchecking room option: Project → Project Options



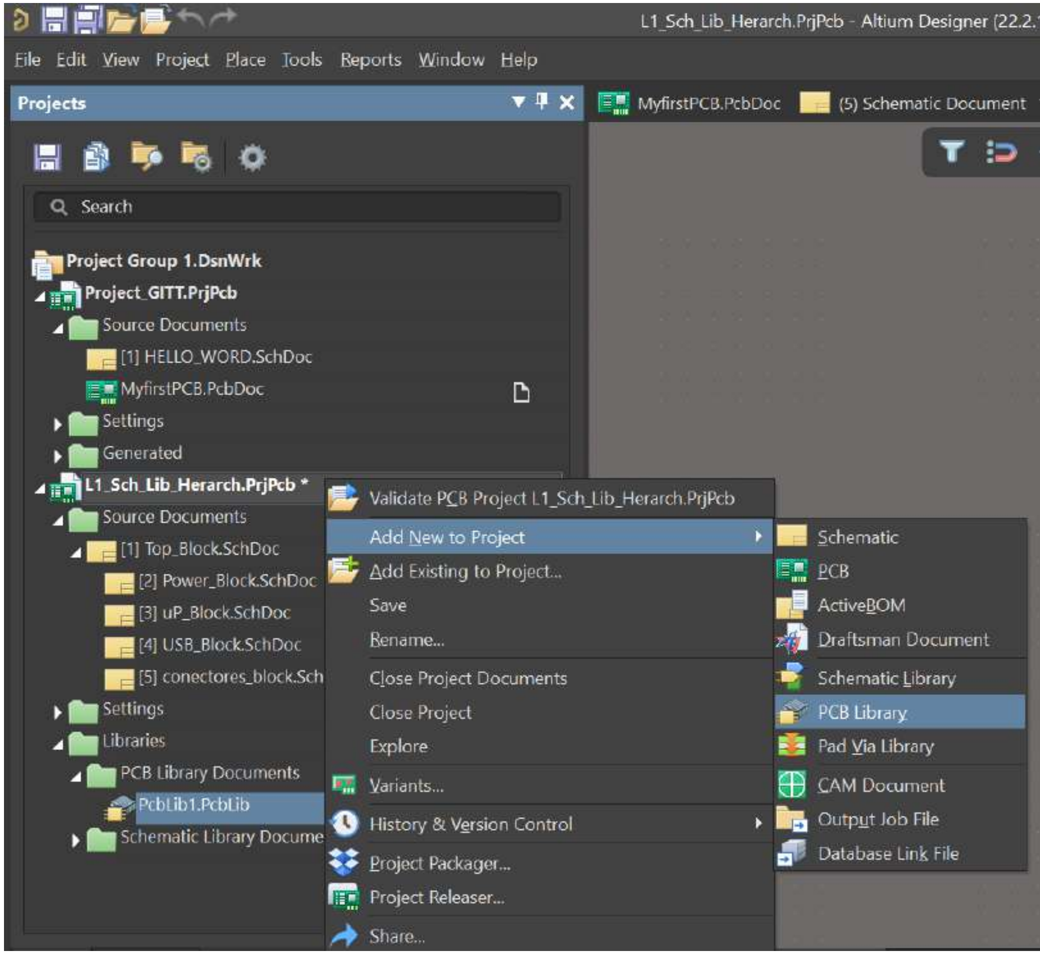
Introduction to Printed Circuit Board (PCB) design with ALTIUM



5.- PCB Library

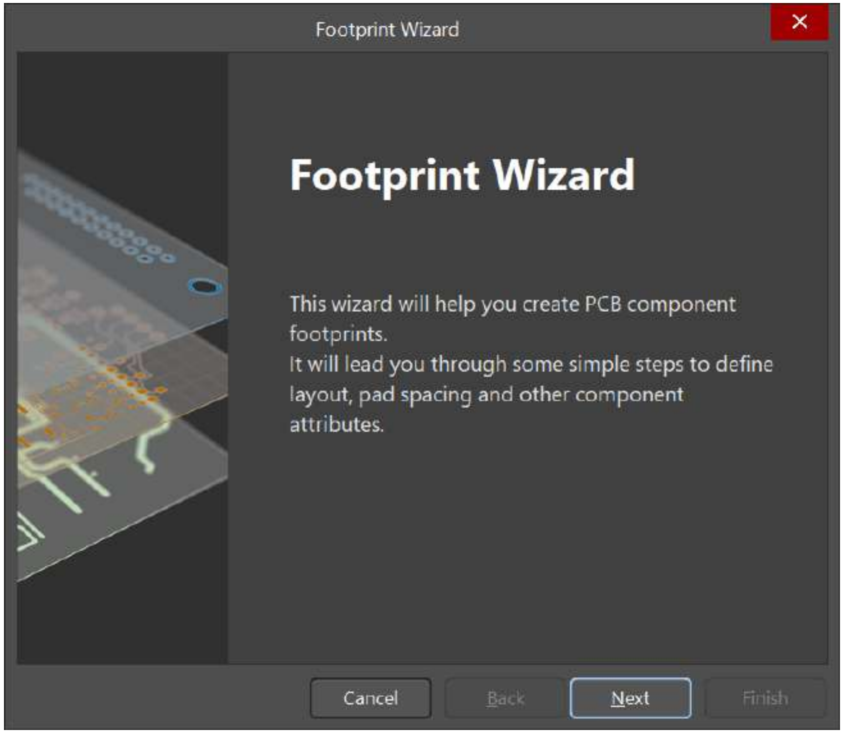
Creating a new PCB_Lib

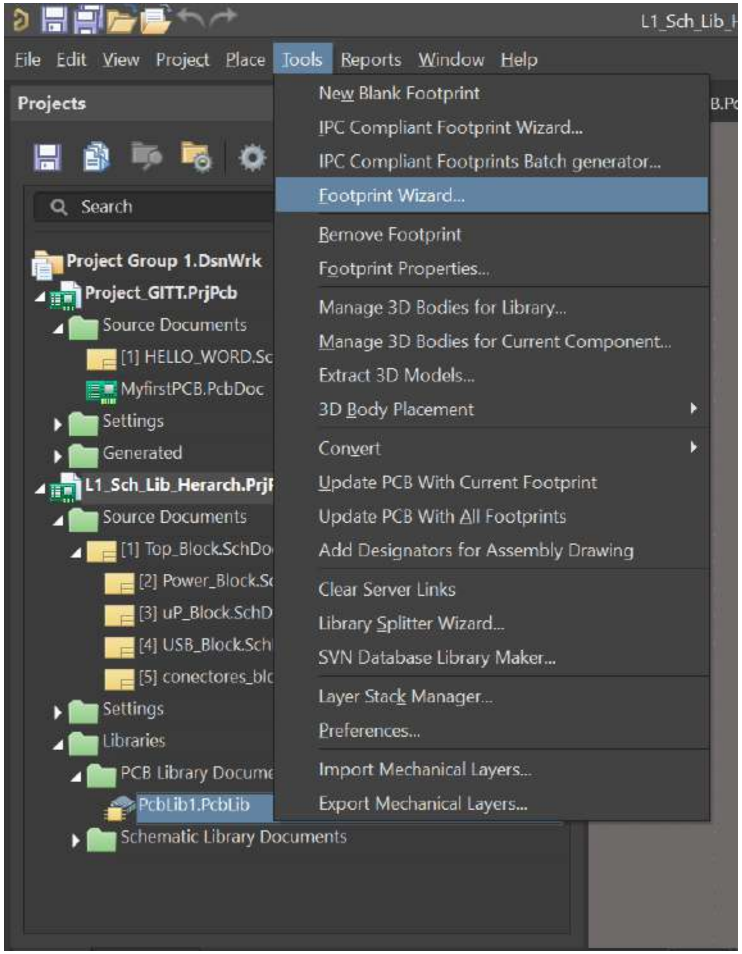
- We are going to produce a Library for PCB tool in a similar way than Schematic Library.
- It is useful when we are not able to find the expected device within Manufacturer Part Search or Component Tab.
- We will create a new Library in the same way that New Sch, PCB o SCH_LIB.



Creating a new PCB_Lib

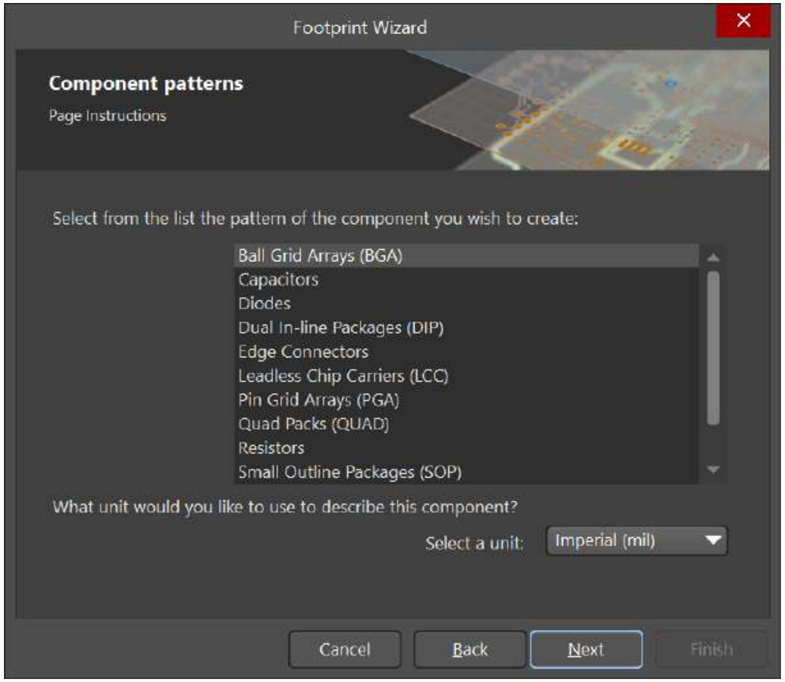
- We will use a Wizard to make our footprint.

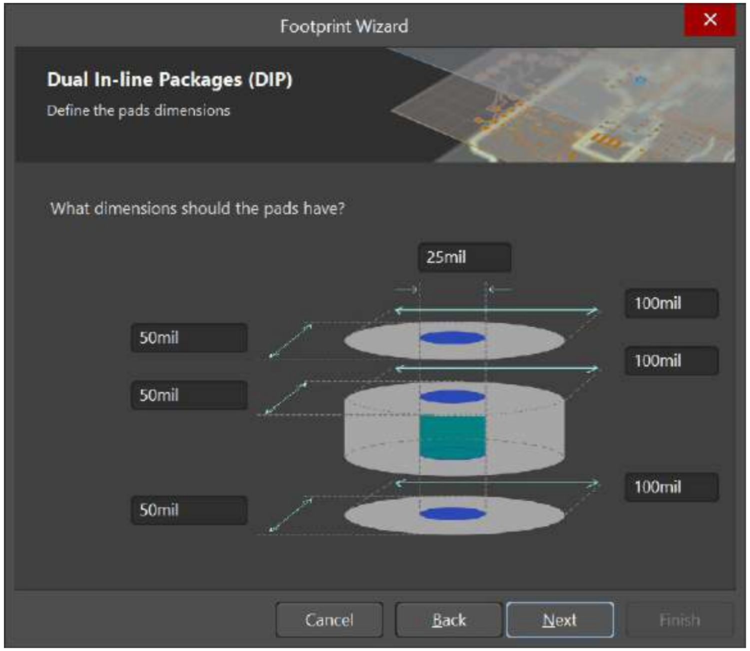




Creating a new Footprint

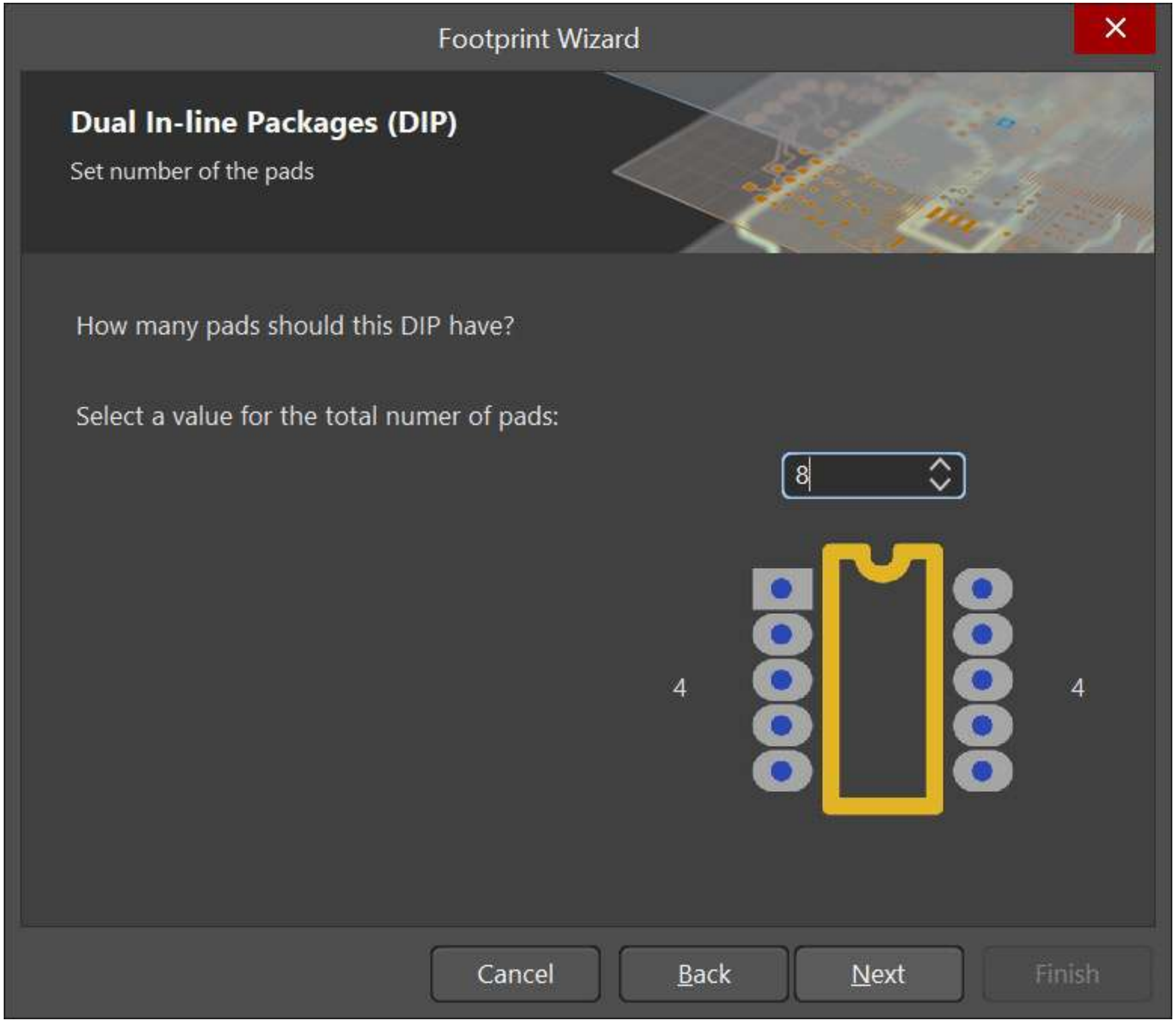
- A menu with a set of typical footprints are showed to choose one.





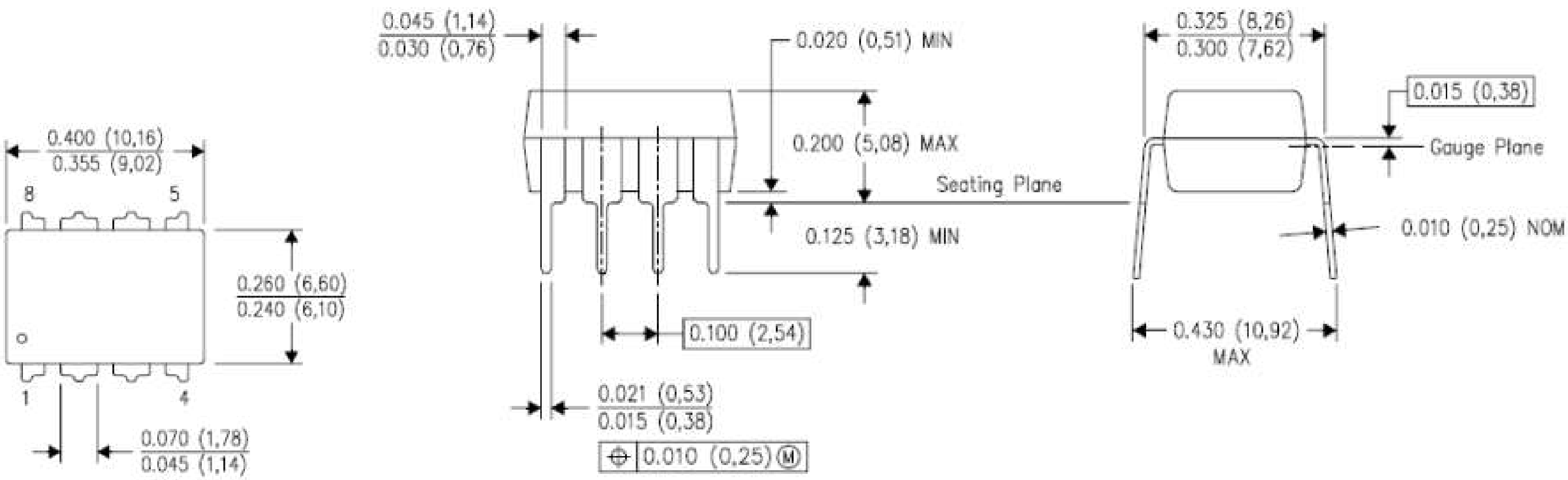
- We select a DIP (e.g.) and a new window with information about the pin features pops-up.

Creating a new Footprint



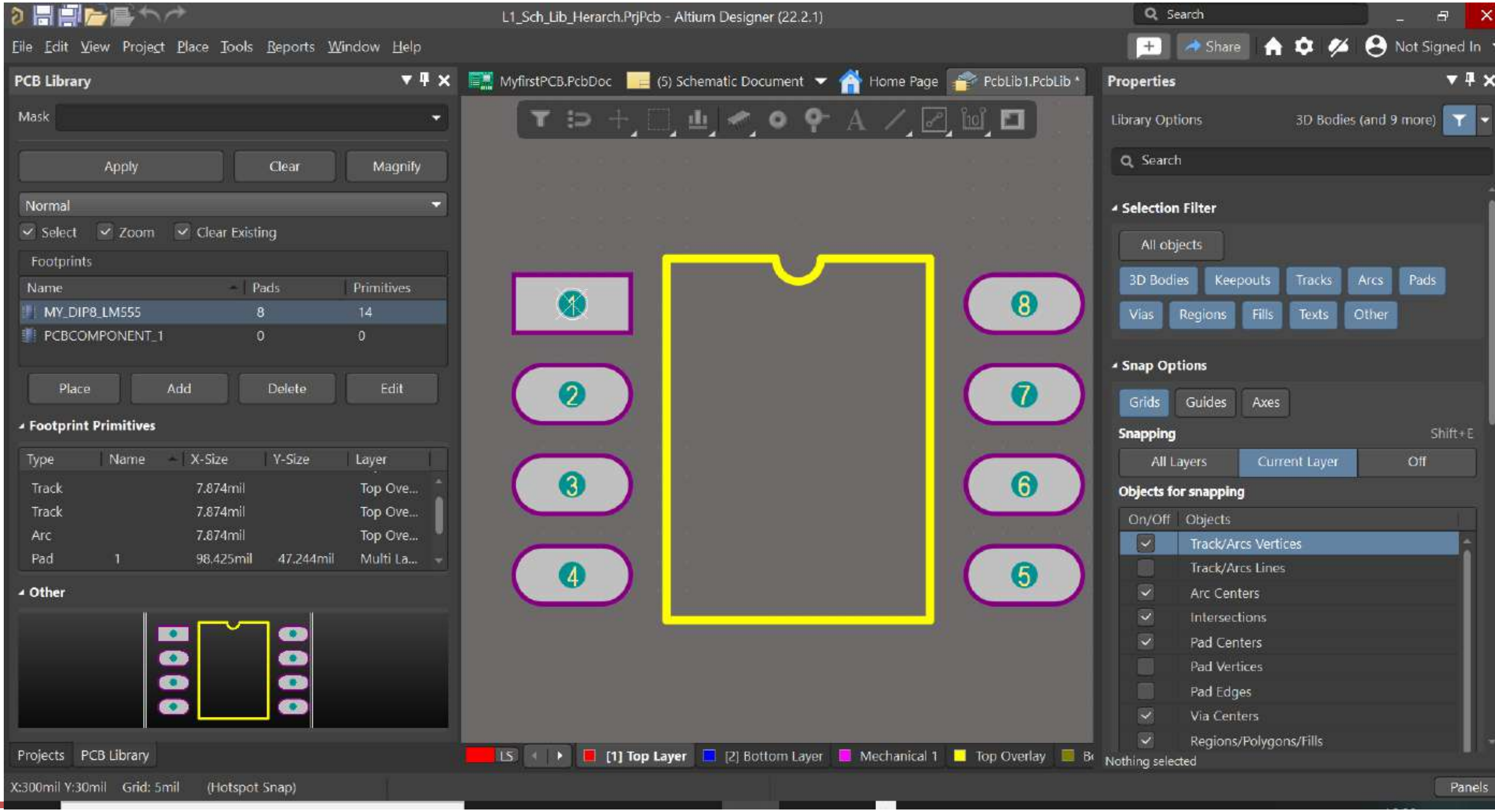
Creating a new Footprint

- The physical dimension of the device must be checked through the datasheet.



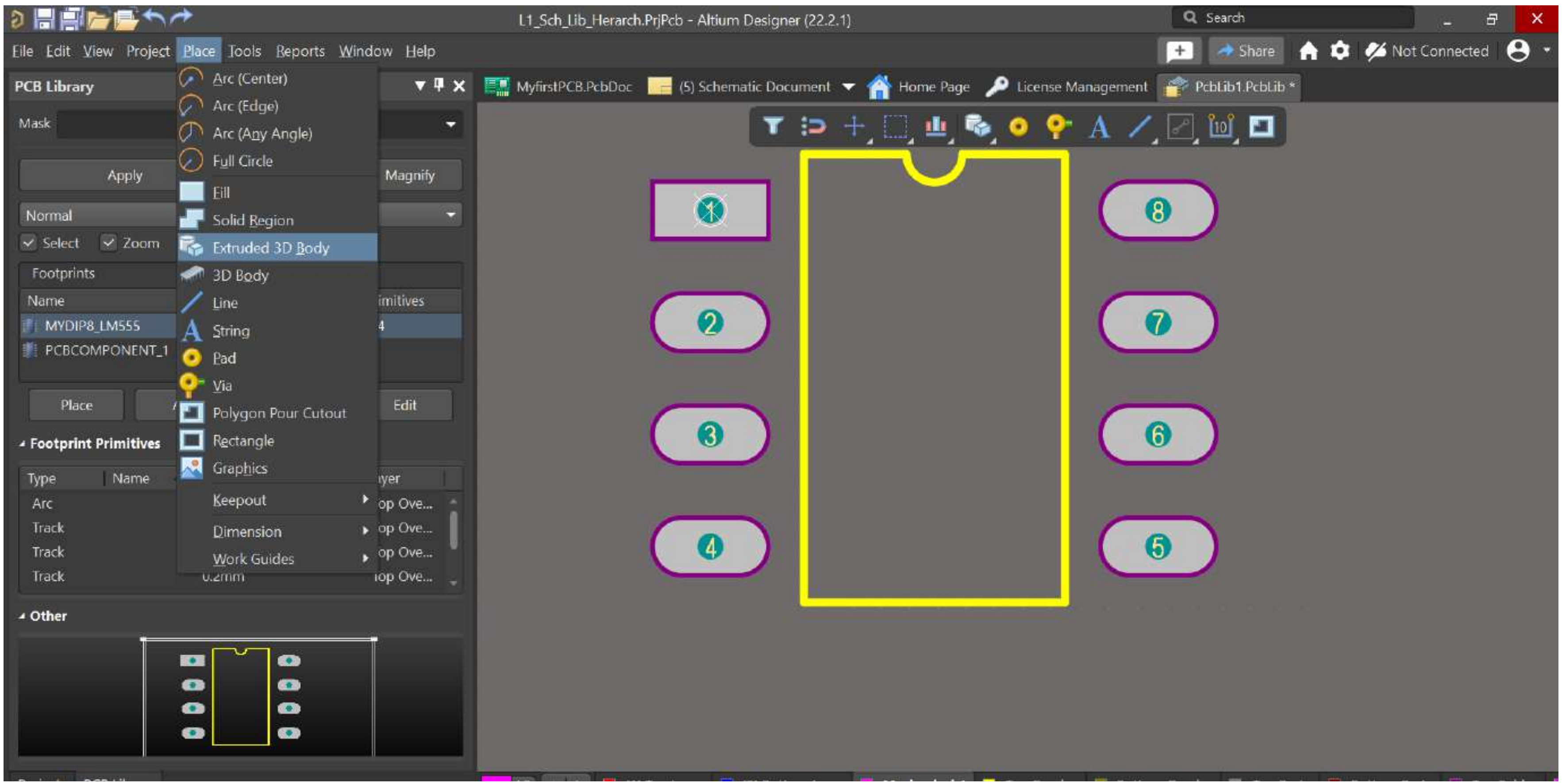
Creating a new Footprint

- A menu with a set of typical footprints are showed to choose one.

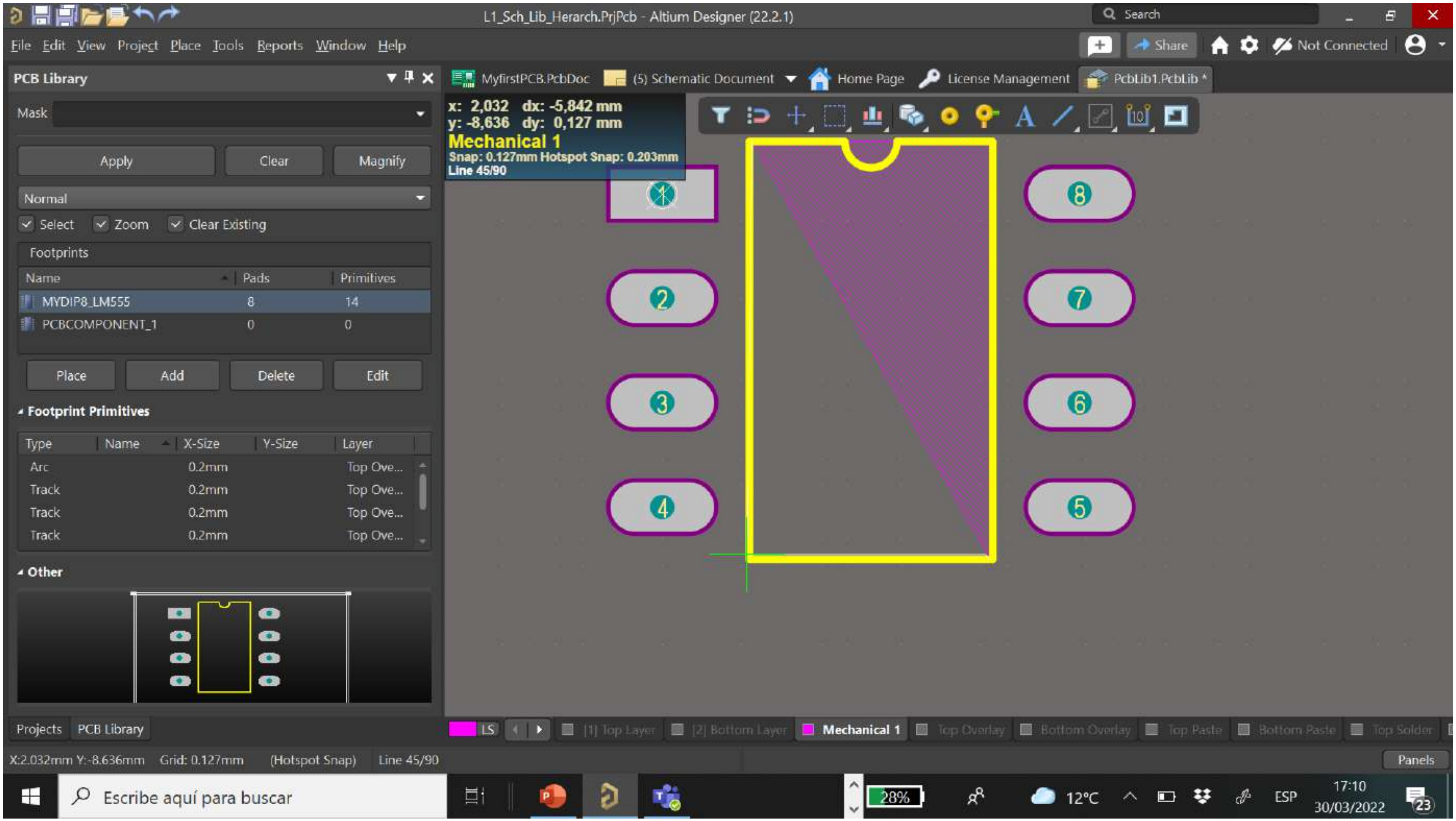


Creating a new Footprint

- Additionally, we could add a 3D model to our footprint. It is useful for the final view.

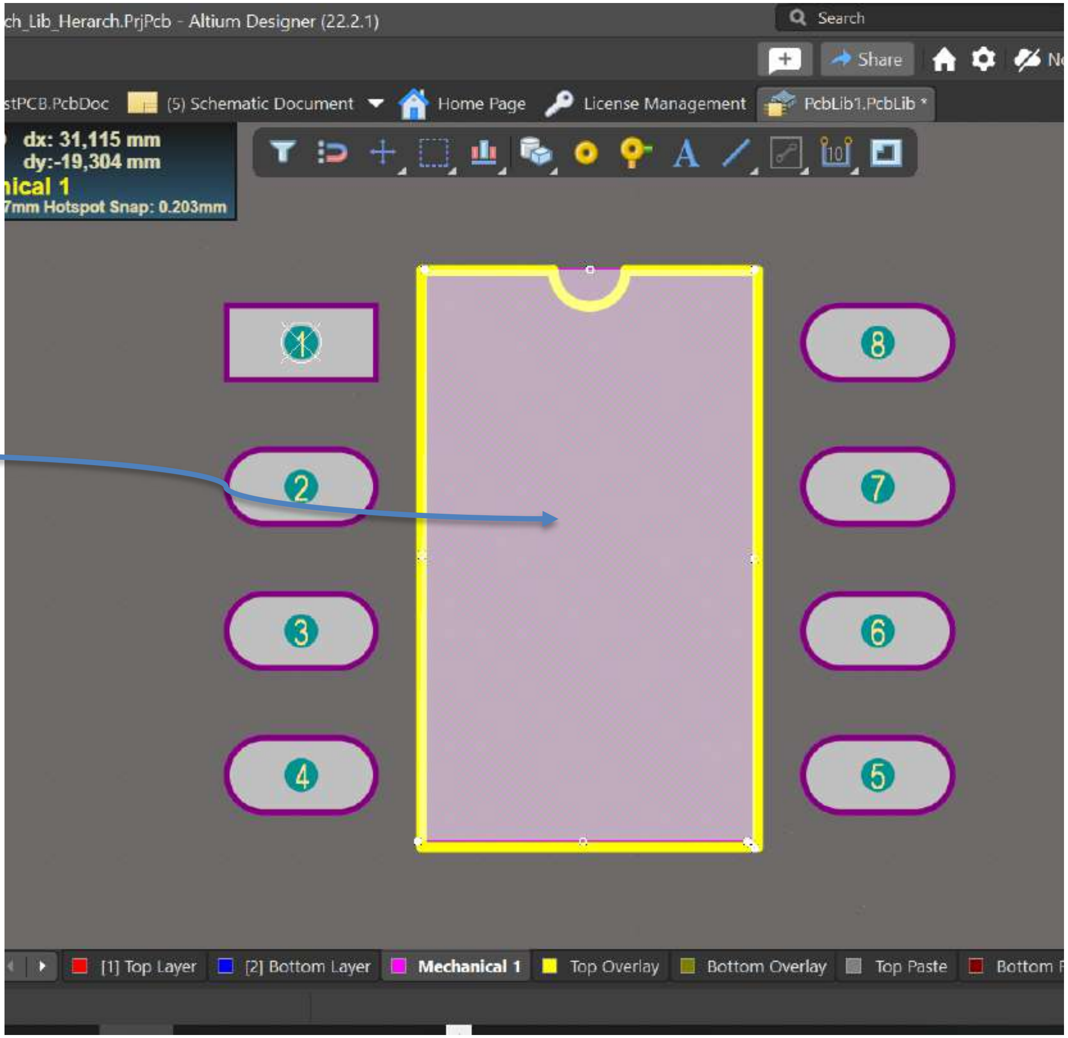


Creating a new Footprint



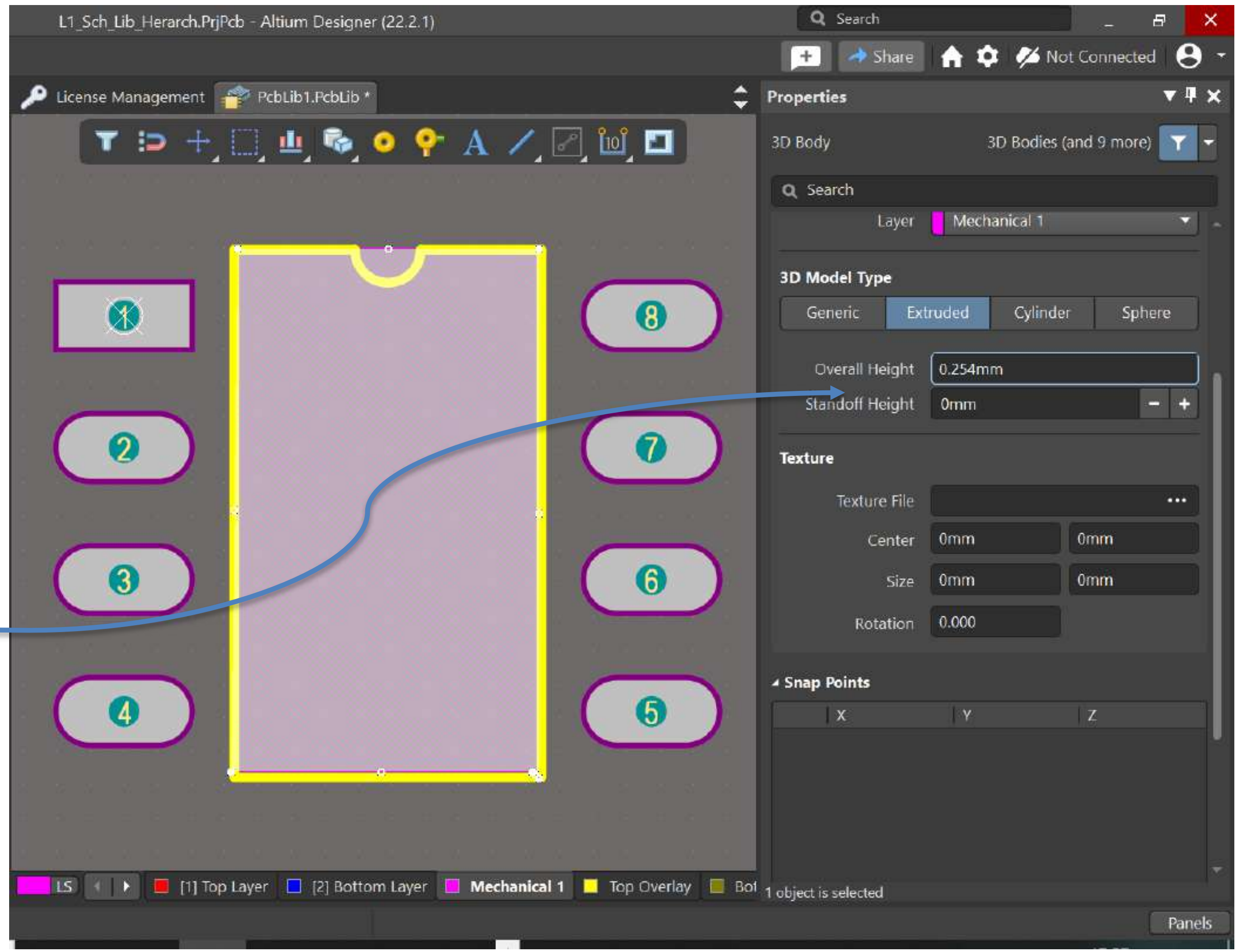
Creating a new Footprint

- ☐ This area is linked with the 3D model



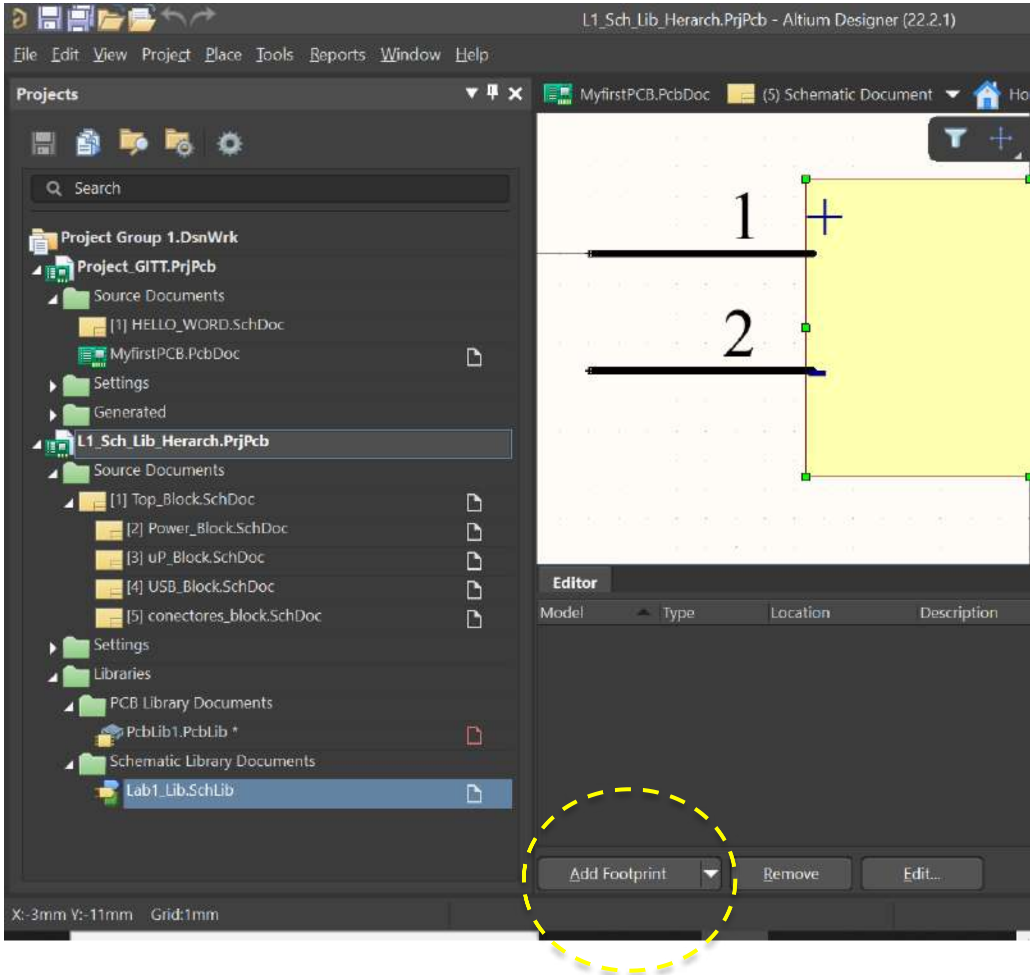
Creating a new Footprint

- We should include a height based on its datasheet



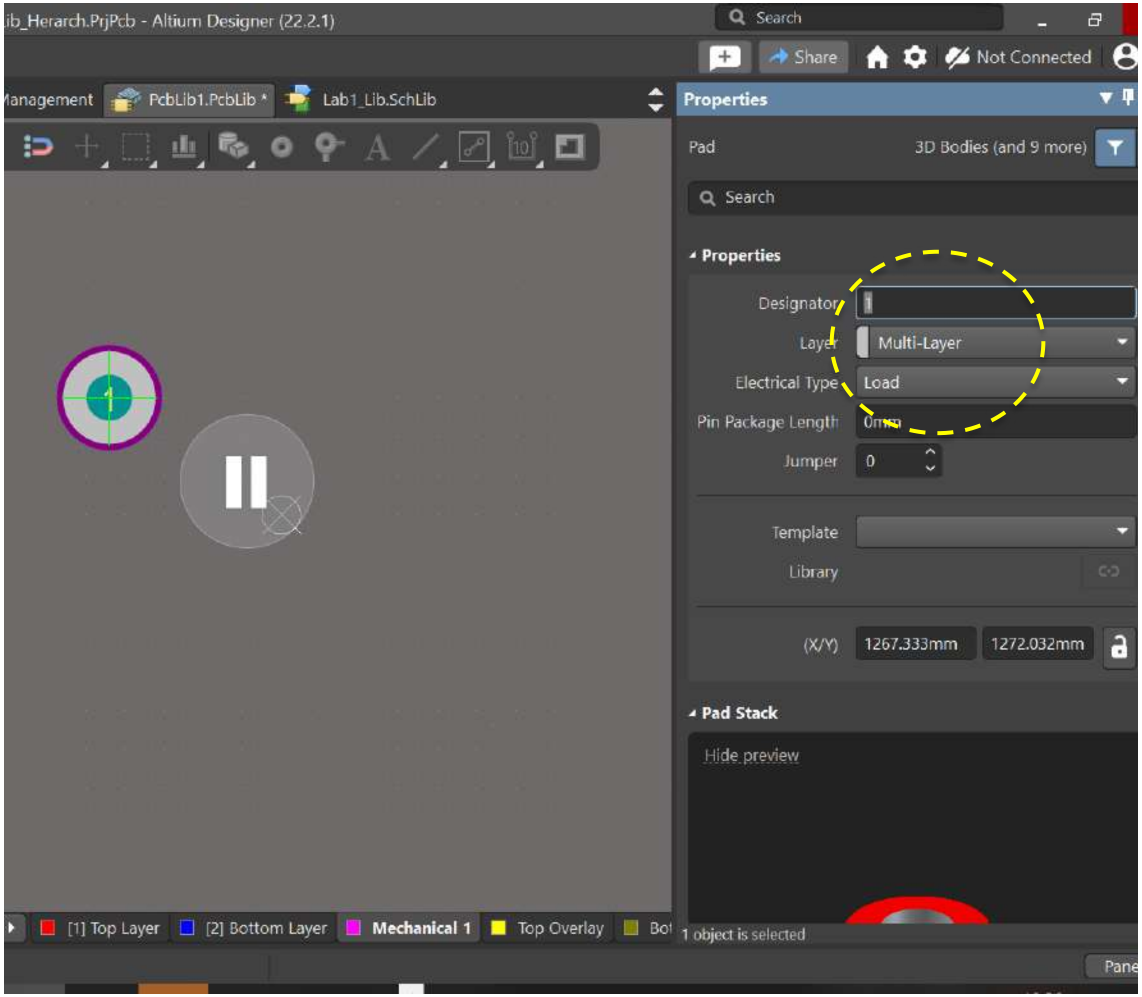
Linking the footprint with the symbol

- Footprint is ready to link with the symbol.
- We must connect the footprint with the symbol within SCHLIB.



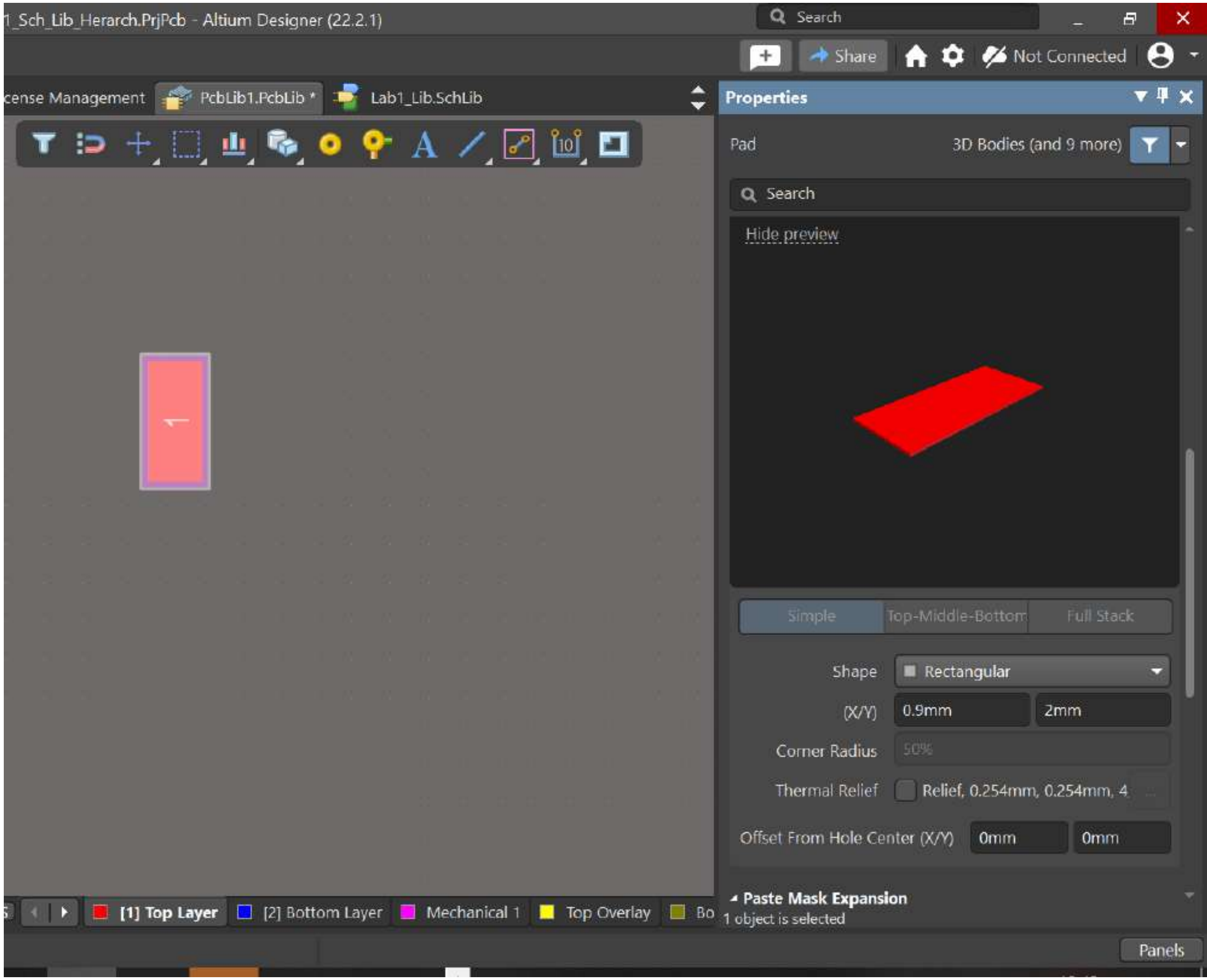
Doing a footprint without wizard

- We can create a footprint from zero without support.
- We use the Place menu to set the PADs
- Properties of the PAD allows to use through-hole or SMD types



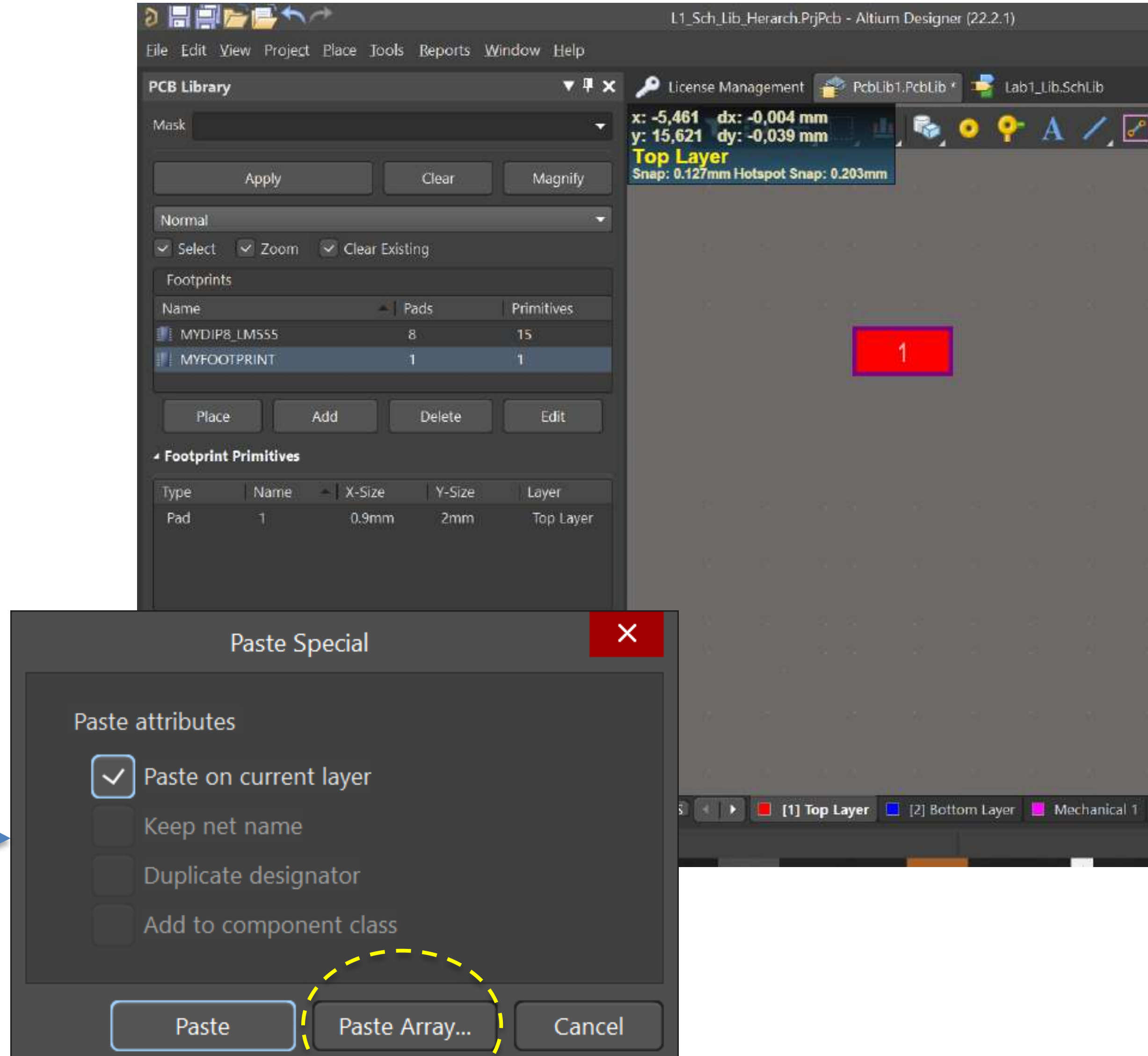
Doing a footprint without wizard

- Also, Properties of the PAD allows to define size and features of the PAD according to the datasheet



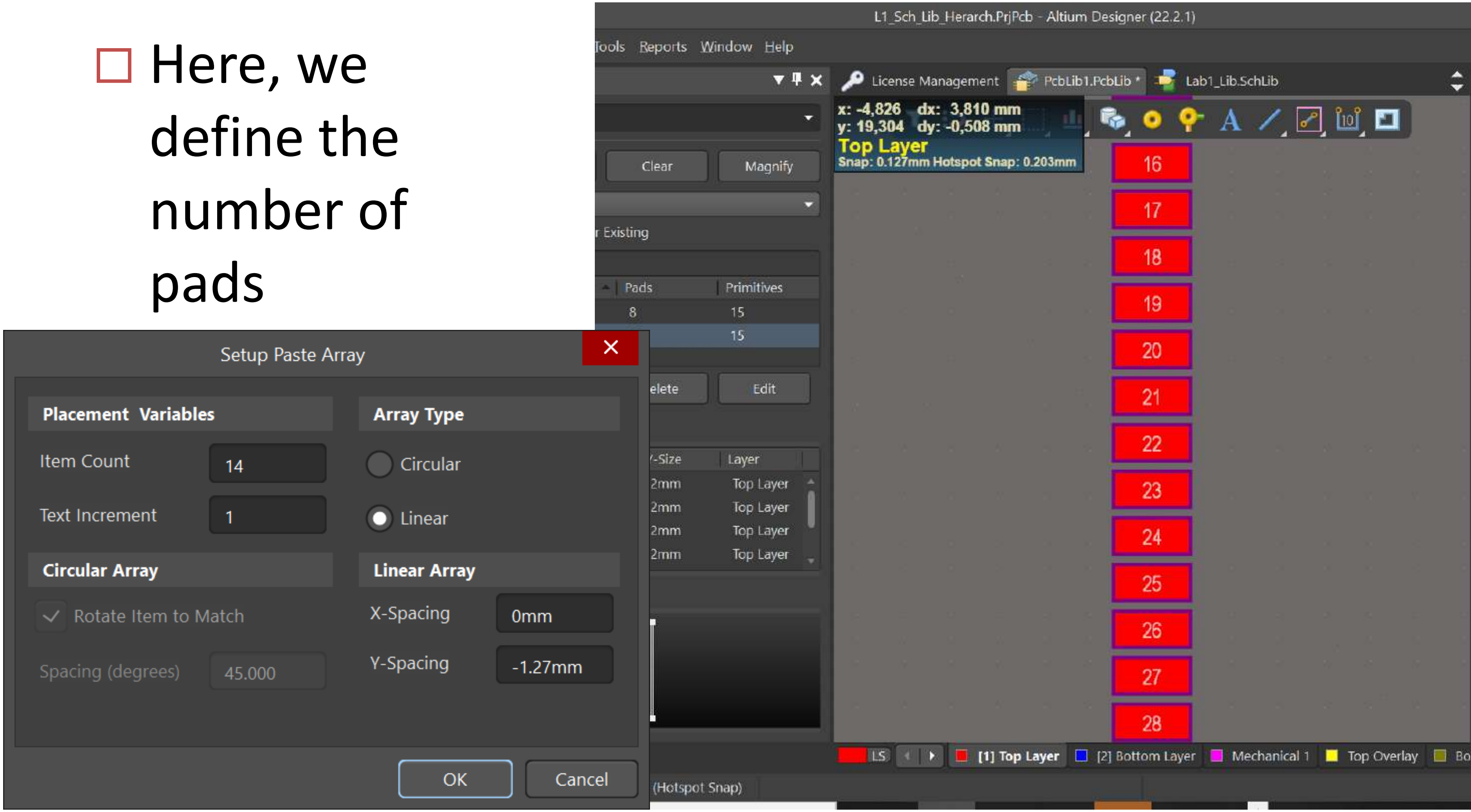
Doing a footprint without wizard

- If we want to create an array of pads:
- Edit->Copy. We match the green cursor in the PAD
- Then, Edit->Paste Special



Doing a footprint without wizard

- Here, we define the number of pads



Doing our footprint21

- You must create a footprint for a LM555 with DIP8 shape.
- Take the datasheets and create this footprint in a PCB Library

Doing our footprint2

- Trying to do the ESP32 footprint.
- You should take care with the lengths and distances.

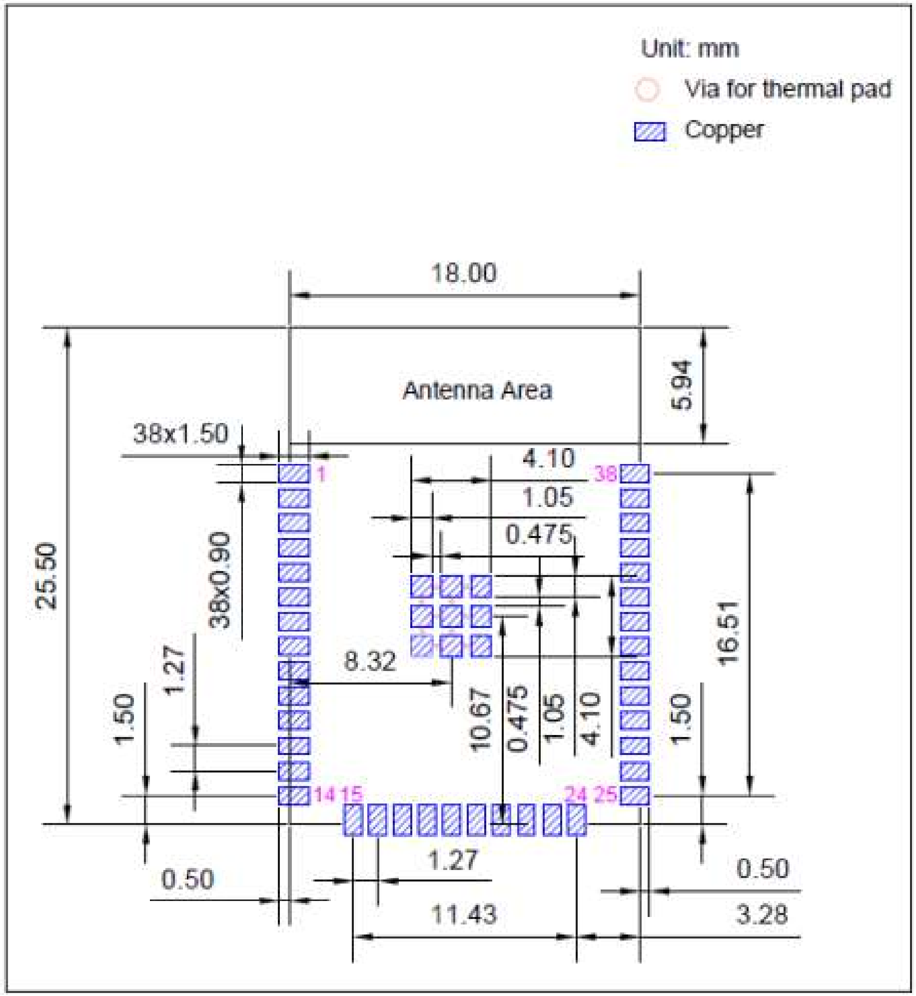
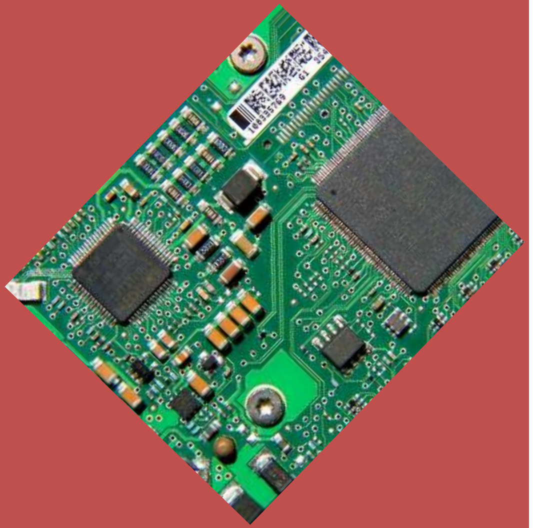


Figure 6: Recommended PCB Land Pattern

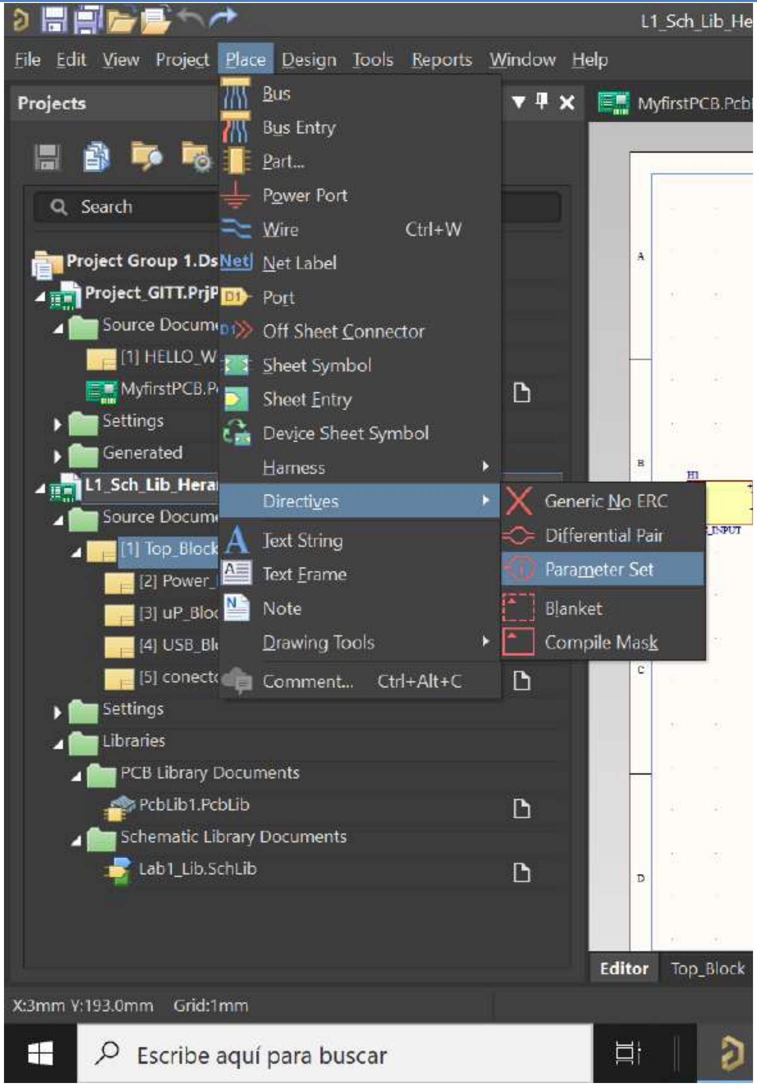


Introduction to Printed Circuit Board (PCB) design with ALTIUM

6.- PCB Advanced

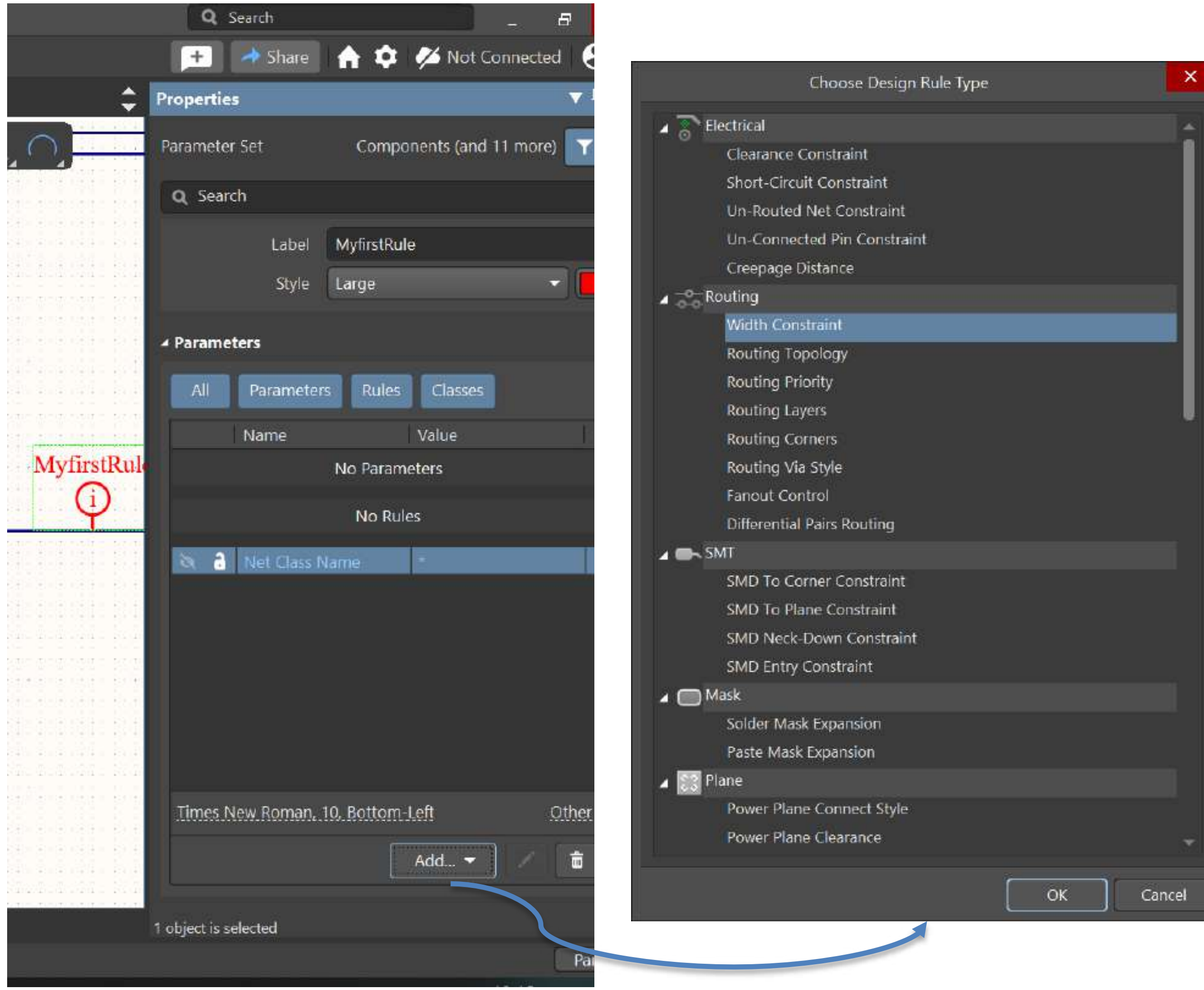
Rules

- The definition of rules for a set of nets can simplify our PCB design.
- It must be asserted in SCH Sheets.
- We will use the Place menu -> Directives-> Parameter Set



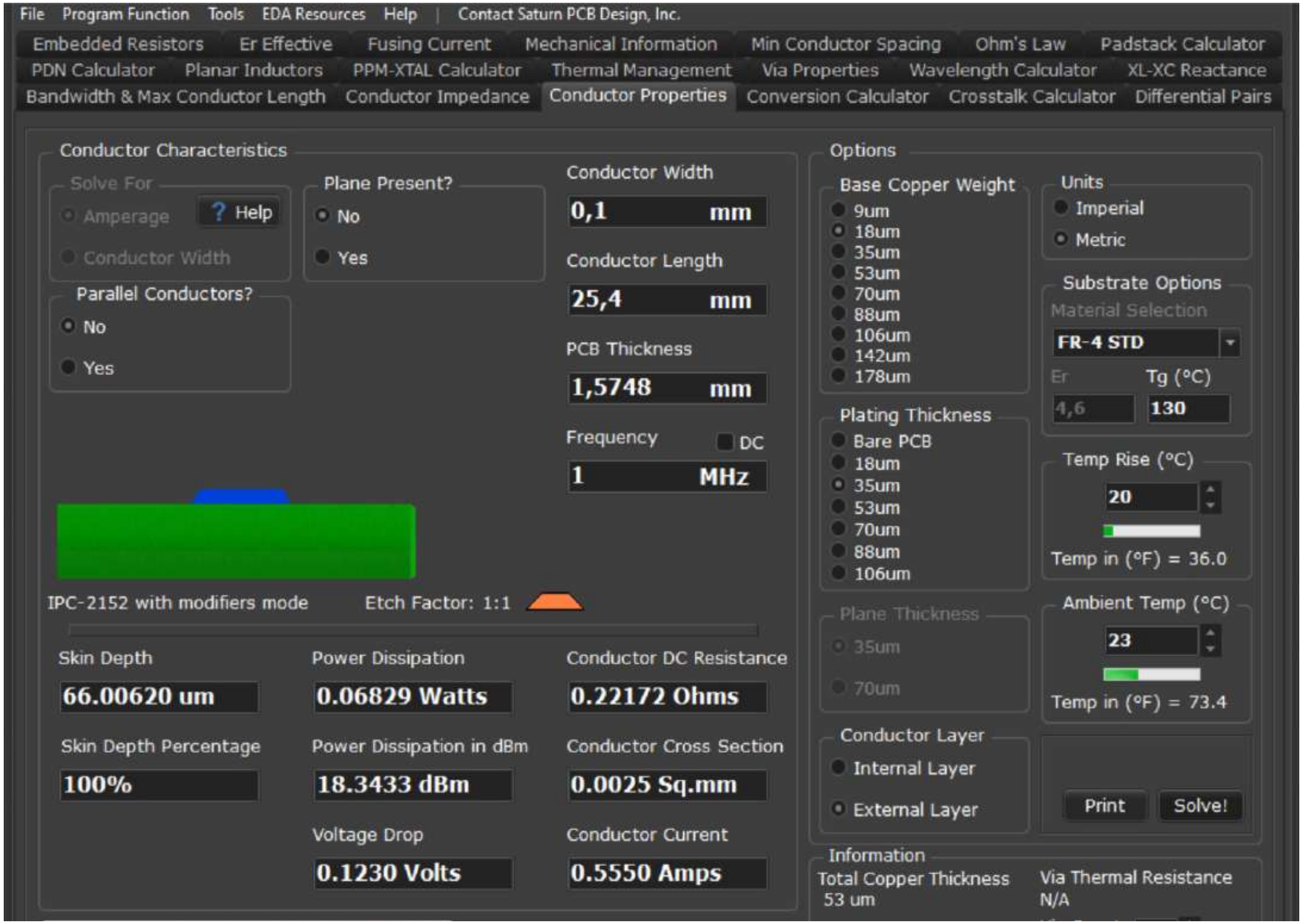
Rules

- We change the rule net.
- Also, we can create a class and we link to the class a Rule.
- We can assign the width of the PCB Track.



Electrical rules

- We can check the maximum current (SATURN PCB)



Autorouting

- **AutoRoute** is an automatic tool for routing.
- We can route the whole PCB, the rules, classes, etc.
- Is it magical? No it could be convenience when we are not able to find solutions in a specific area.